Model Structure (26.2_FAA):

This is a Fleets−as−Areas model (do_spatial == 1).
There are 2 sexes, 2 fishing fleets, 5 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey did NOT use off−year extrapolations for the BS (even years) or AI (odd years).
Only the non−extrapolated area RPNs were included in the index (drop_LLS_extrap == 1 & do_LLS_carry == 0).
The FAA model is not using a carryover version of the LLS RPN index (do_spatial == 1 & do_carry_FAA == 0).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

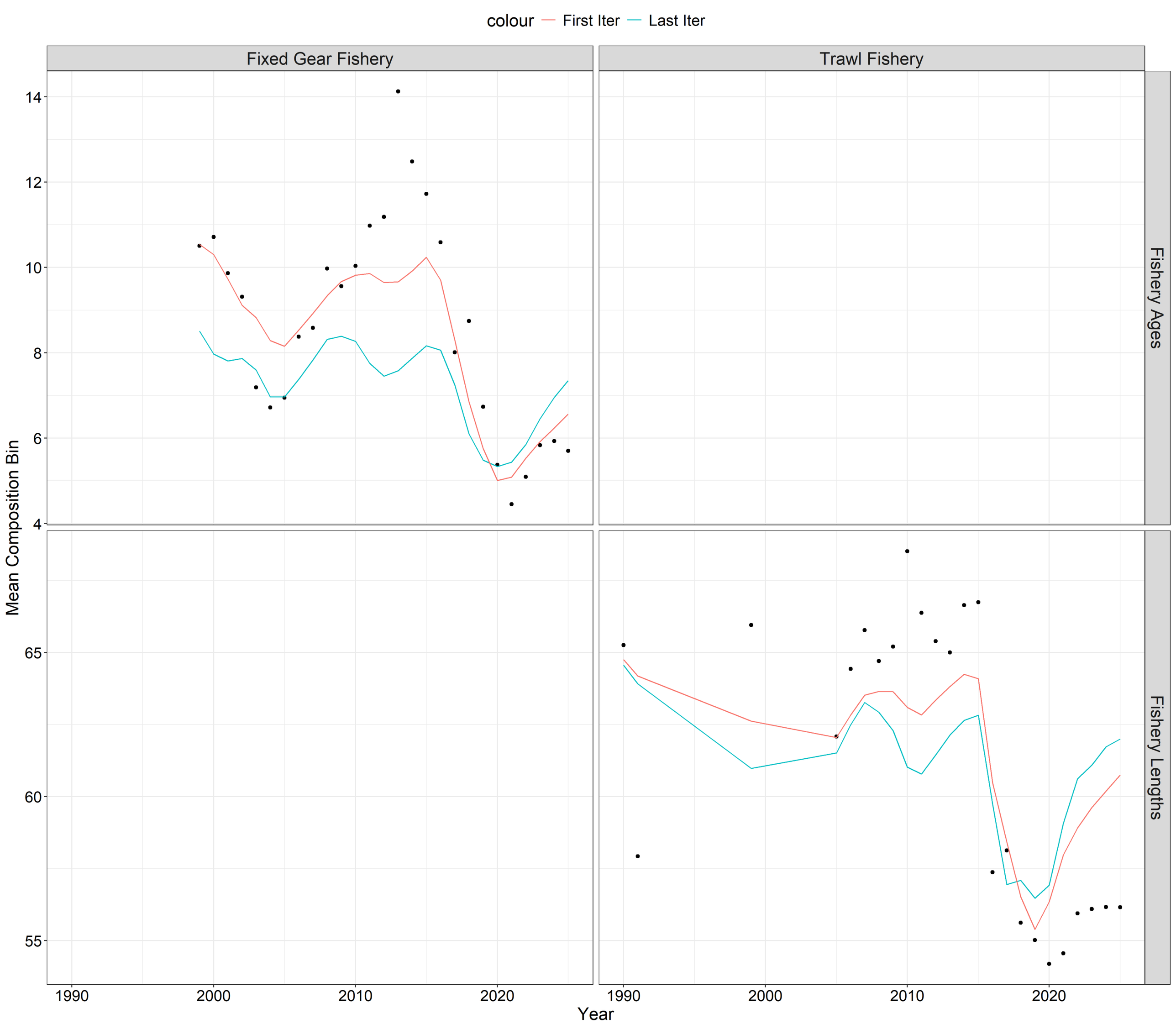
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial > 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes/fleets to address interannual variability (do_spatial == 1 & do_scale_ISS == 1).

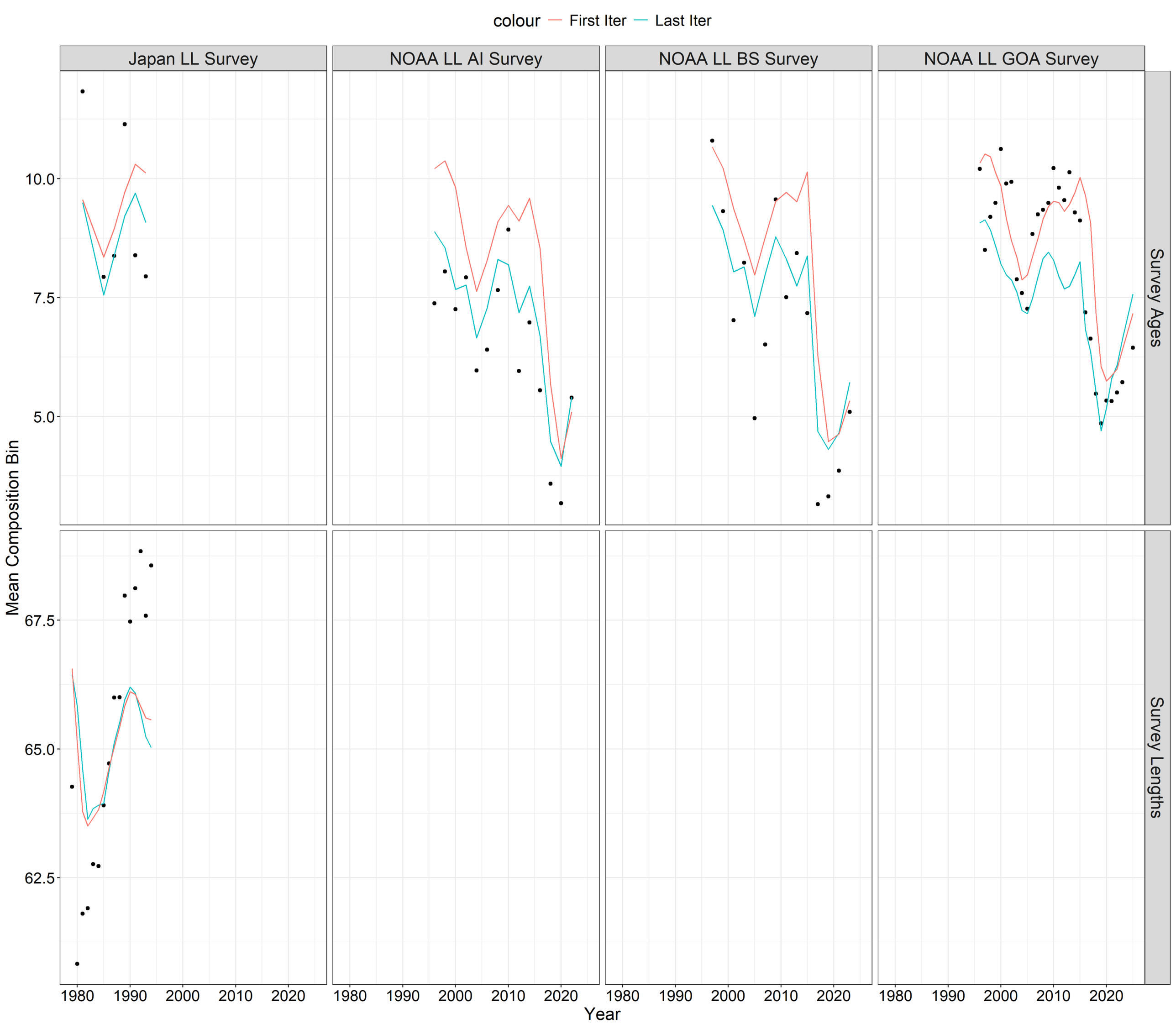
Length Composition Data Use:

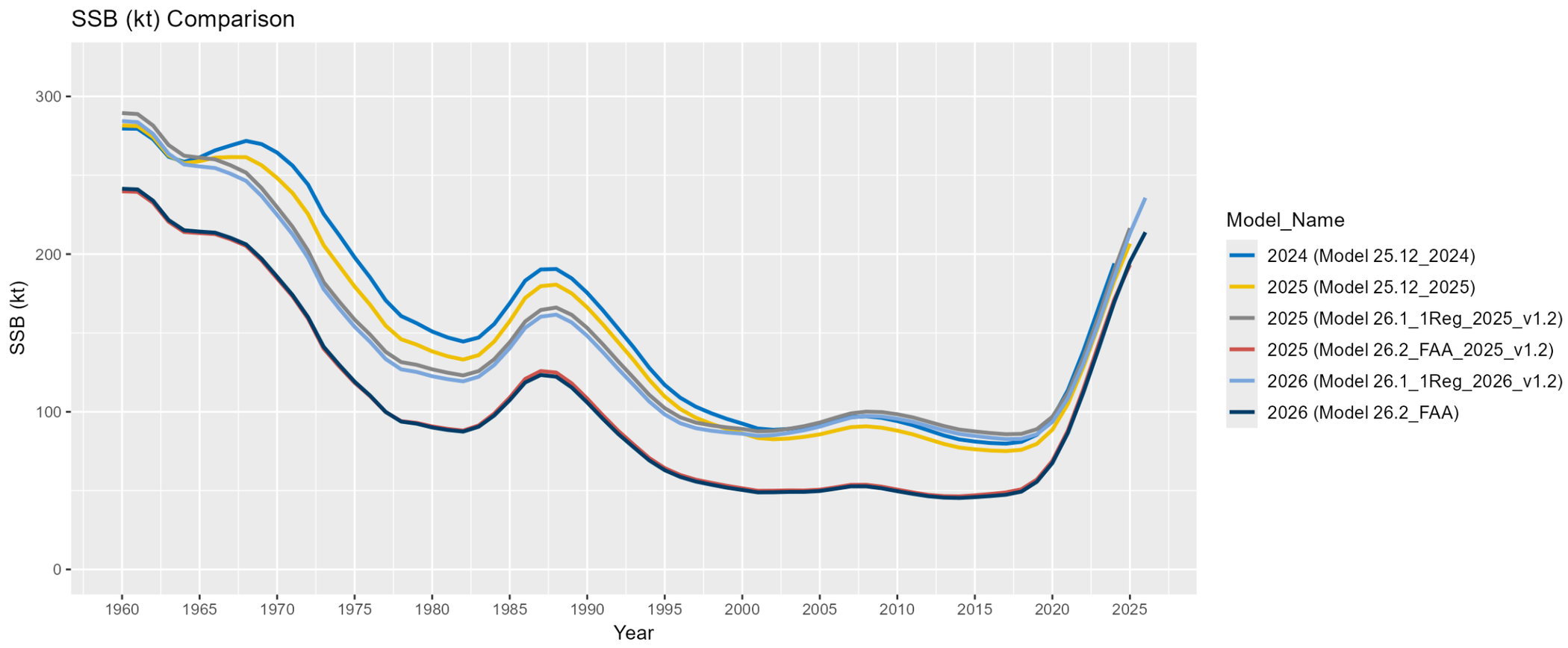
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

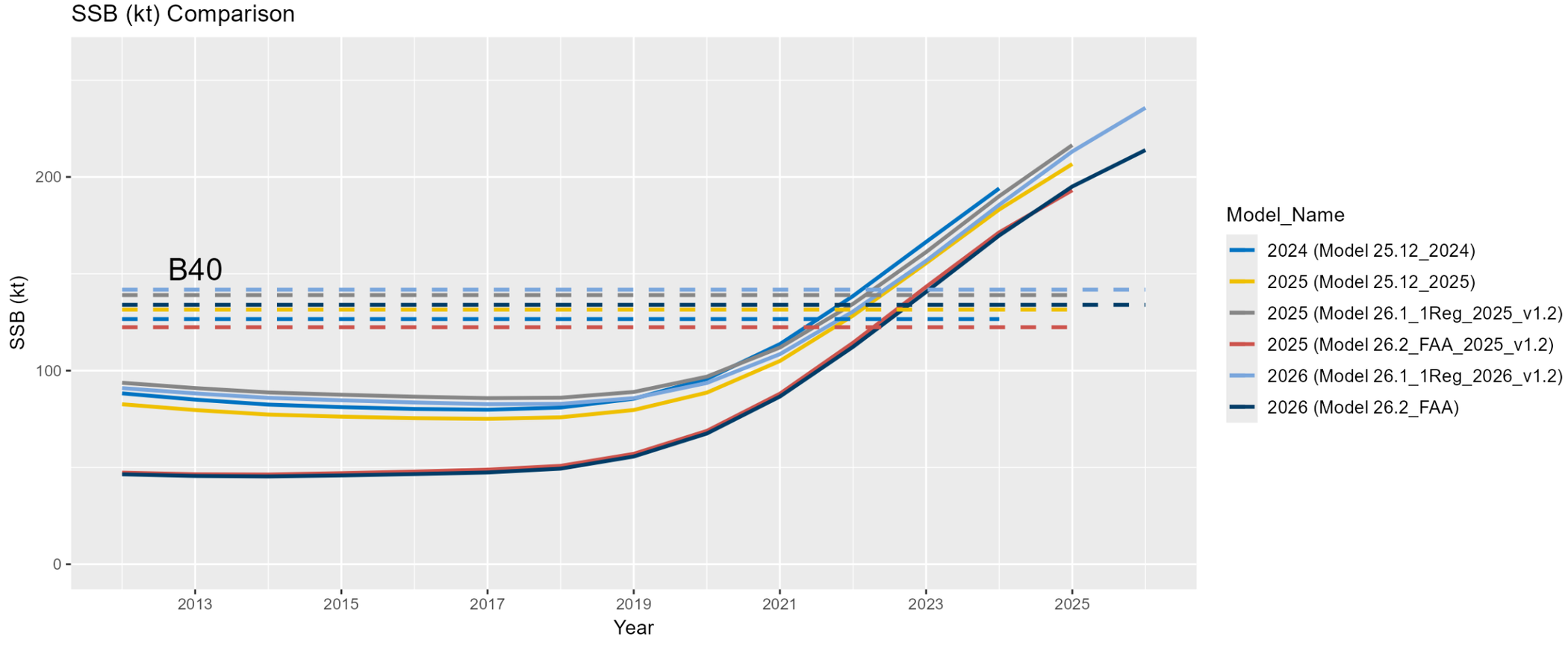
General Data Fitting Notes:

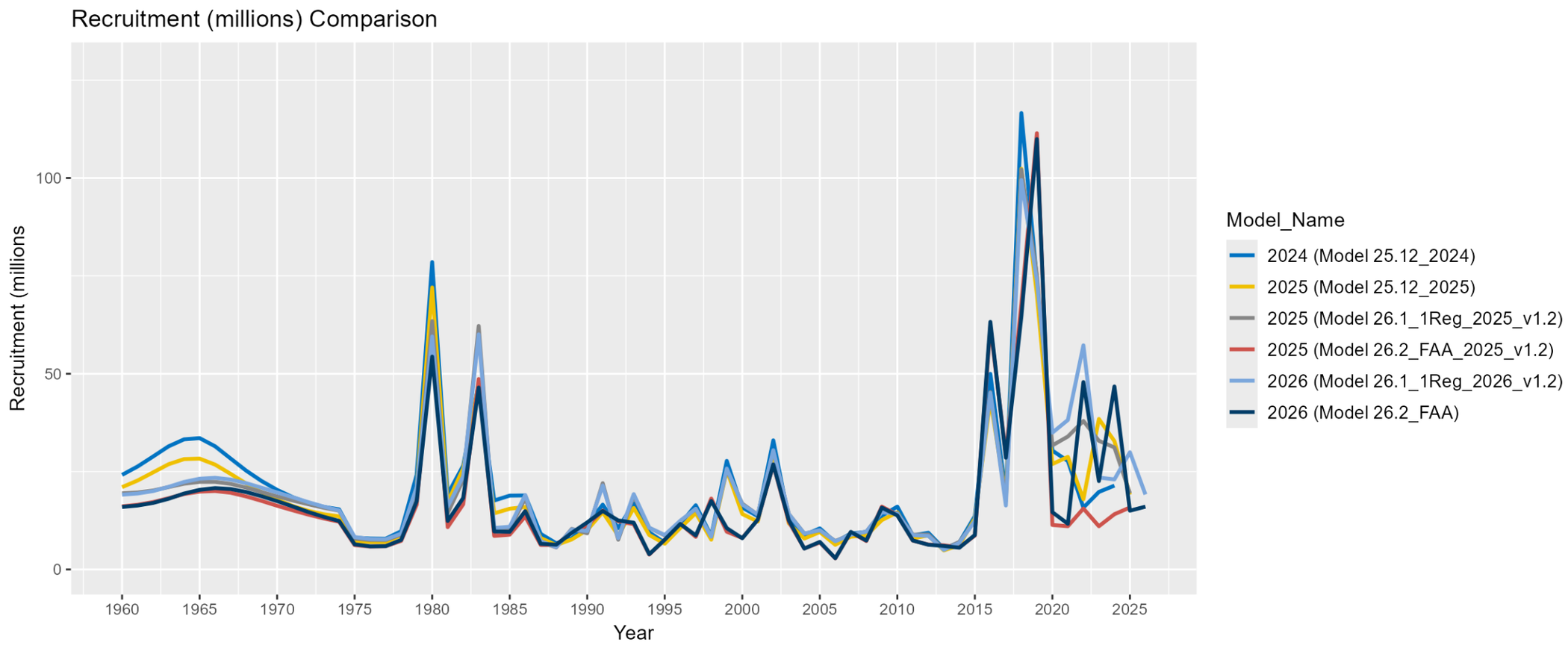
Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where all sources of removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 1).

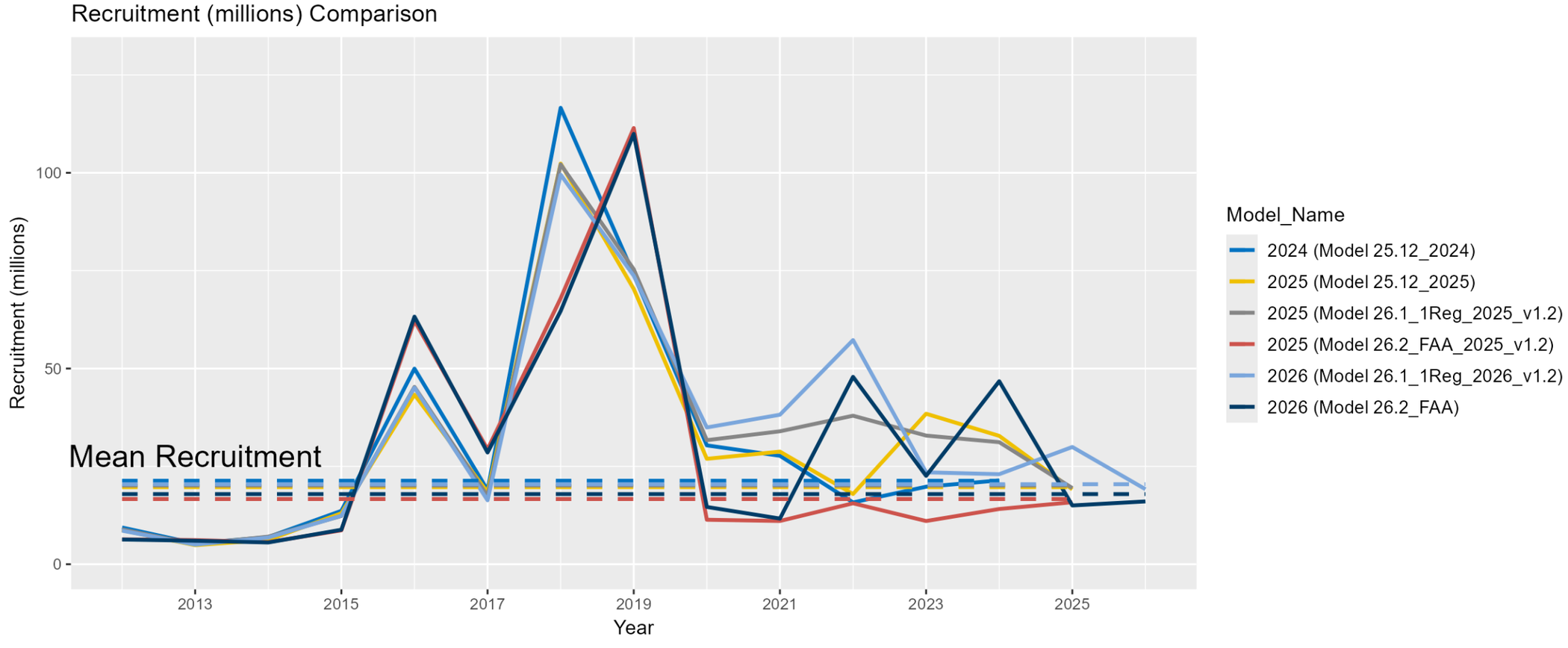

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year

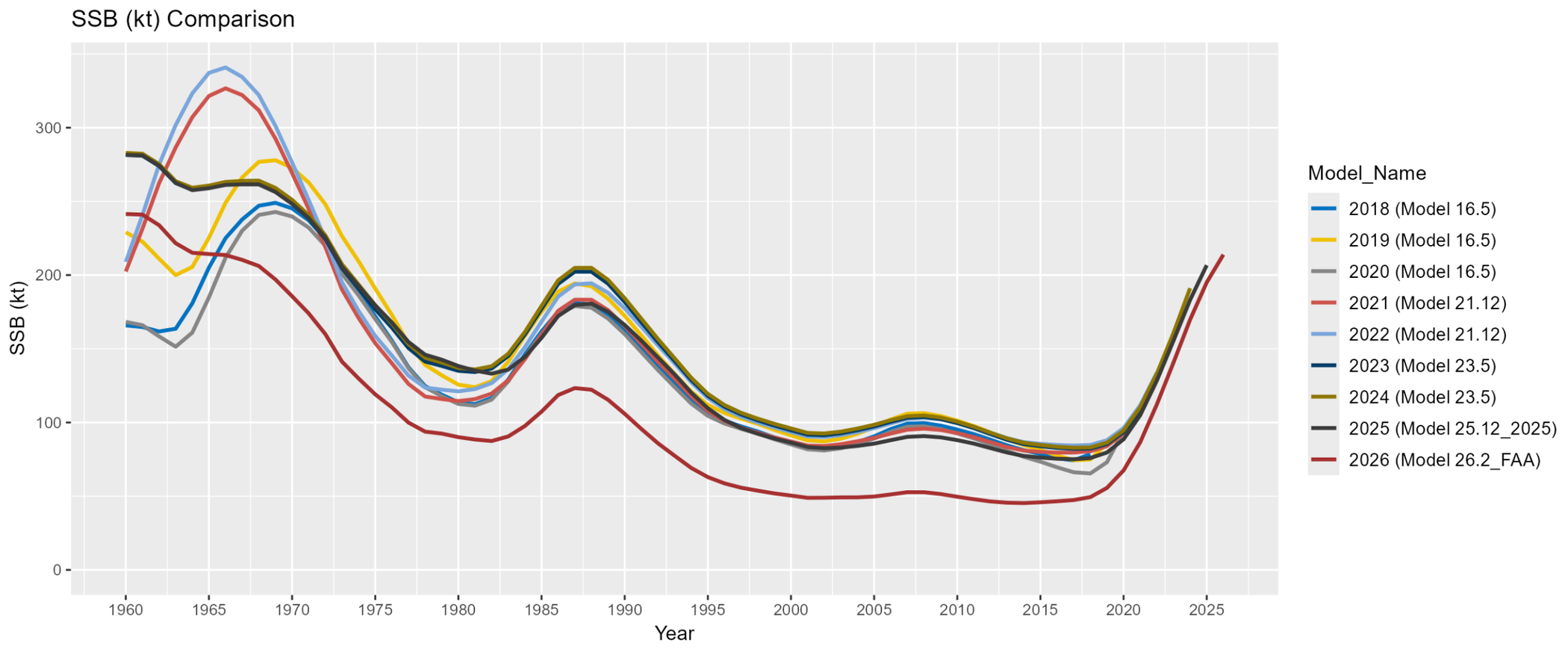

colour
First Iter
Last Iter
Japan LL Survey
NOAA LL AI Survey
NOAA LL BS Survey
NOAA LL GOA Survey
Survey Ages
Survey Lengths
Mean Composition Bin
10.0
7.5
5.0
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year

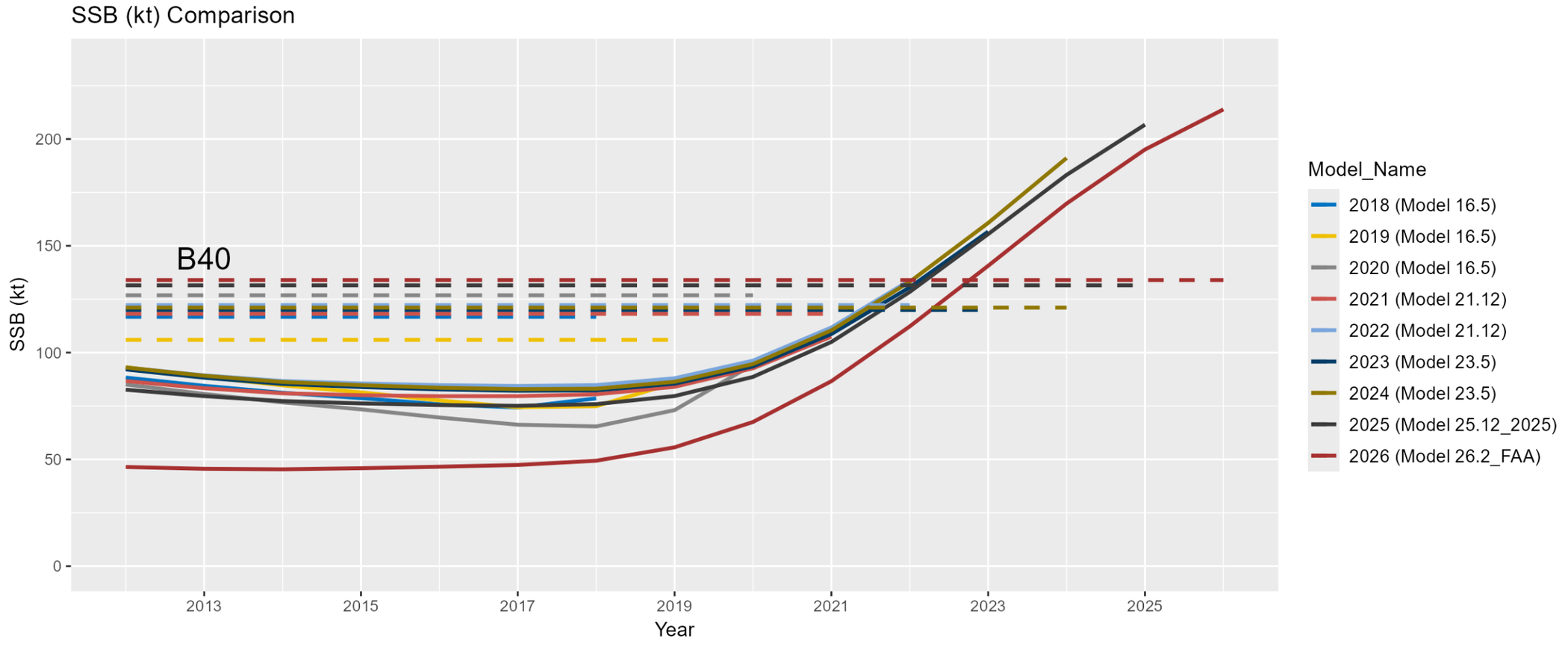

SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA)

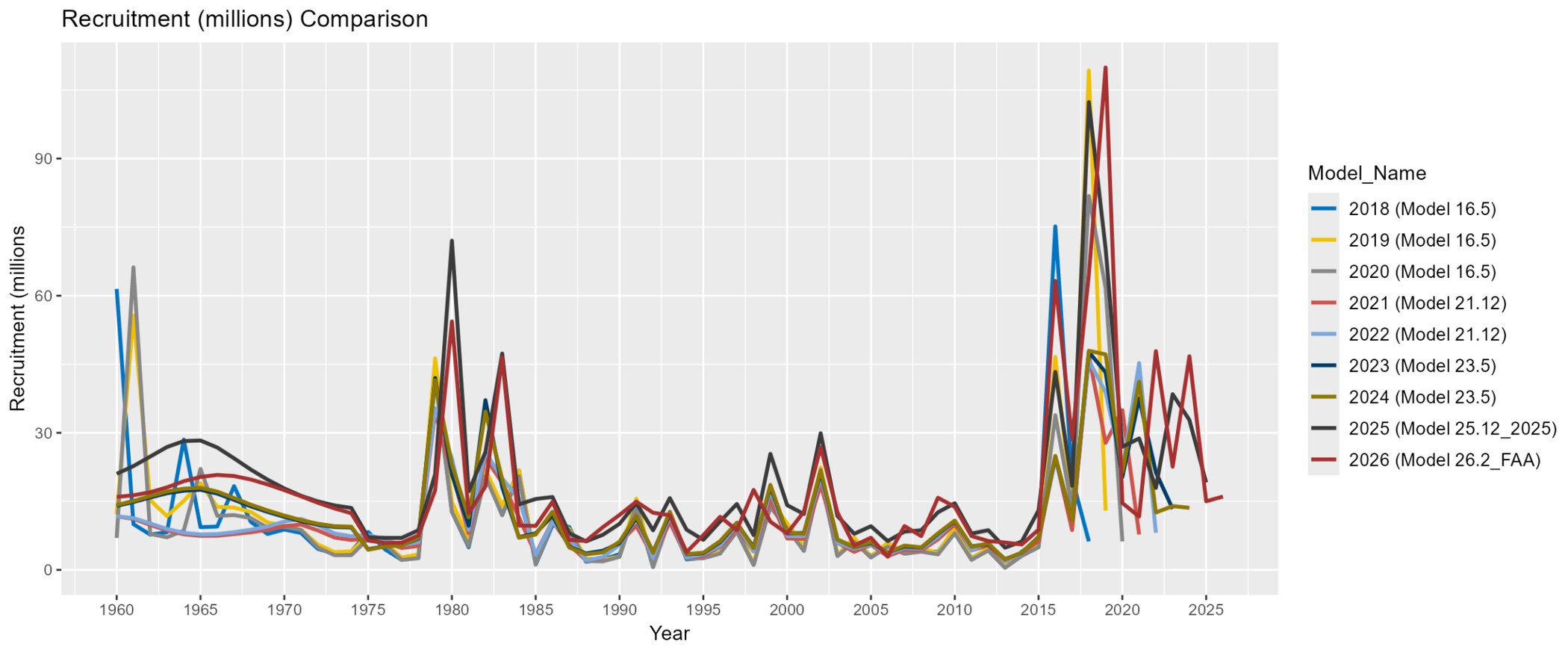

SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA)

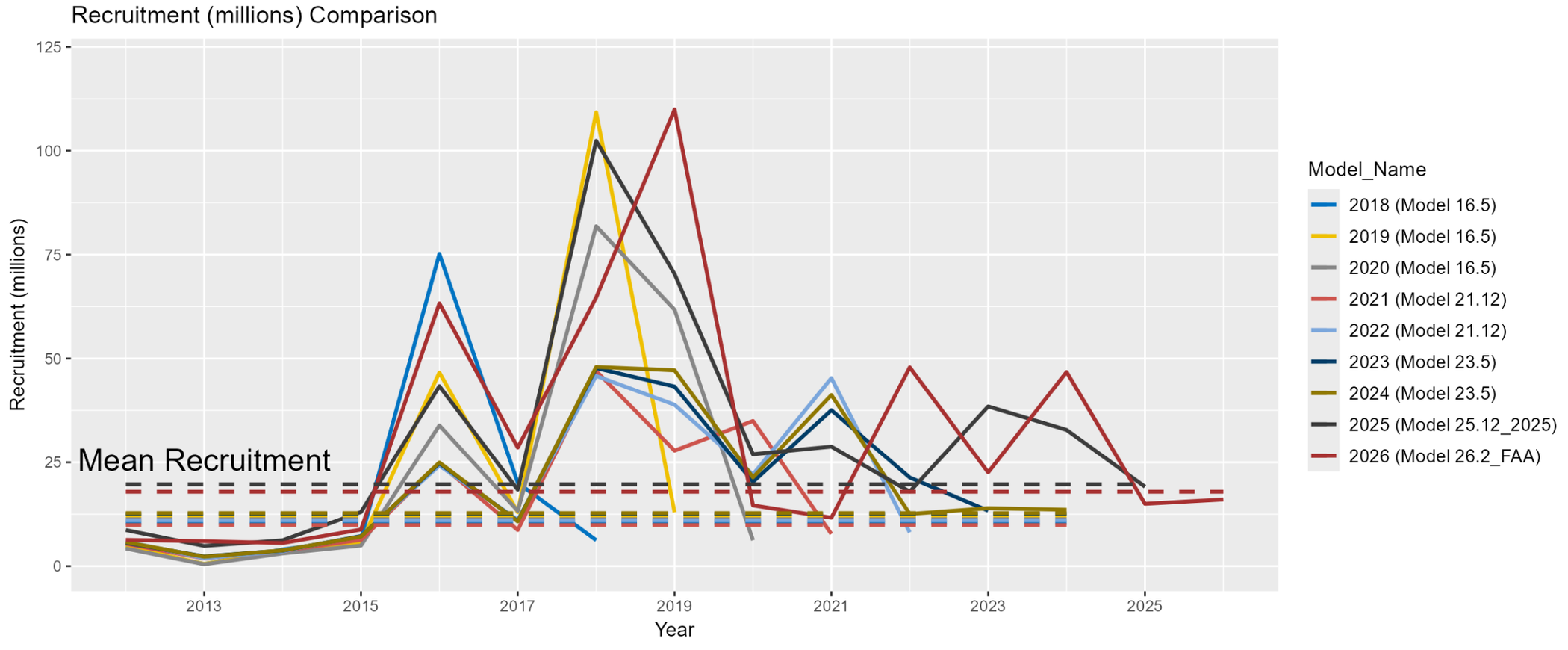

Recruitment (millions) Comparison
Recruitment (millions
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA)


Recruitment (millions) Comparison
Recruitment (millions)
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA)


SSB (kt) Comparison
B40
SSB (kt)
200
150
100
50
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA)


Recruitment (millions) Comparison
Recruitment (millions
90
60
30
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.3009

SSB

Mohns Rho: 0.0247

AK

Relative Difference from Terminal Year

0.0

-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

90
60
30
0

250
200
150
100
50
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

90

60

30

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

2019

2020

2021

2022

2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.3008

SSB

Mohns Rho: 0.0558

AK

Relative Difference from Terminal Year

0.0

-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

AK

Recruitment (millions)

90

60

30

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

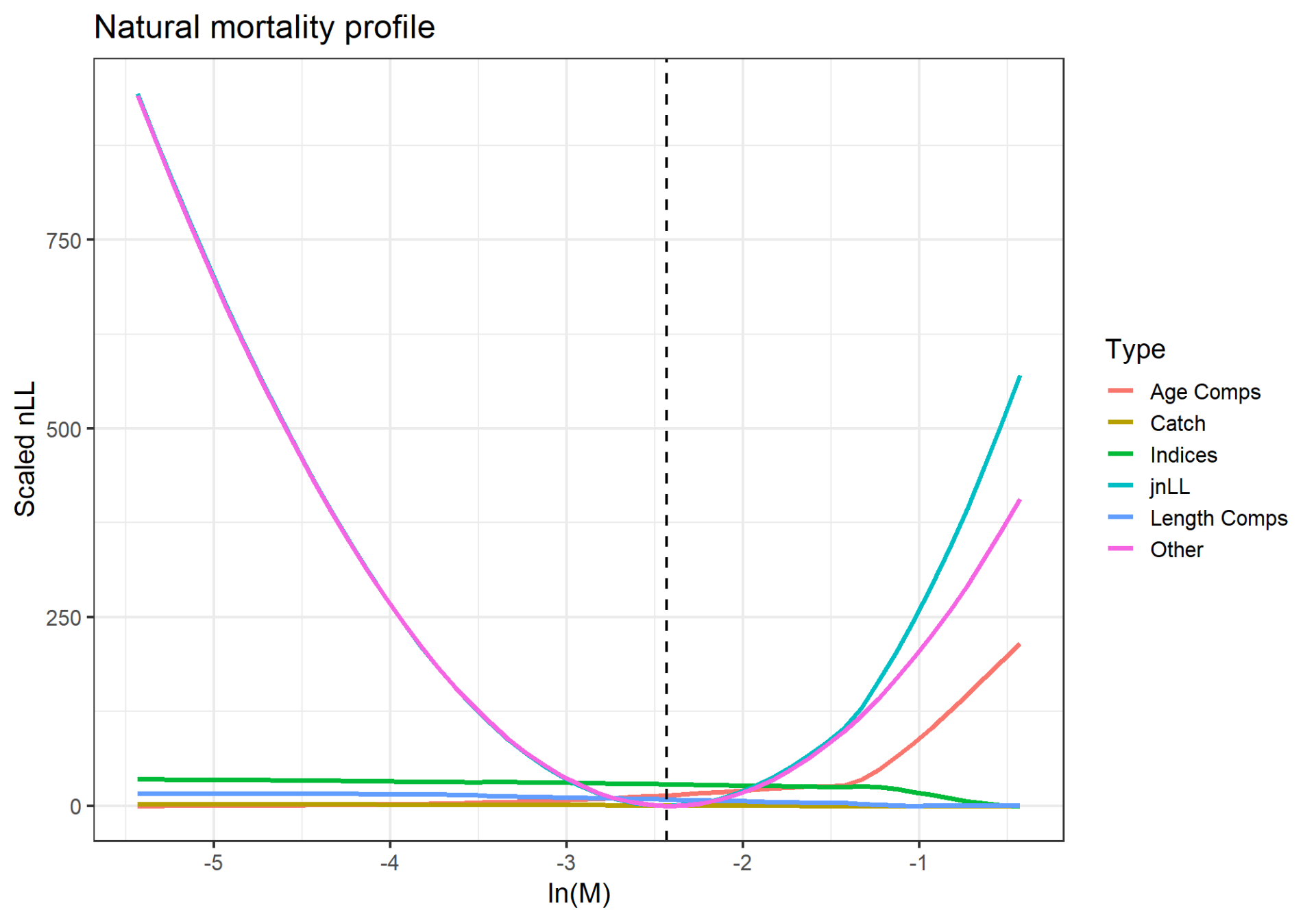
2019

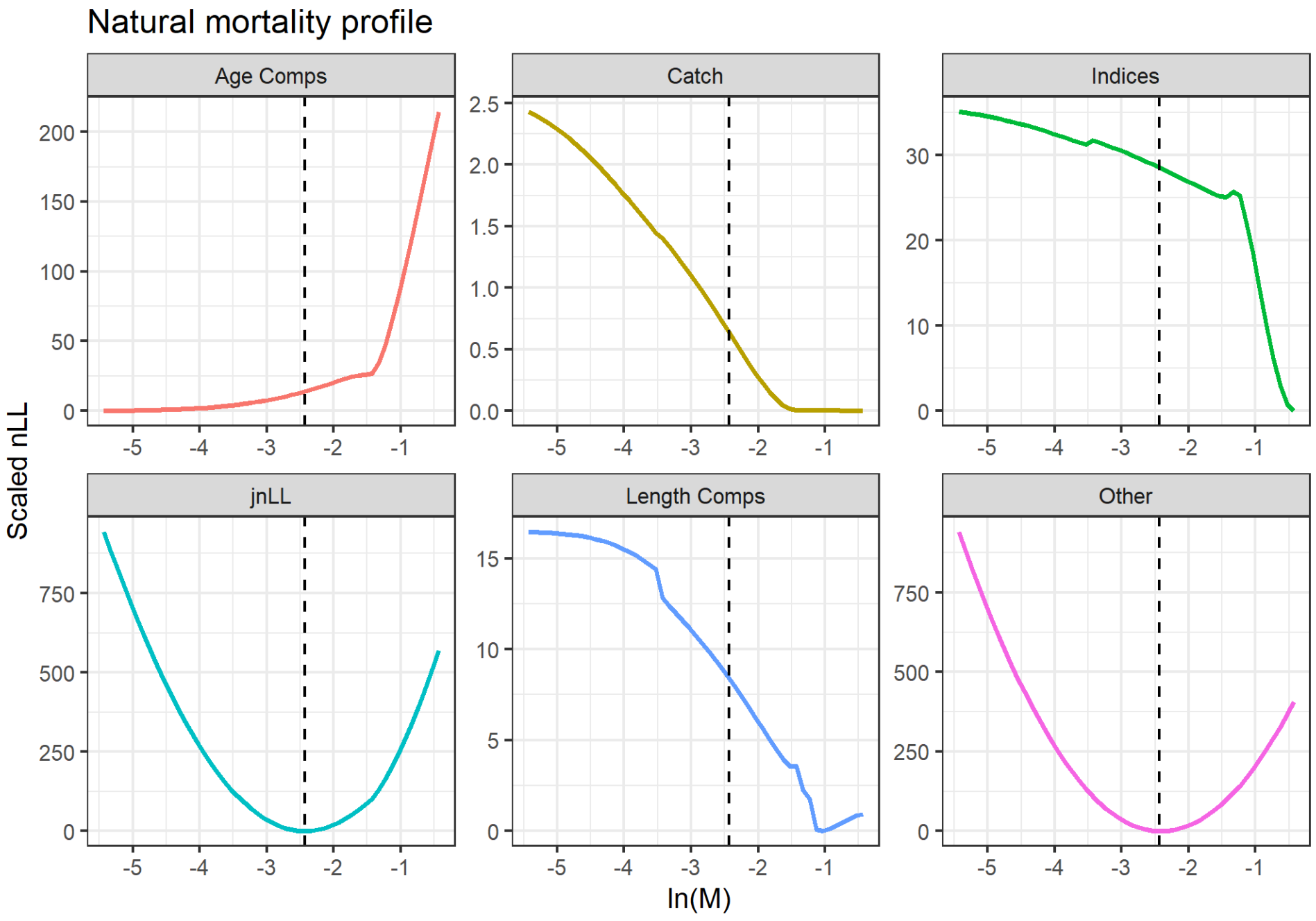
2020

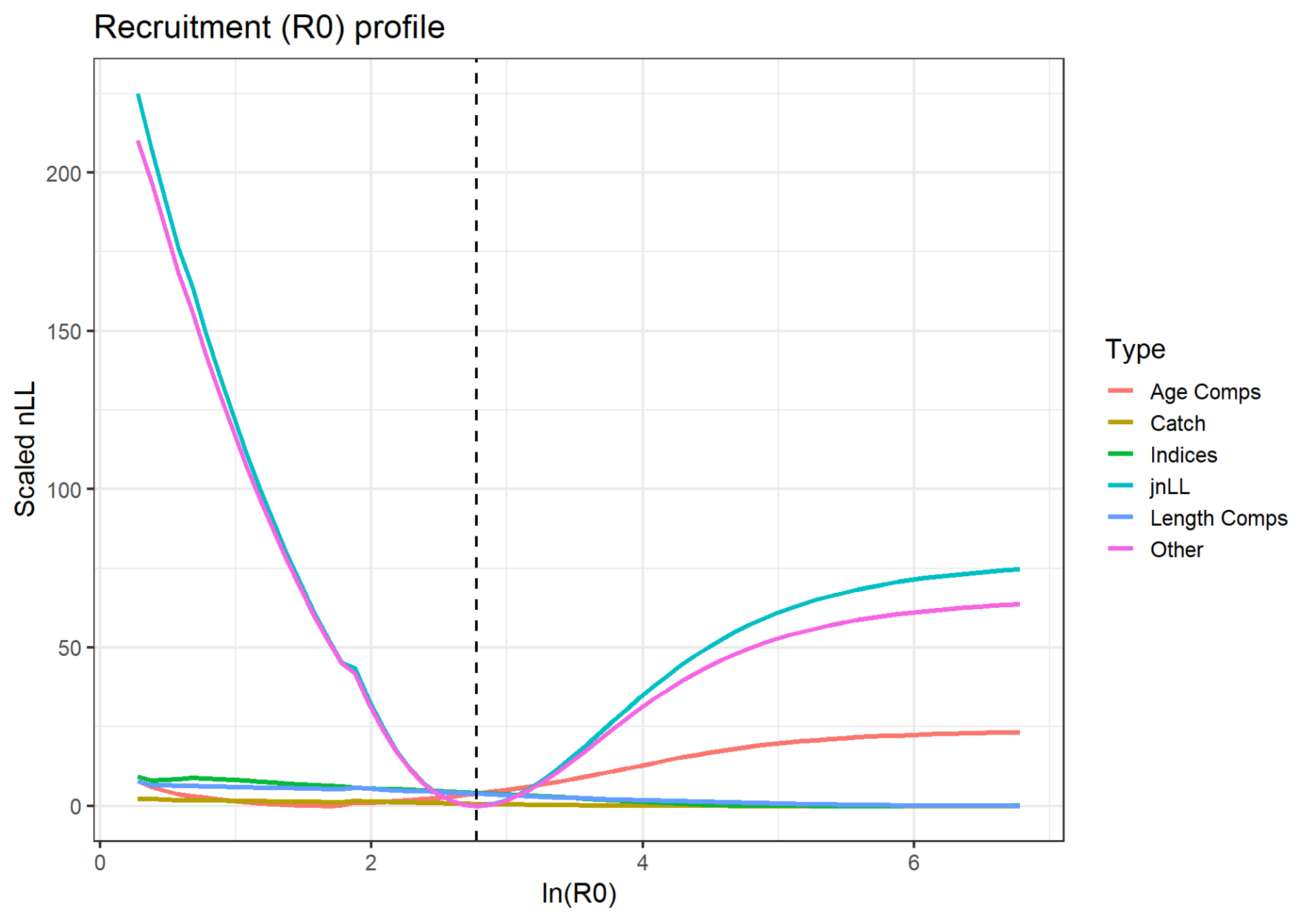
2021

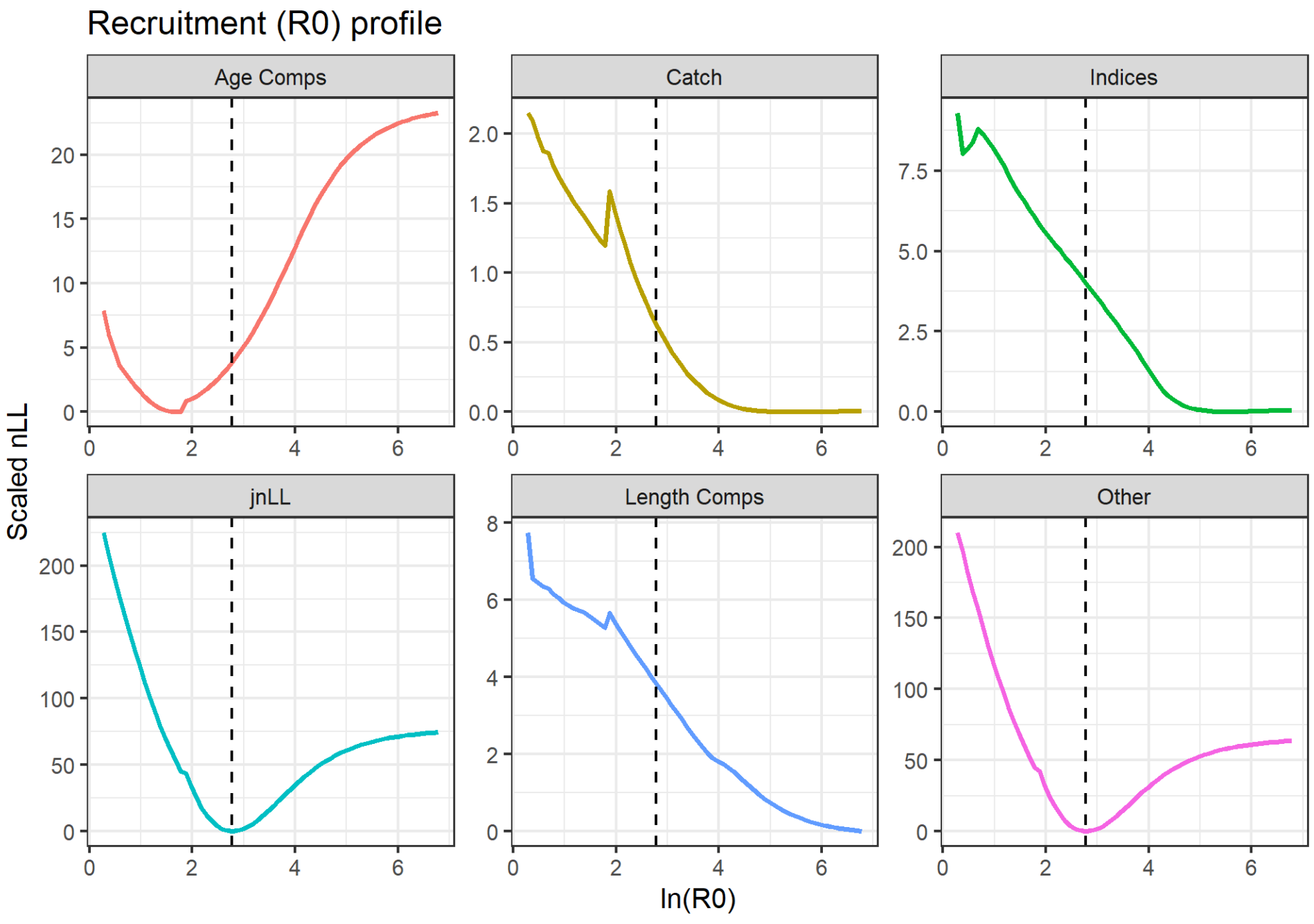
2022

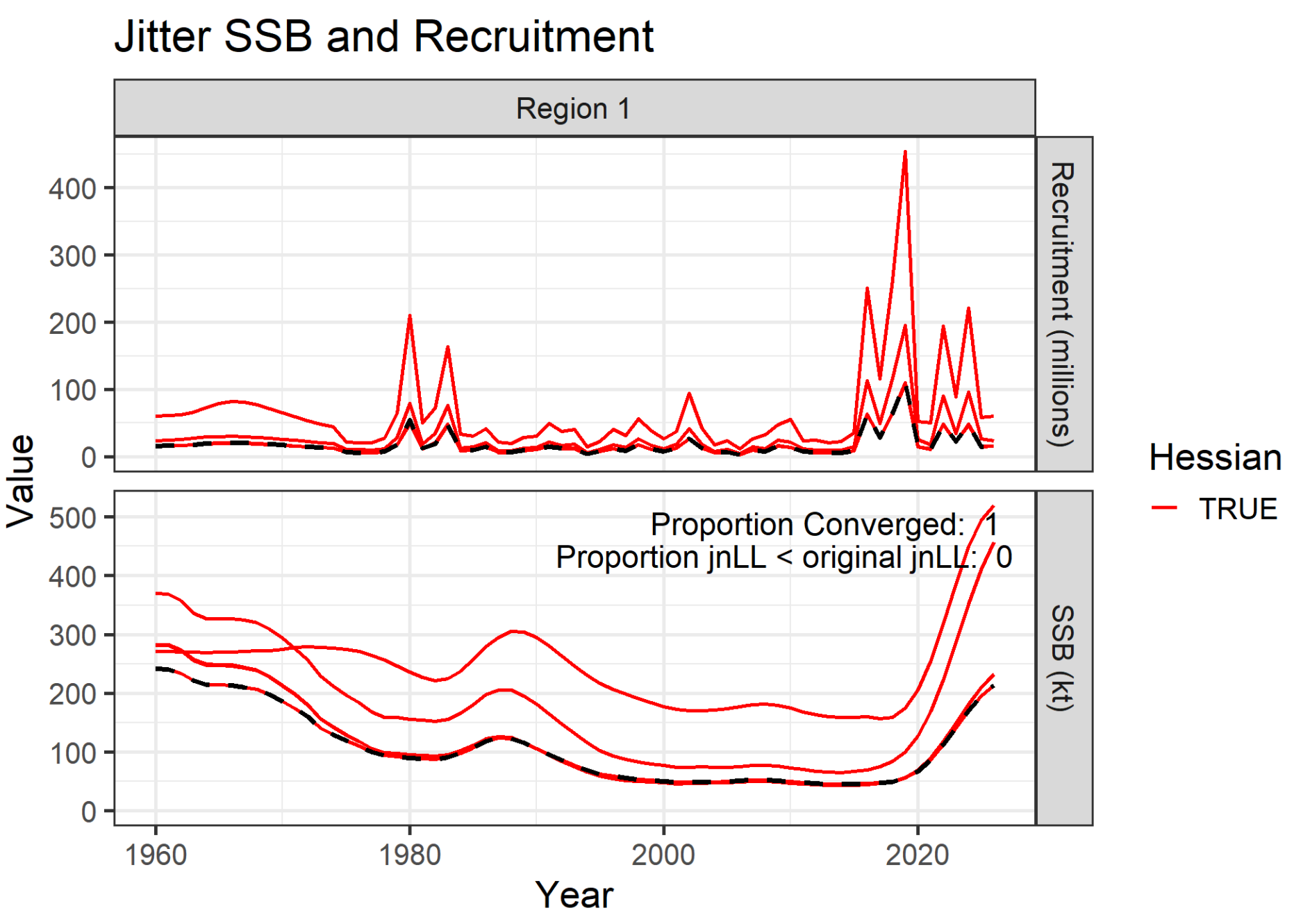
2023

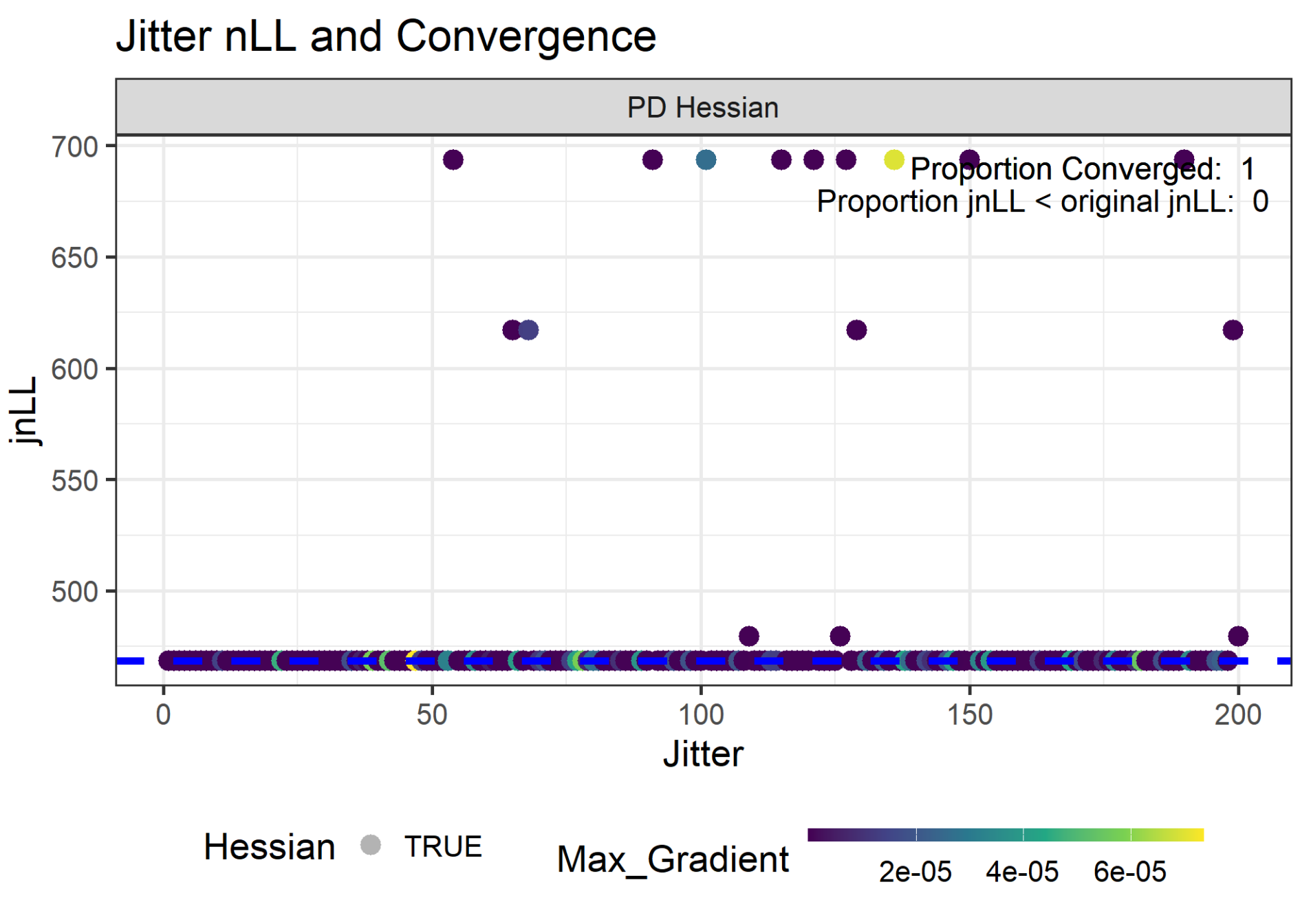

Natural mortality profile
Scaled nLL
750
500
250
0
-5
-4
-3
-2
-1
ln(M)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

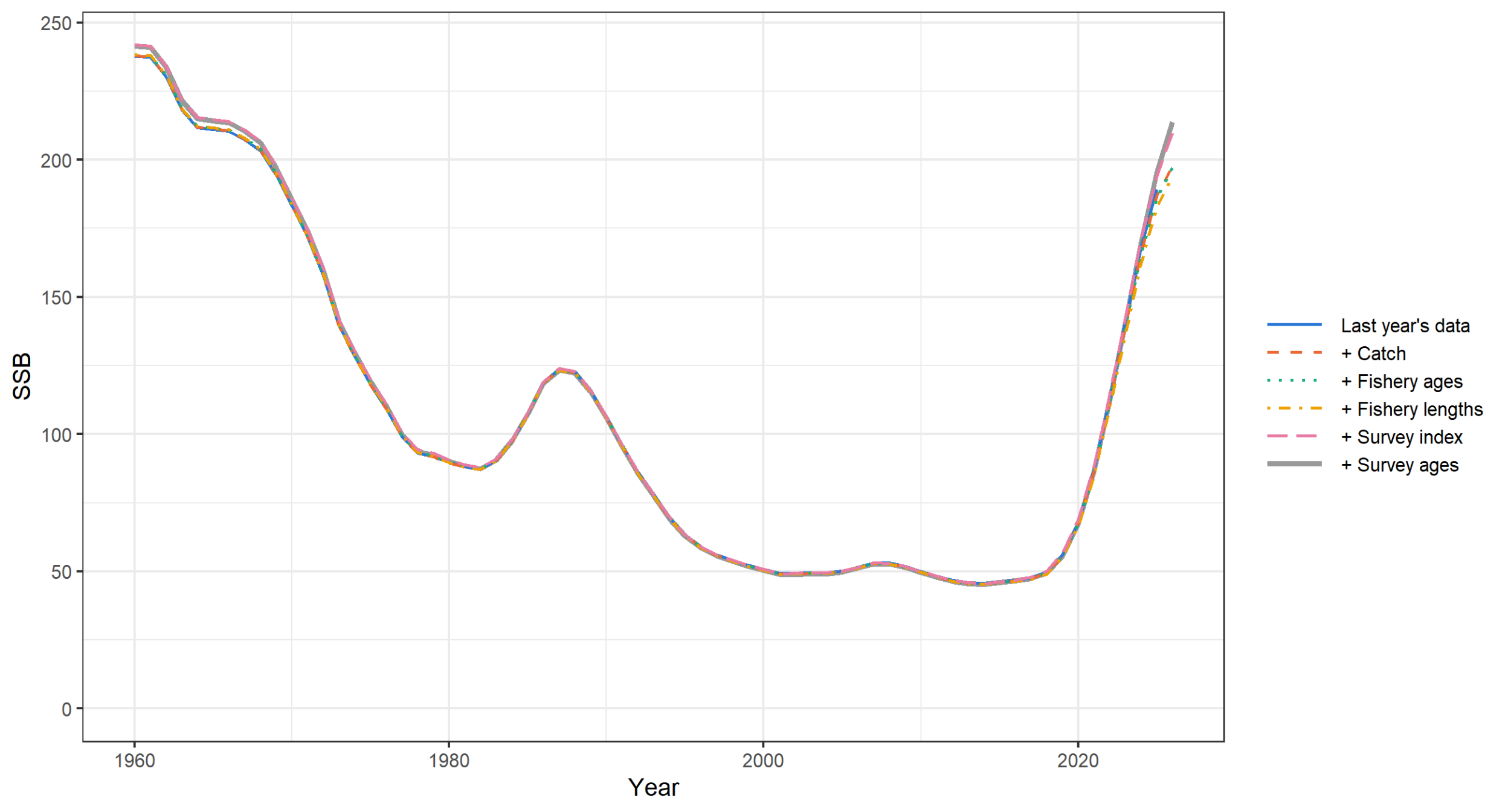

Natural mortality profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(M)

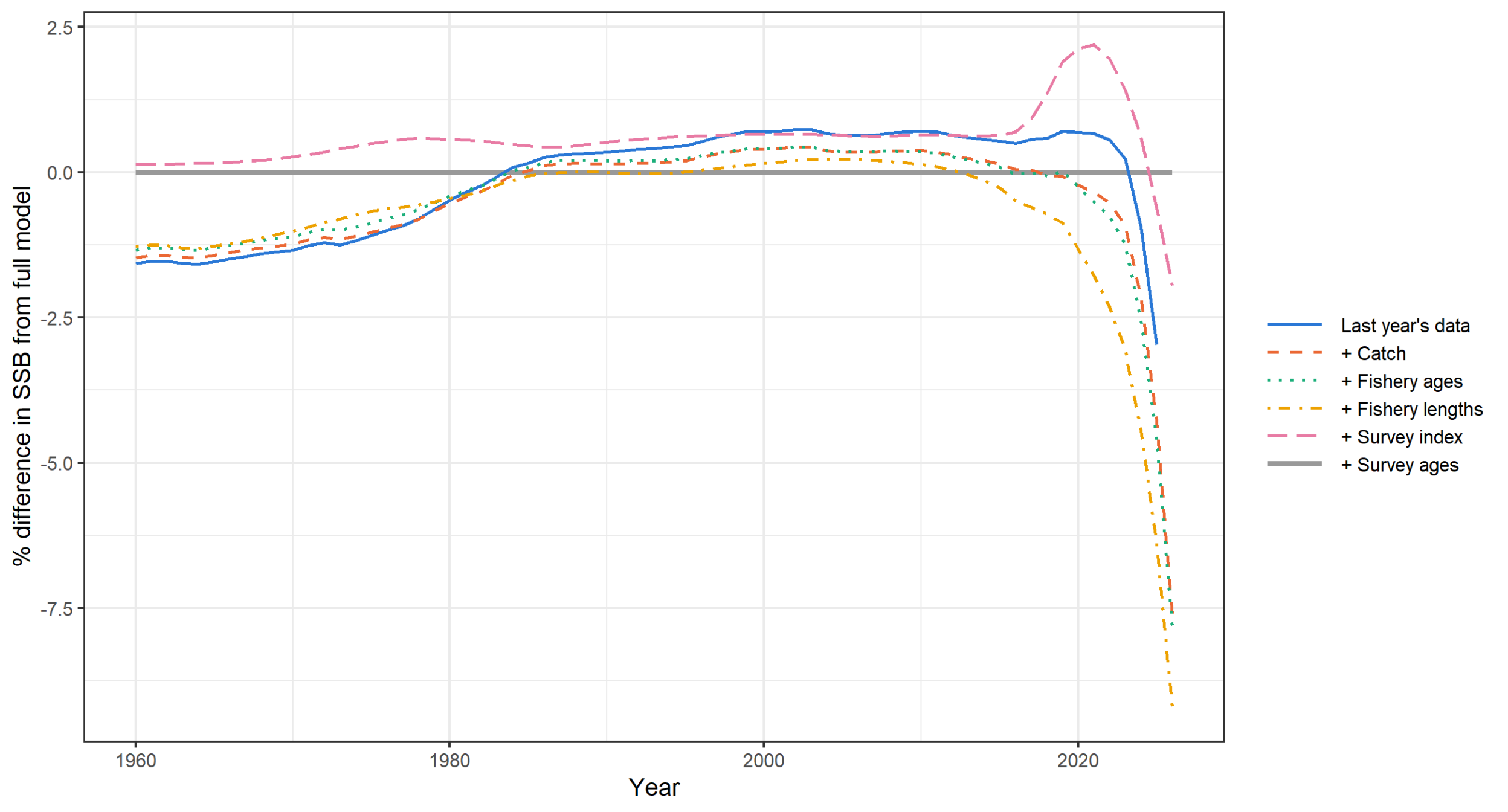

Recruitment (R0) profile
Scaled nLL
200
150
100
50
0
0
2
4
6
ln(R0)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

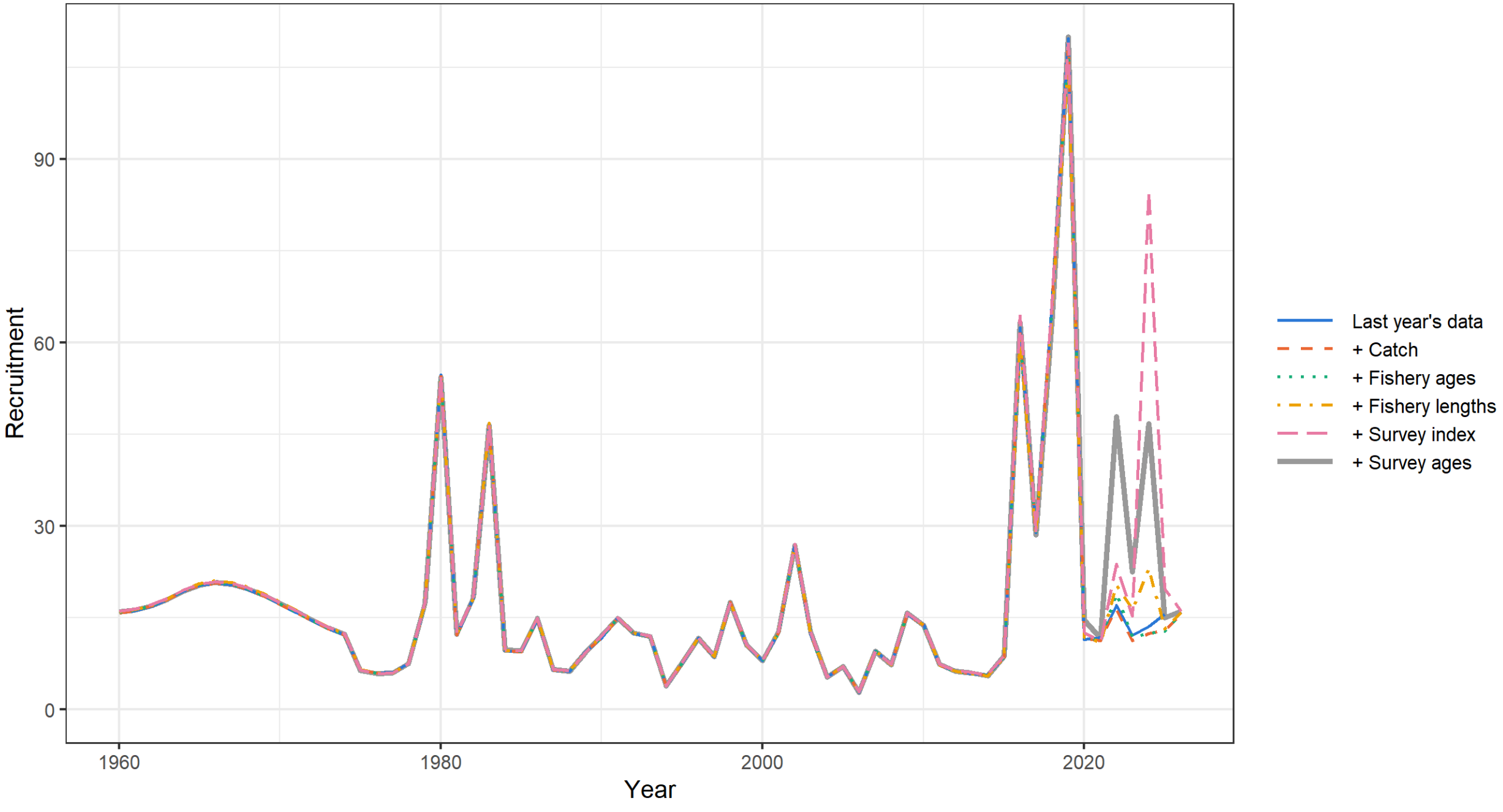

Recruitment (R0) profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(R0)

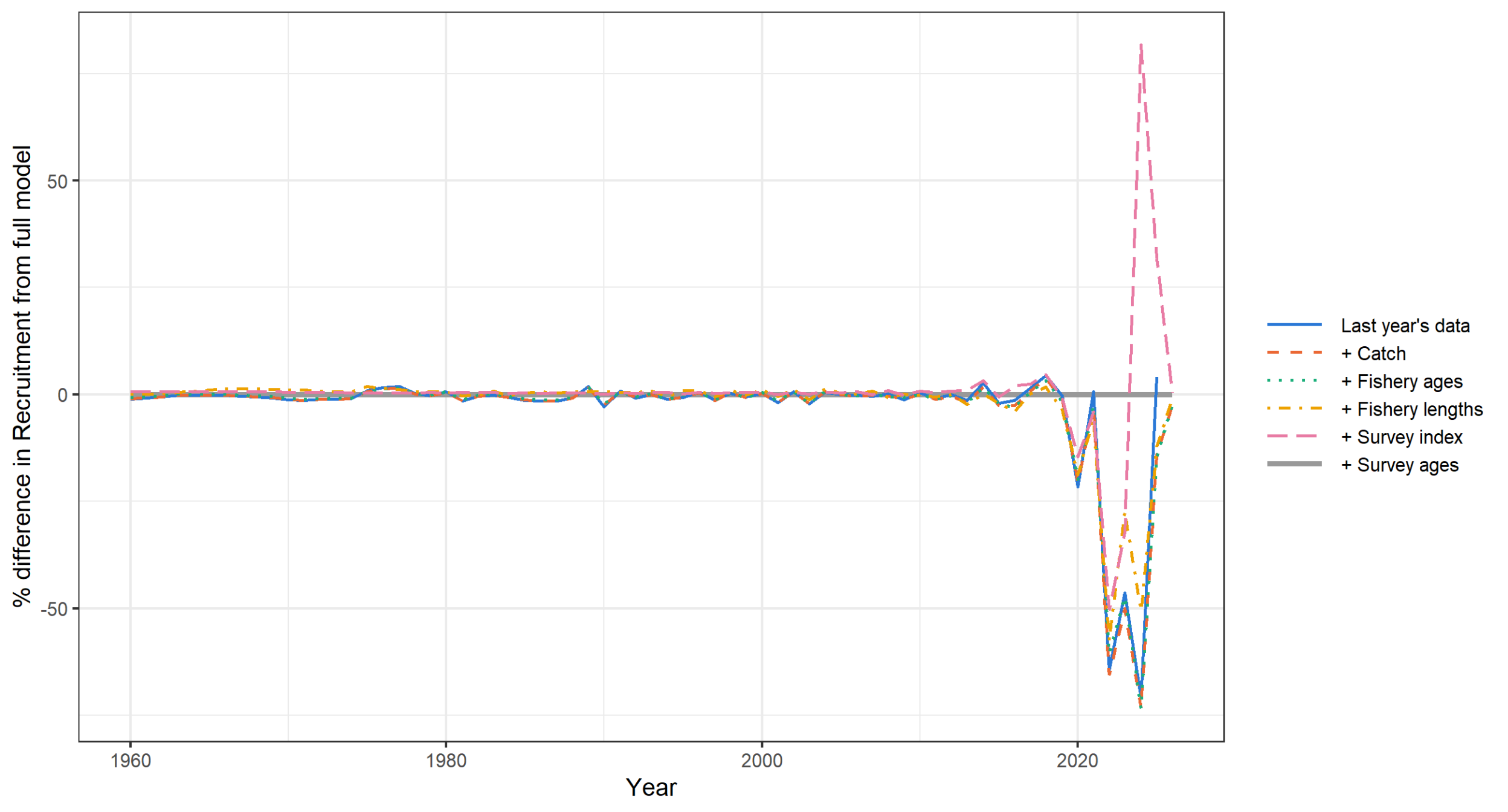

Jitter SSB and Recruitment
Region 1
Recruitment (millions)
SSB (kt)
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
Value
Year
0
100
200
300
400
500
1960
1980
2000
2020
Hessian
TRUE

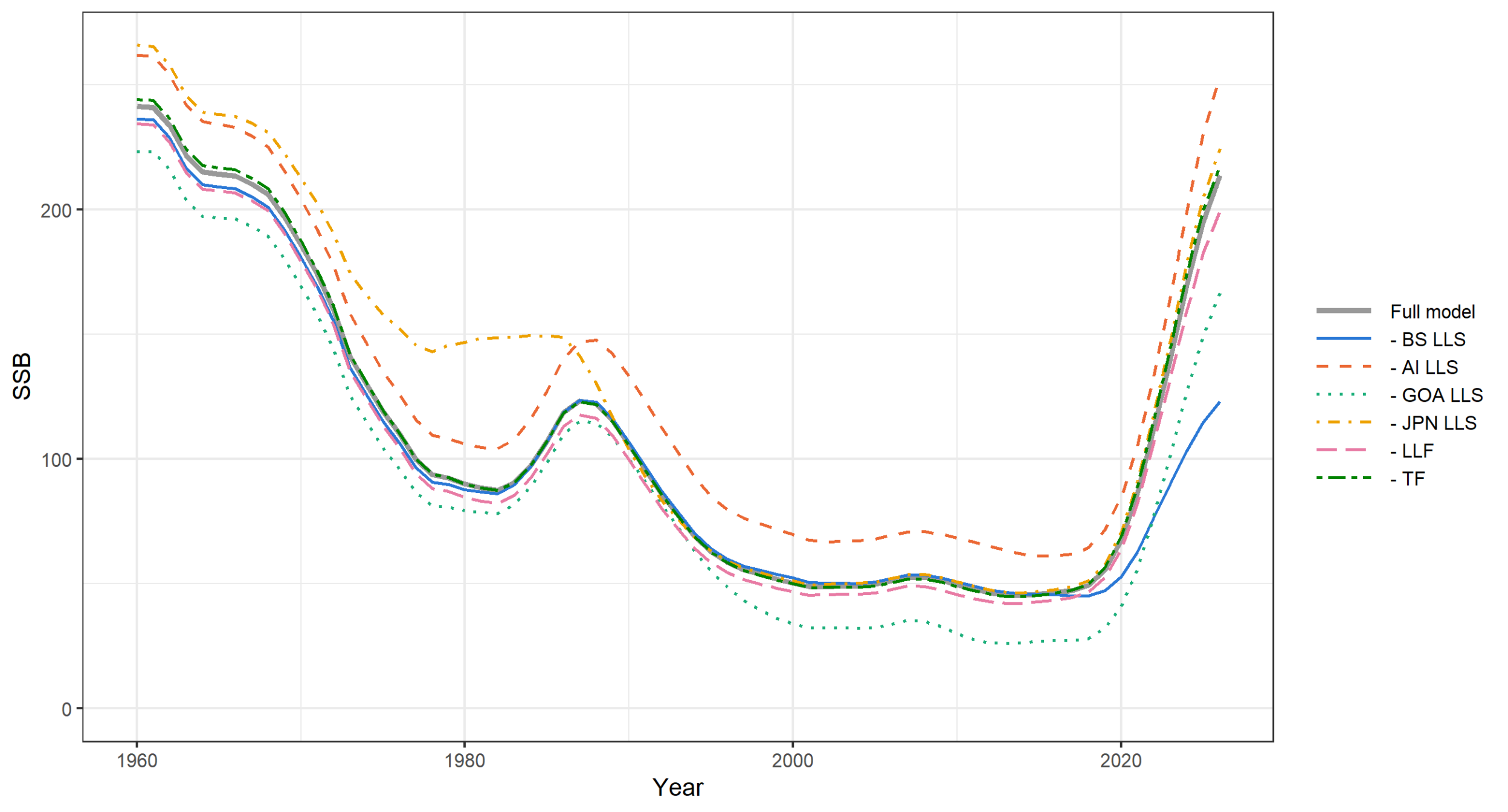

Jitter nLL and Convergence
PD Hessian
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
jnLL
700
650
600
550
500
0
50
100
150
200
Jitter
Hessian
TRUE
Max_Gradient
2e-05
4e-05
6e-05

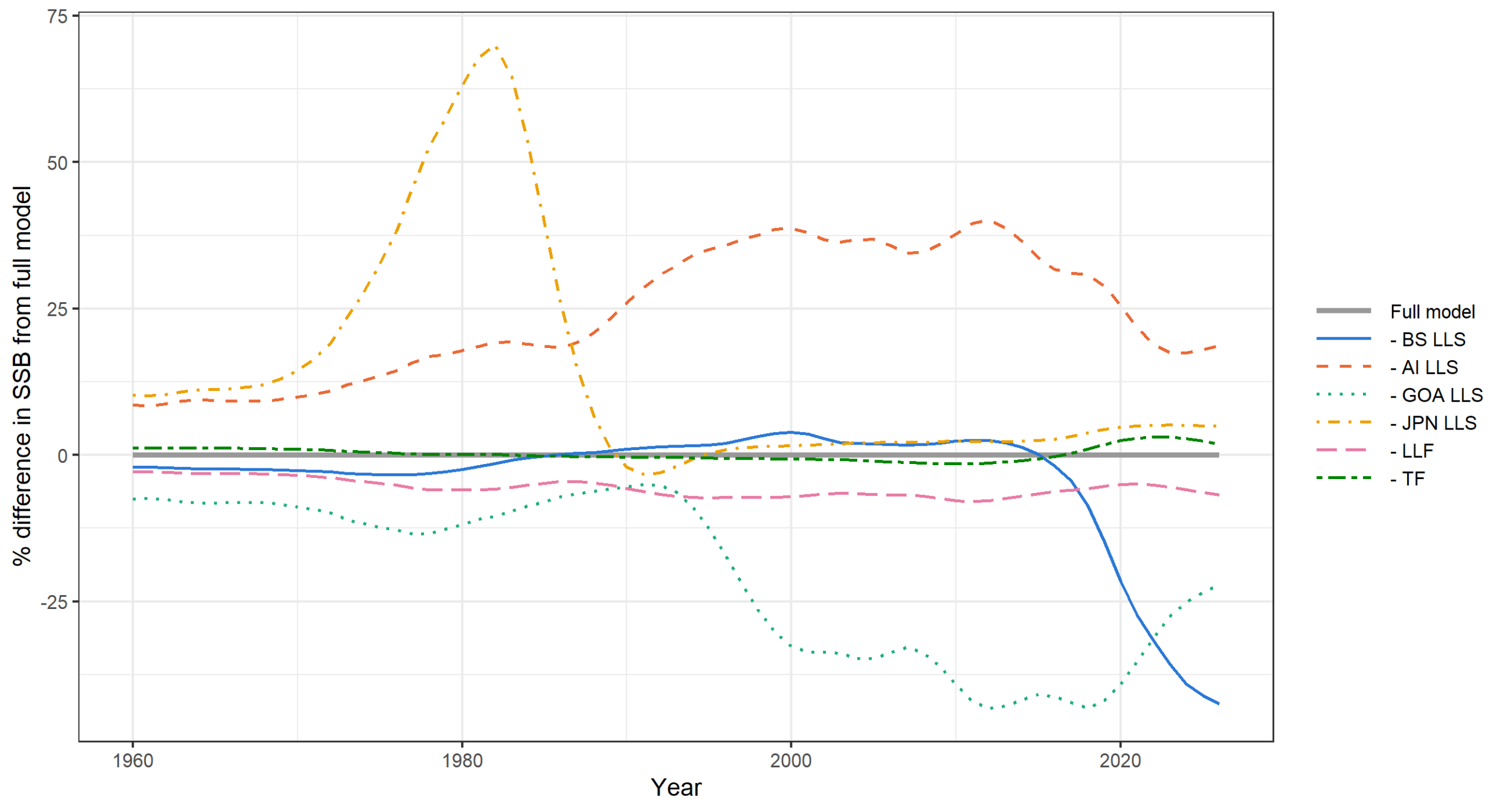

250
200
150
100
50
0
SSB
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

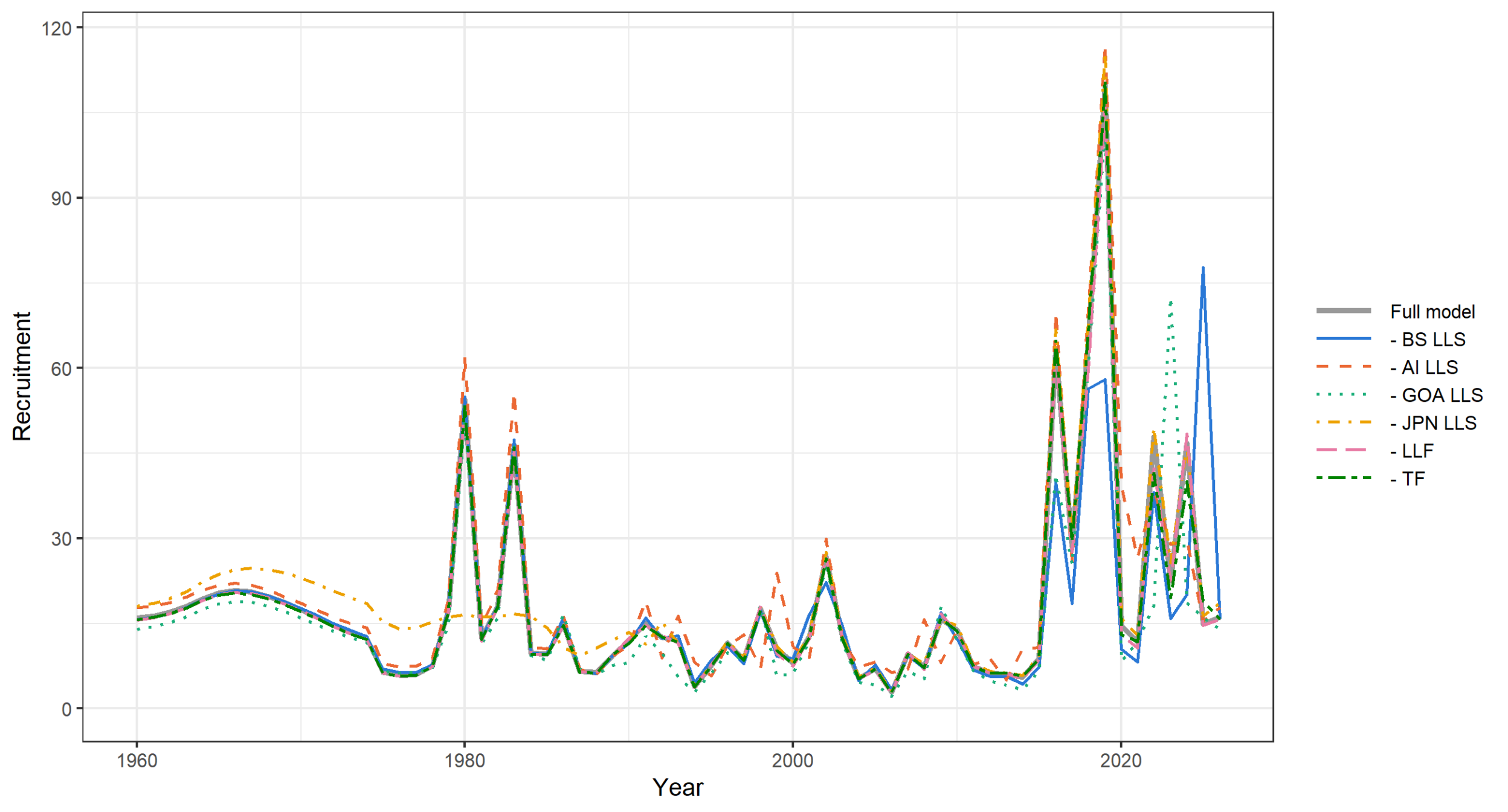

2.5
0.0
-2.5
-5.0
-7.5
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

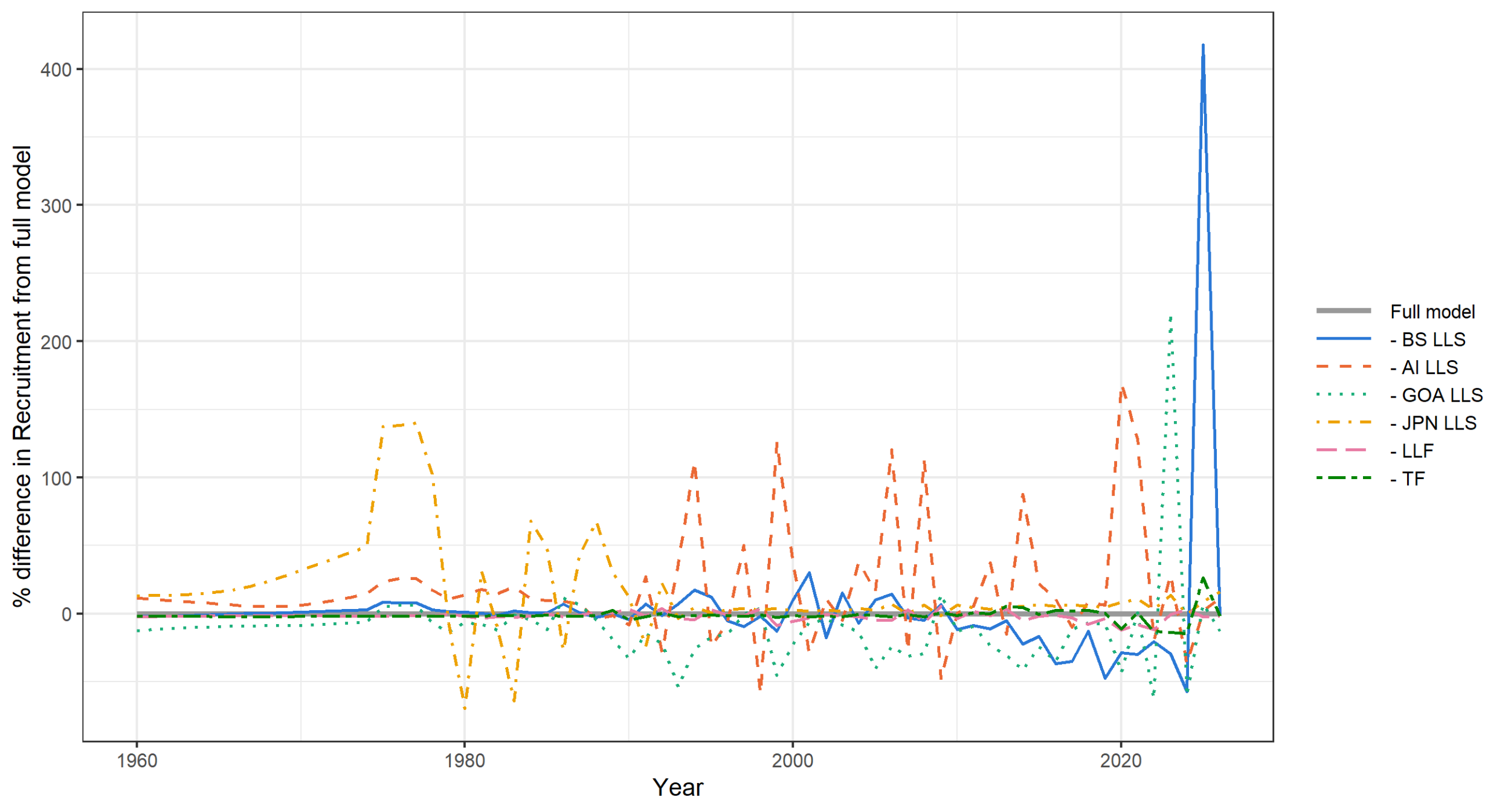

Recruitment
90
60
30
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in Recruitment from full model
50
0
-50
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

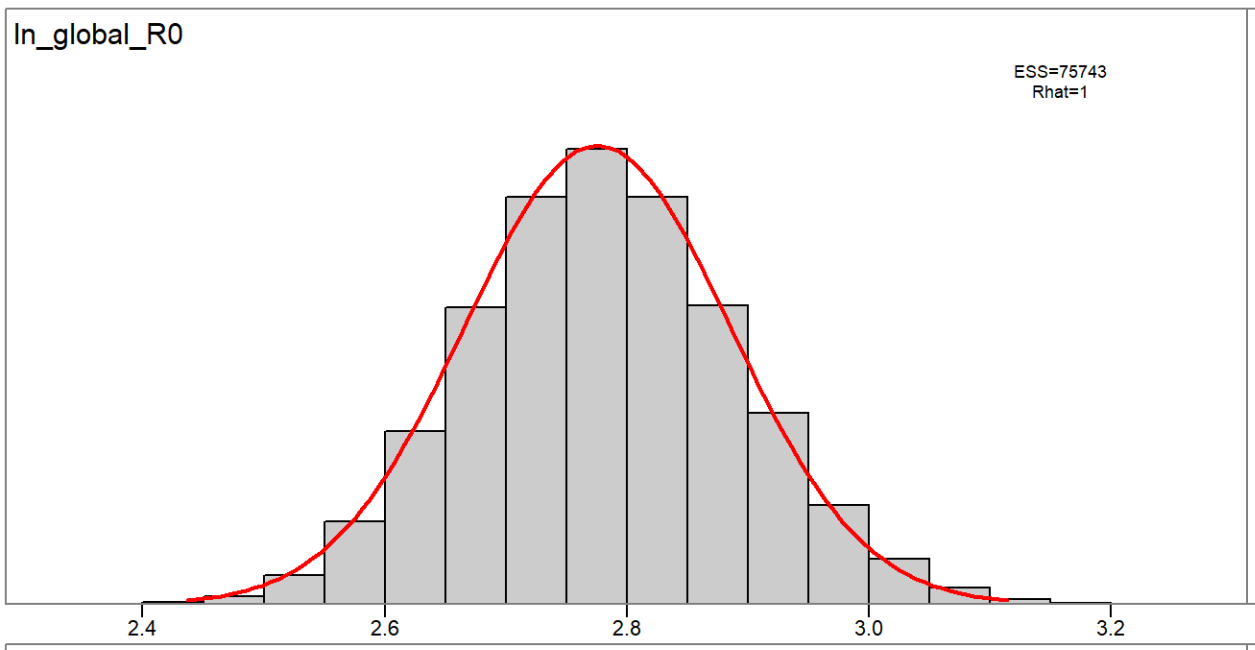

SSB
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


75
50
25
0
-25
% difference in SSB from full model
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

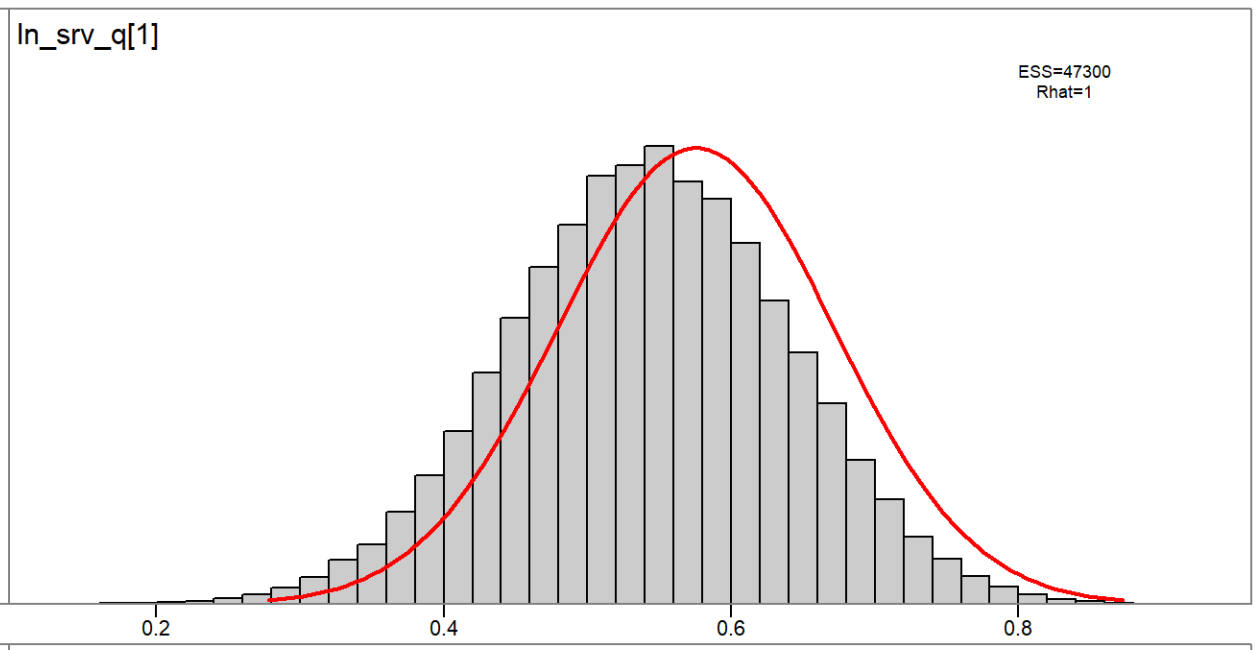

120
90
60
30
0
Recruitment
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


% difference in Recruitment from full model
400
300
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

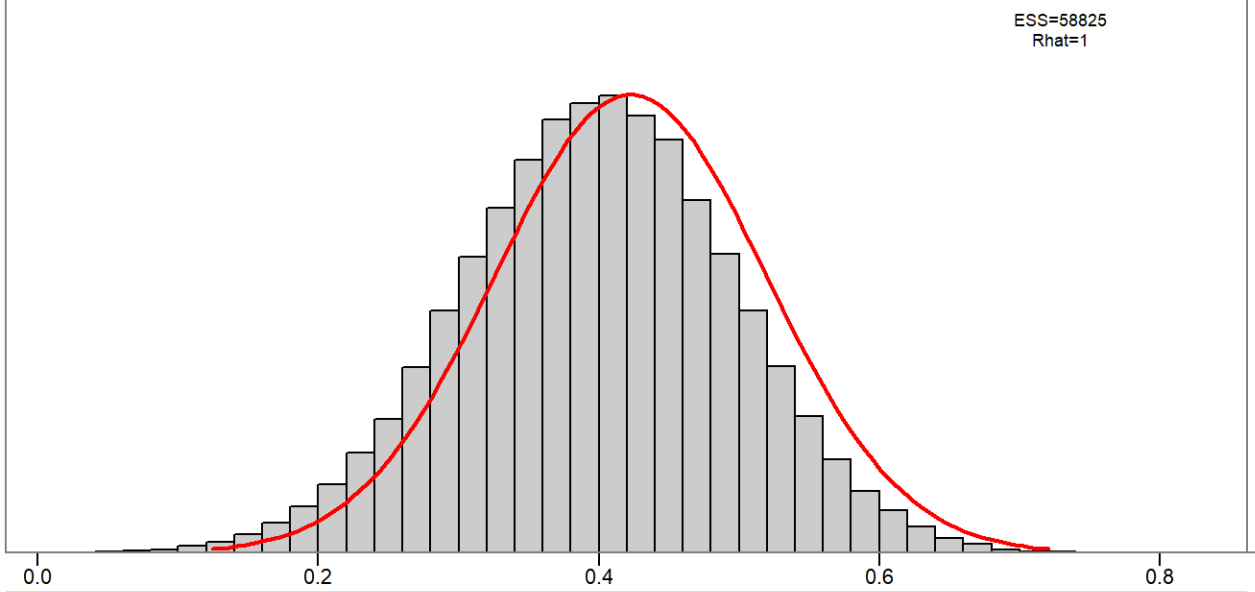
ln_global_R0



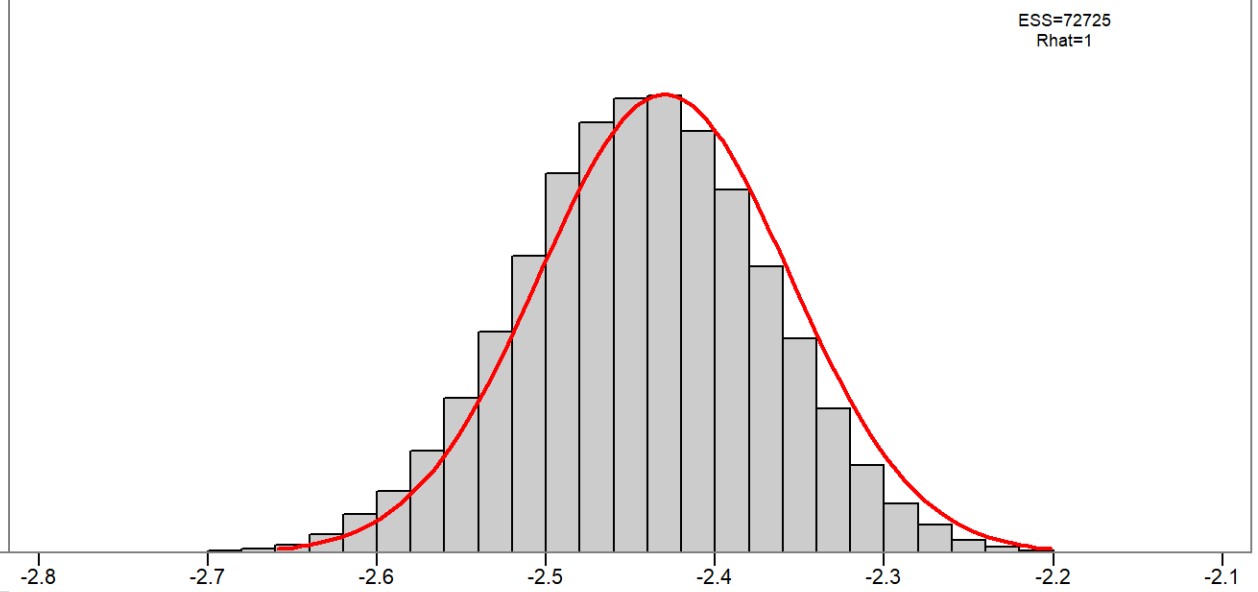
ln_srv_q[1]



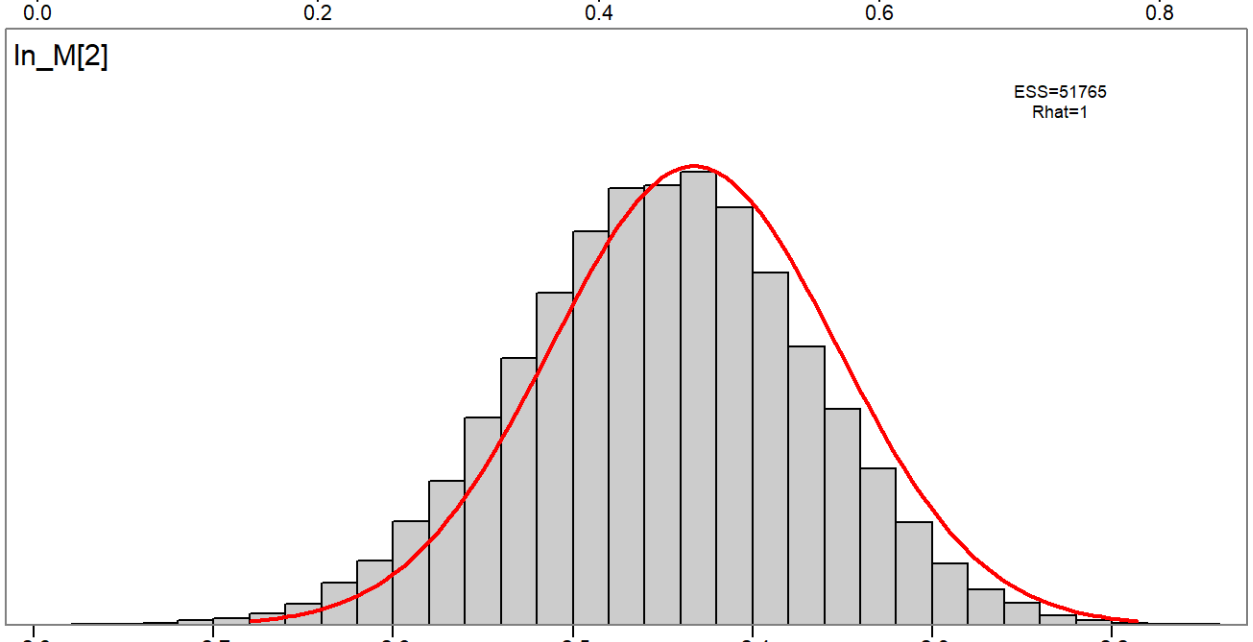
ln_srv_q[2]



ln_M[1]



ln_M[2]



ln_global_R0 | ln_srv_q[1] | ln_srv_q[2] | ln_M[1] | ln_M[2]

ESS=75743 Rhat=1.00

-0.46 | -0.46 | 0.69 | 0.65

ESS=47300 Rhat=1.00

0.72 | -0.4 | -0.43

ESS=58825 Rhat=1.00

-0.39 | -0.43

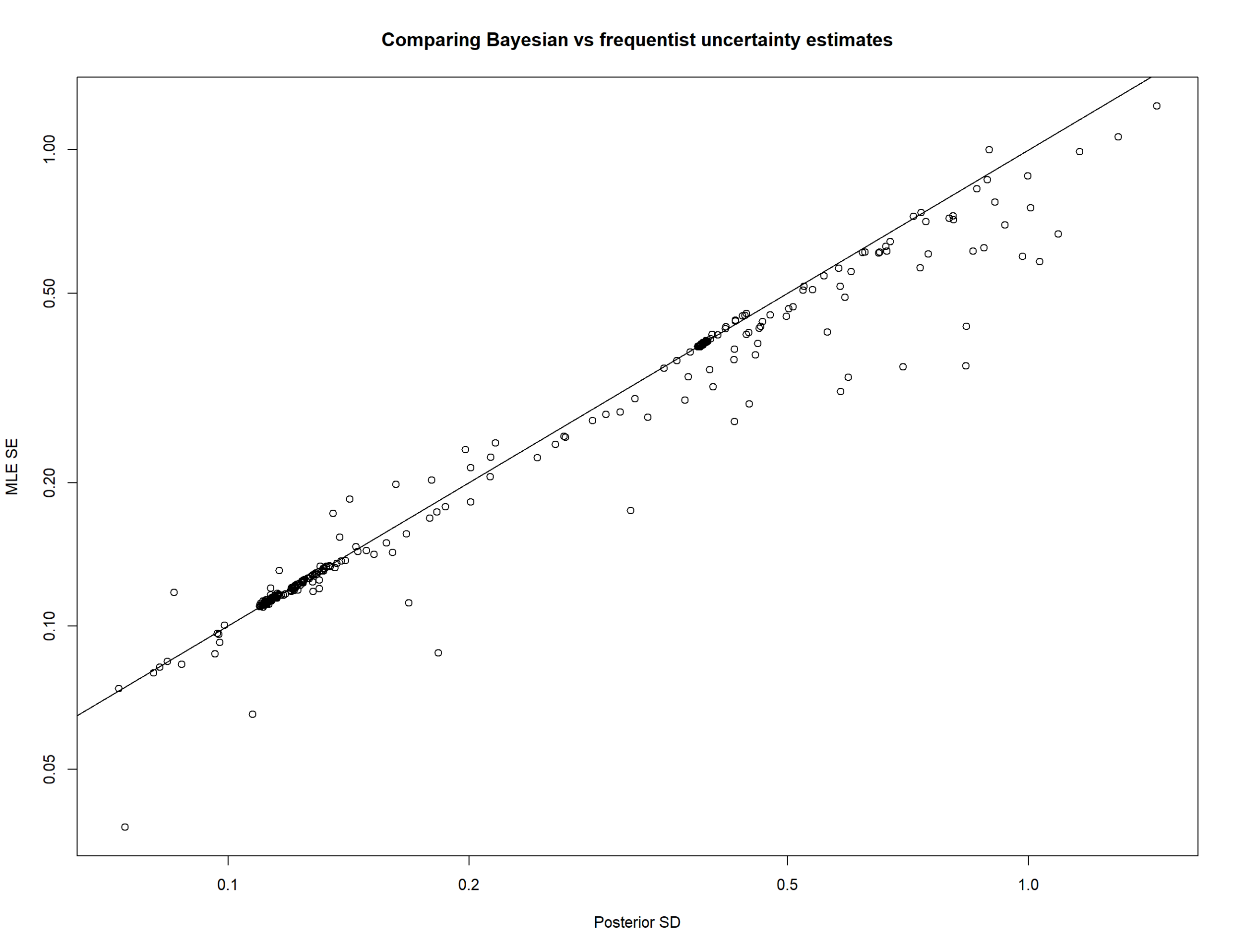
ESS=72725 Rhat=1.00

0.56

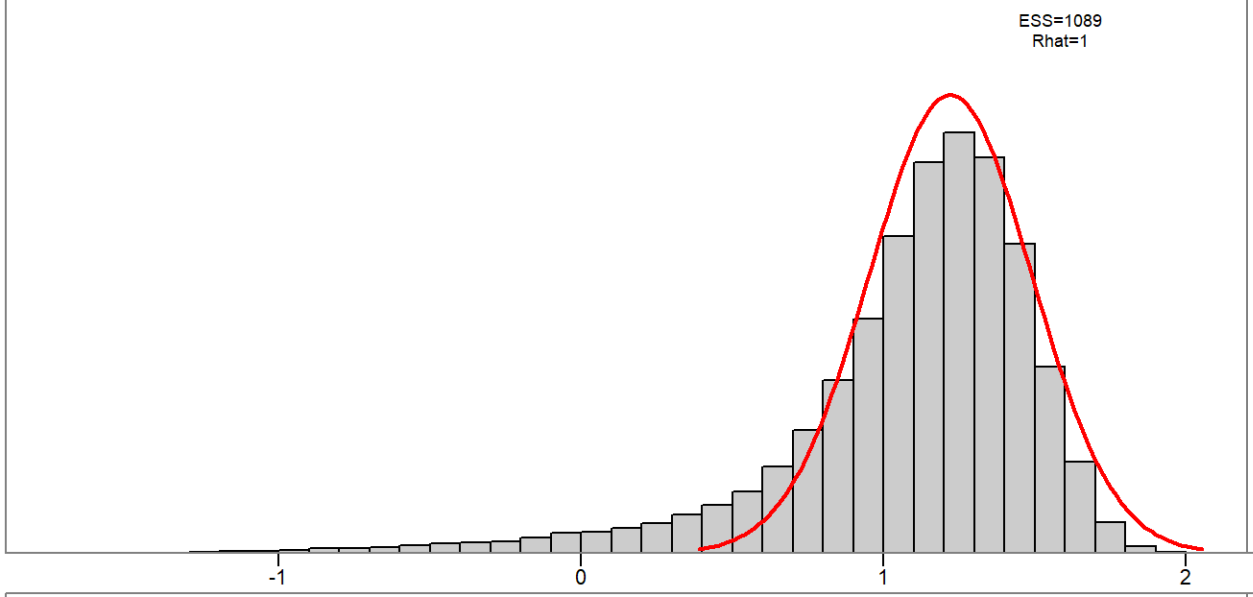
ESS=51765 Rhat=1.00

ln_global_R0 | ln_srv_q[1] | ln_srv_q[2] | ln_M[1] | ln_M[2]

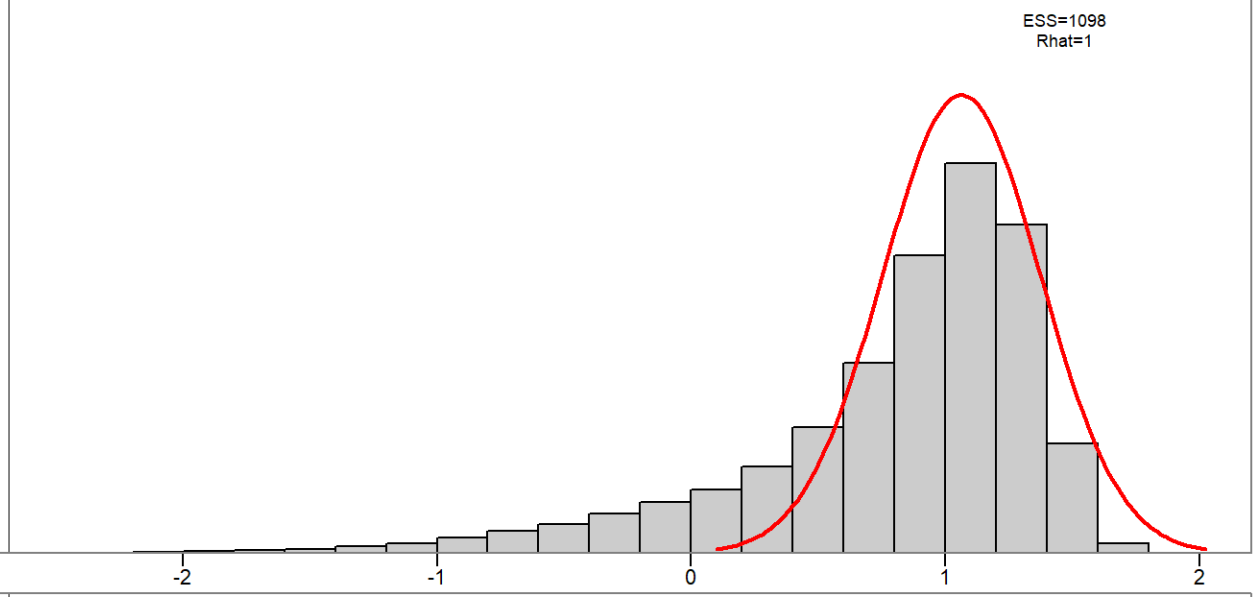
2.2 2.4 2.6 2.8 3.0 3.2 3.4 | 0.0 0.2 0.4 0.6 0.8 1.0 | 0.0 0.2 0.4 0.6 0.8 | -2.8 -2.6 -2.4 -2.2 -2.0 | 5000 10000 15000 20000


Comparing Bayesian vs frequentist uncertainty estimates
MLE SE
0.05
0.10
0.20
0.50
1.00
0.1
0.2
0.5
1.0
Posterior SD

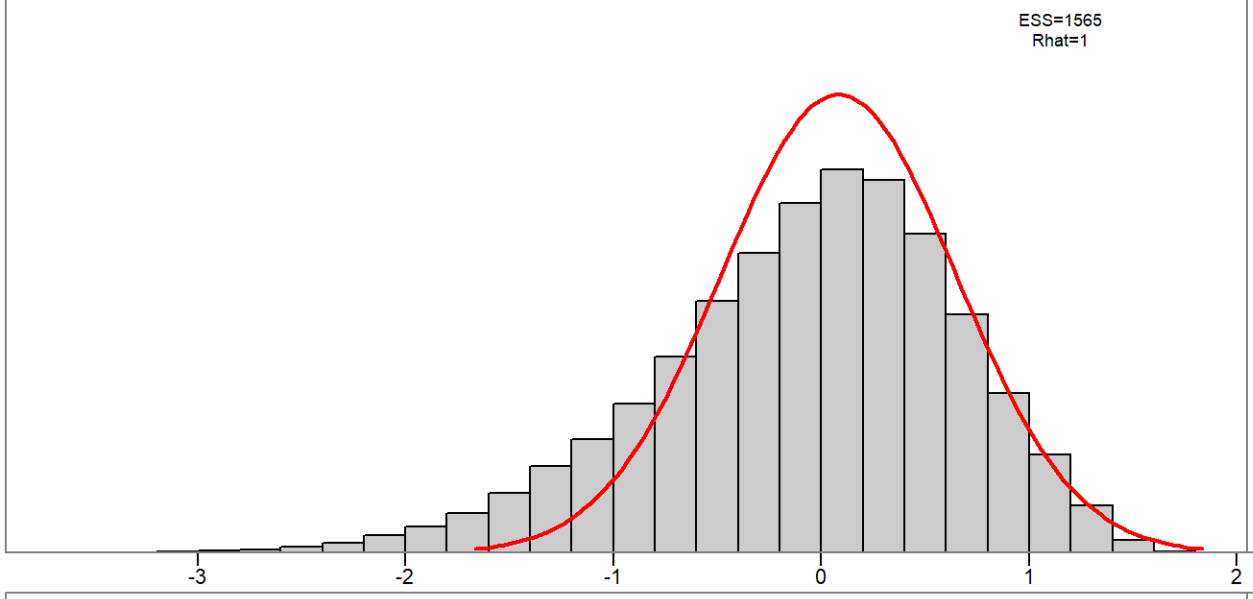
ln_RecDevs[21]



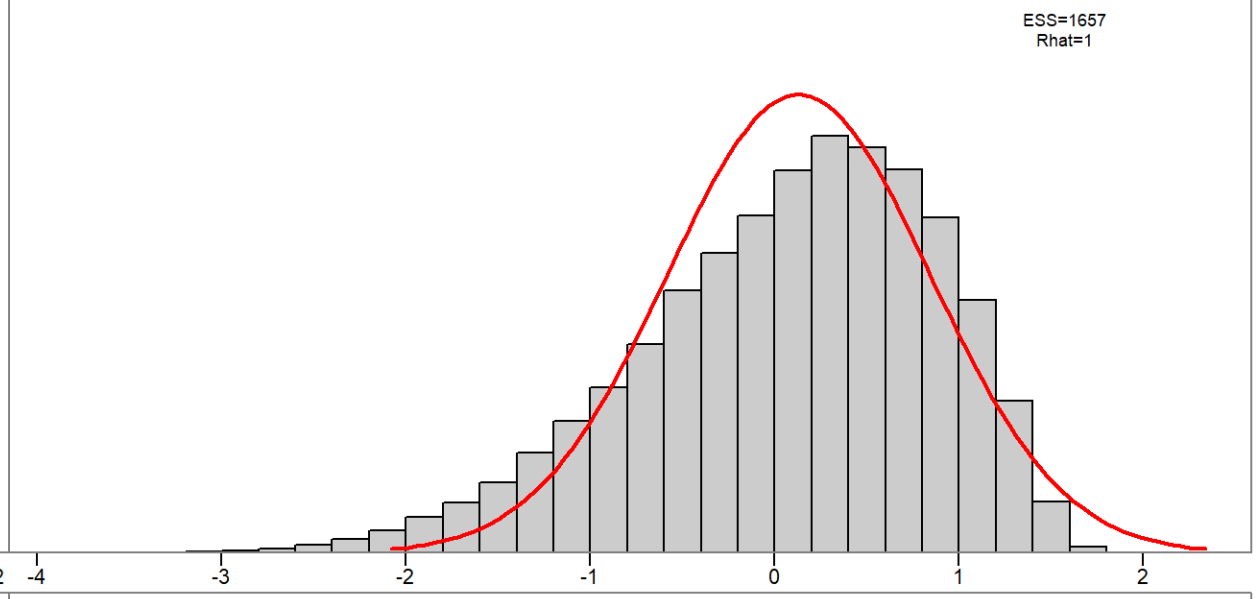
ln_RecDevs[24]



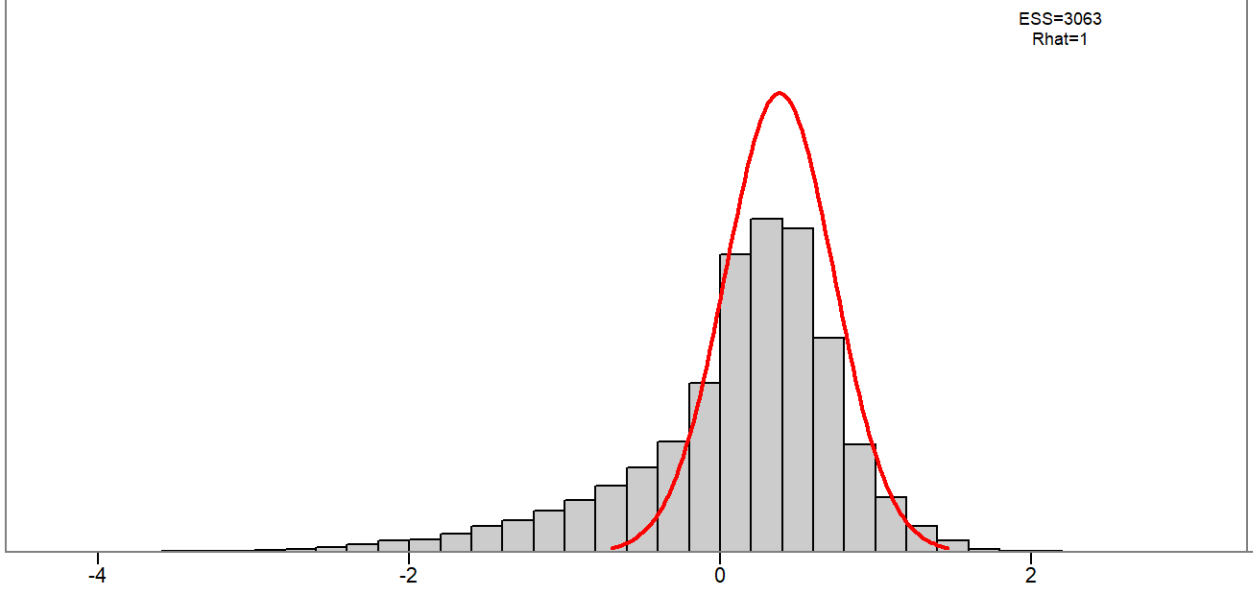
ln_RecDevs[20]



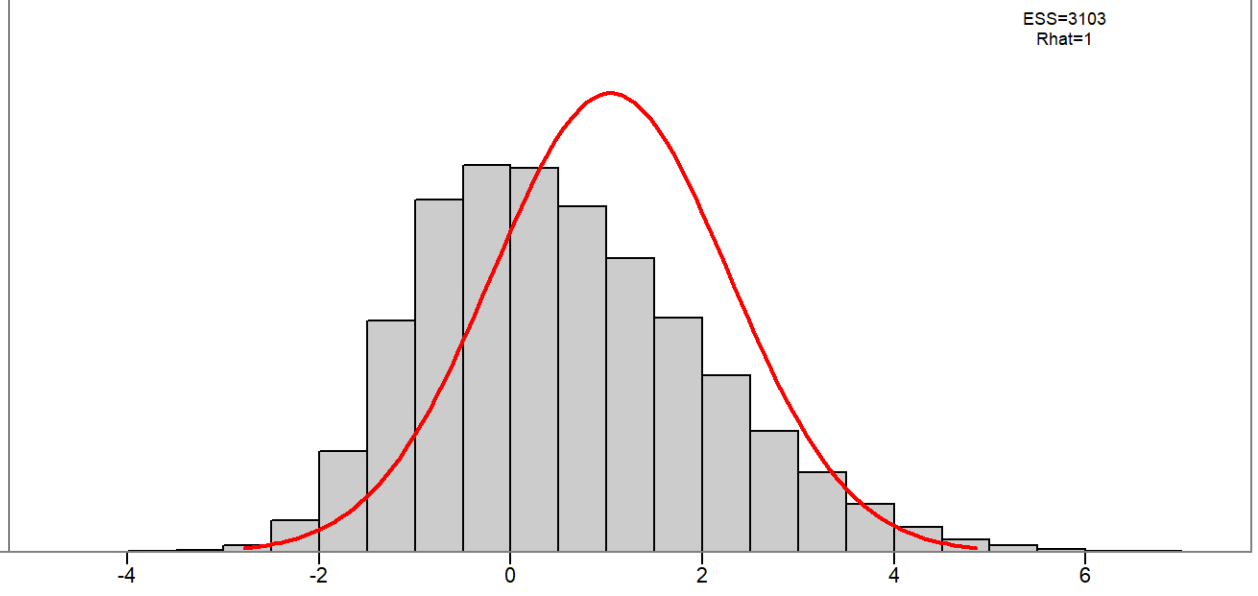
ln_RecDevs[23]



fish_fixed_sel_pars[9]

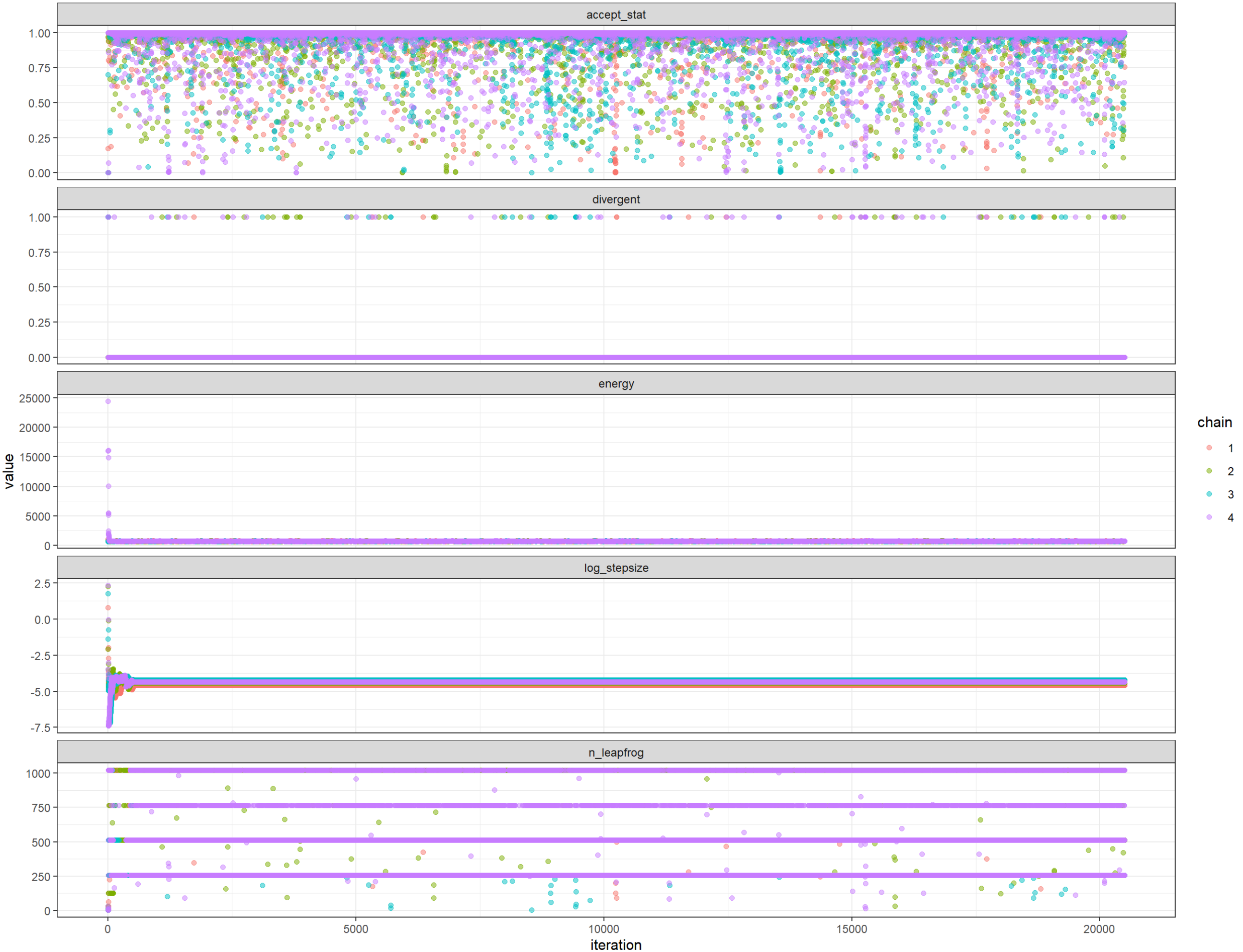


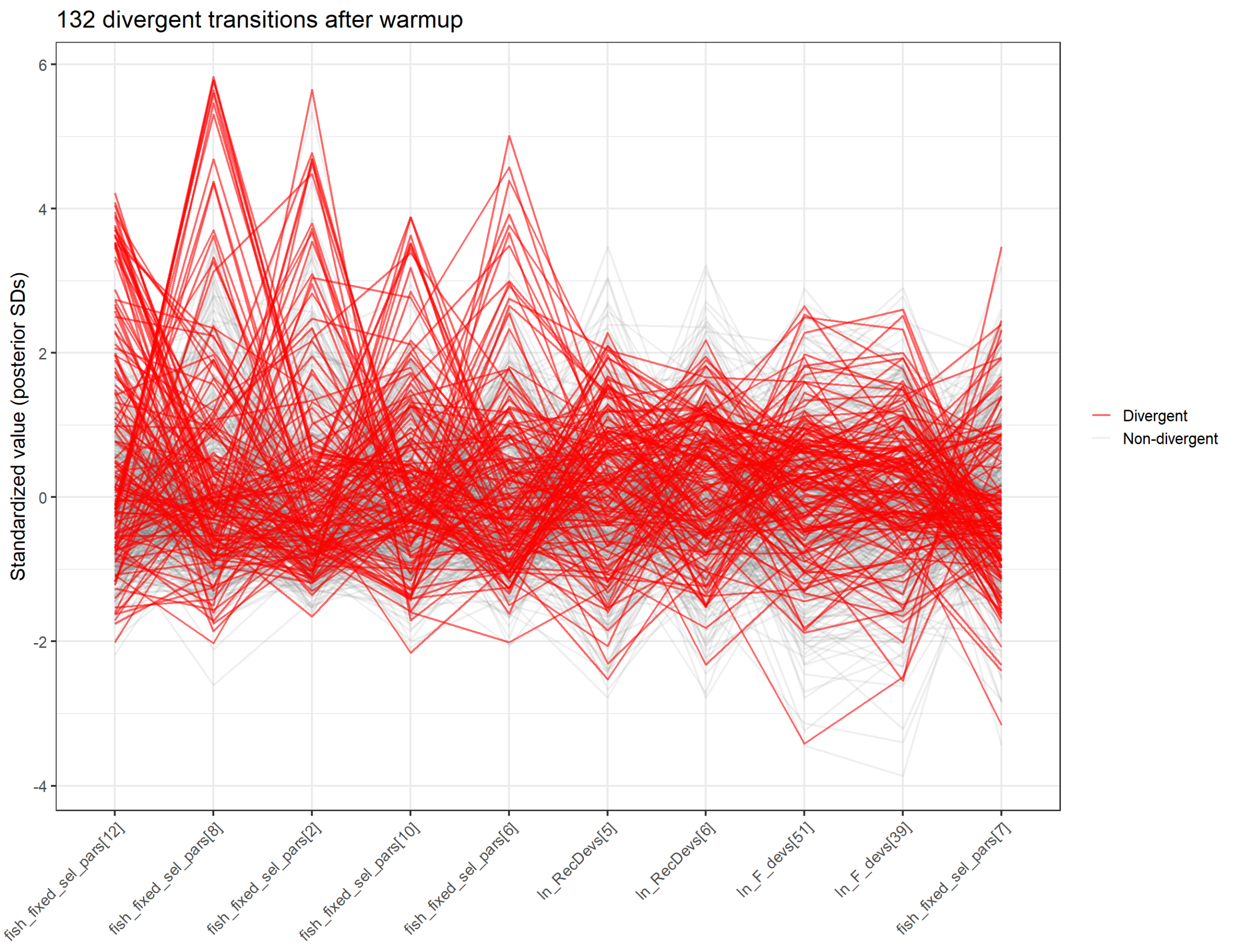
fish_fixed_sel_pars[10]

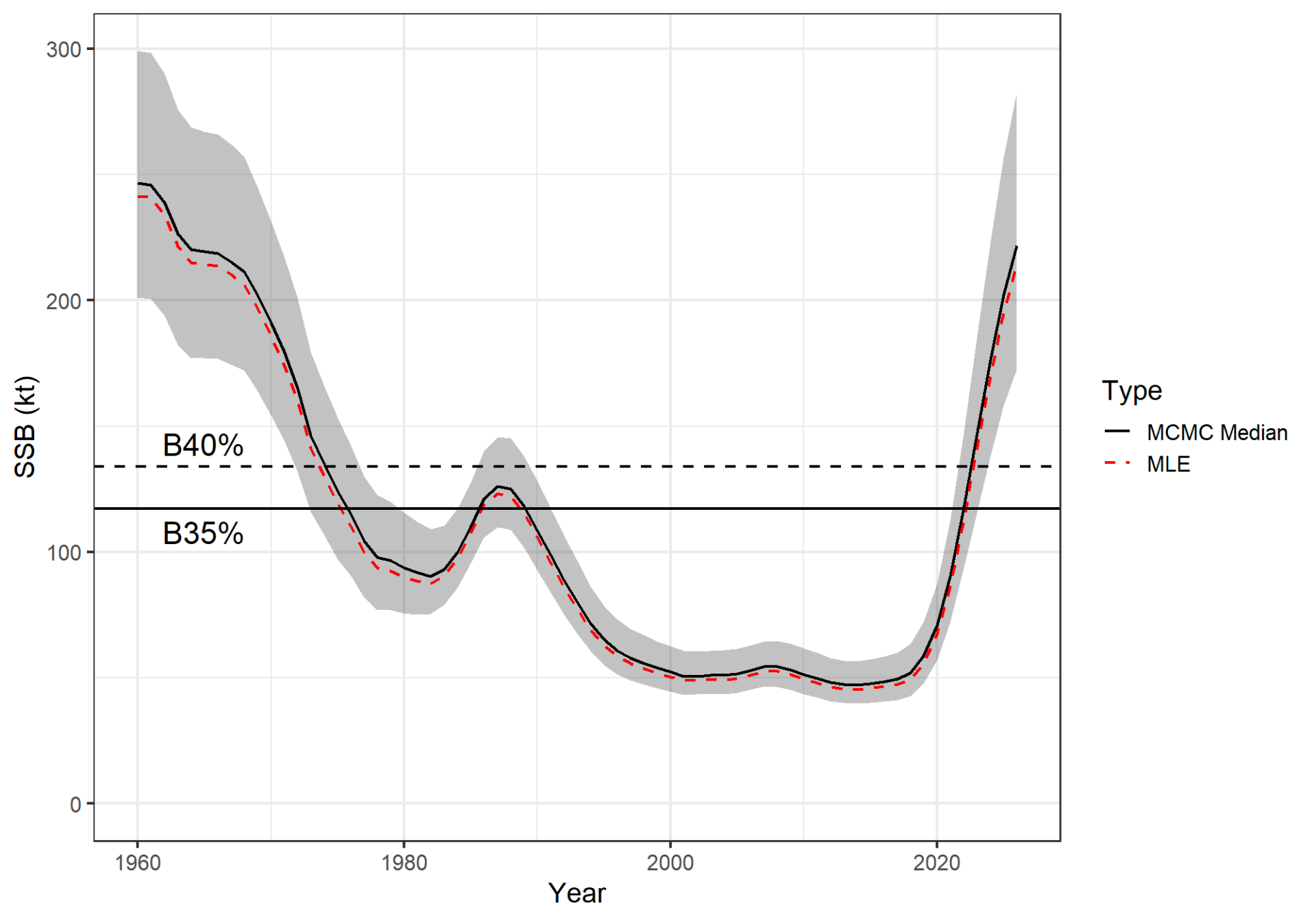


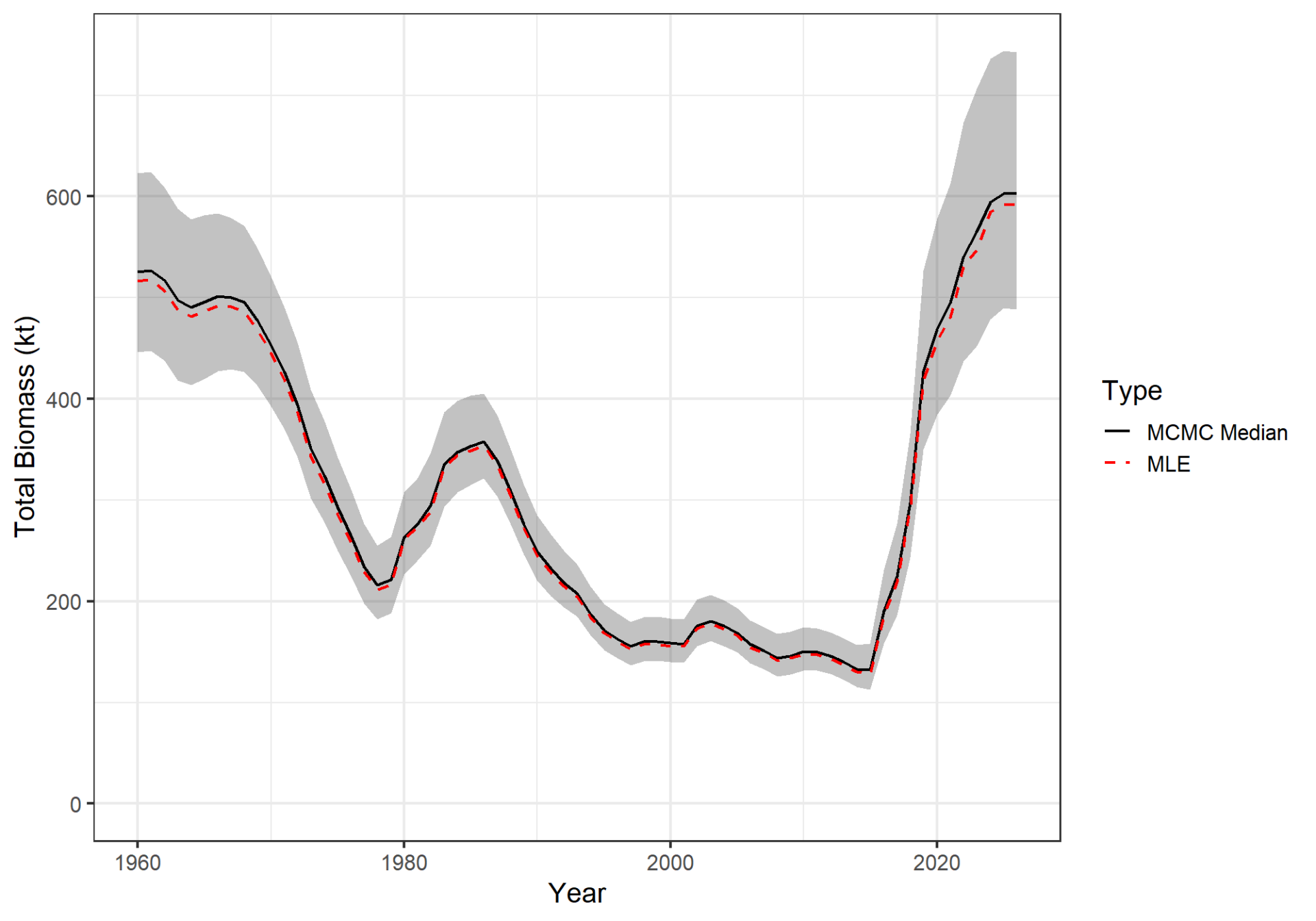
ln_RecDevs[21]
ln_RecDevs[24]
ln_RecDevs[20]
ln_RecDevs[23]
fish_fixed_sel_pars[9]
fish_fixed_sel_pars[10]

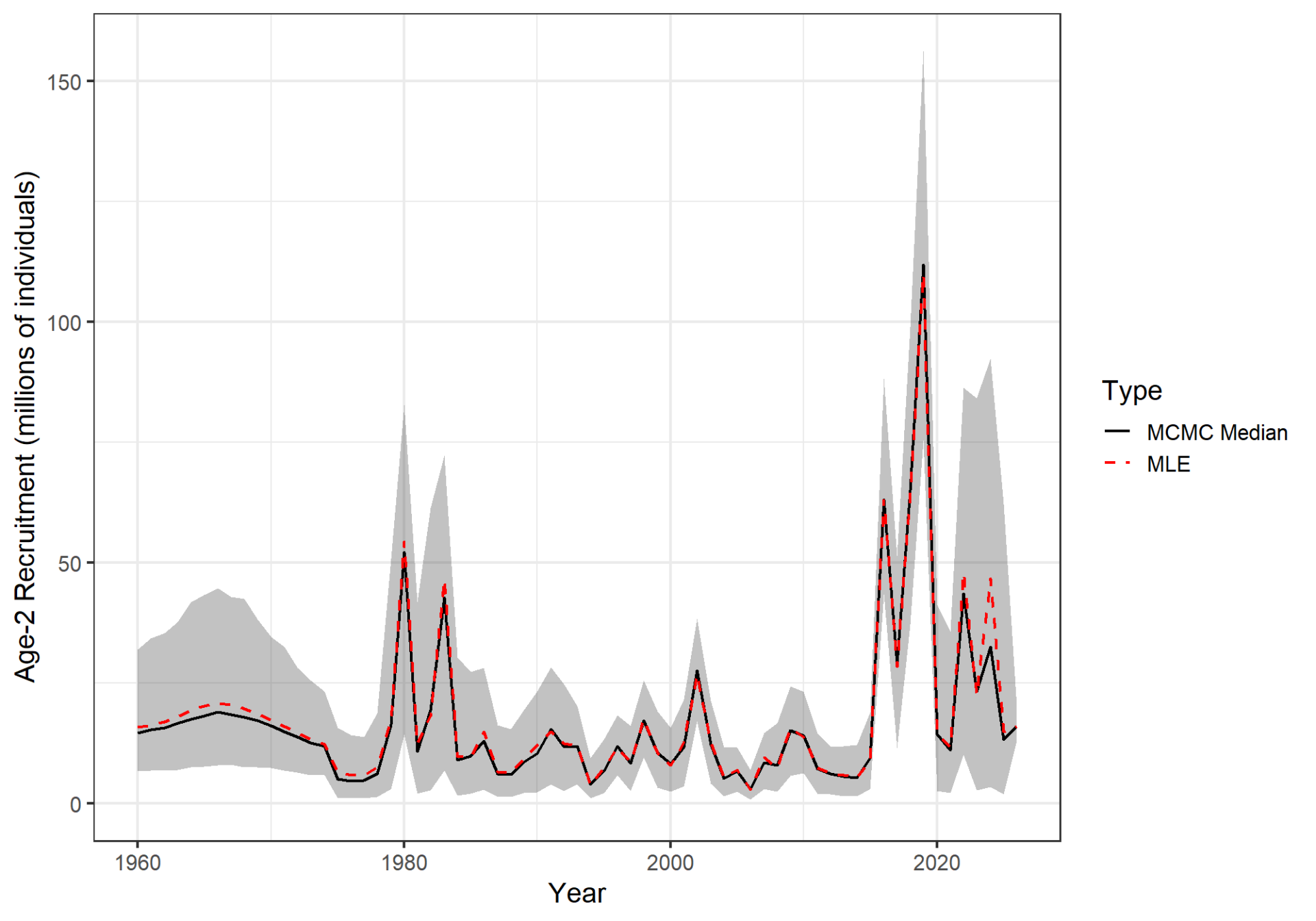
ESS=1089 Rhat=1.00
0.21
-0.61
-0.25
ESS=1098 Rhat=1.00
-0.15
-0.6
ESS=1565 Rhat=1.00
0.22
ESS=1657 Rhat=1.00
ESS=3063 Rhat=1.00
ESS=3103 Rhat=1.00

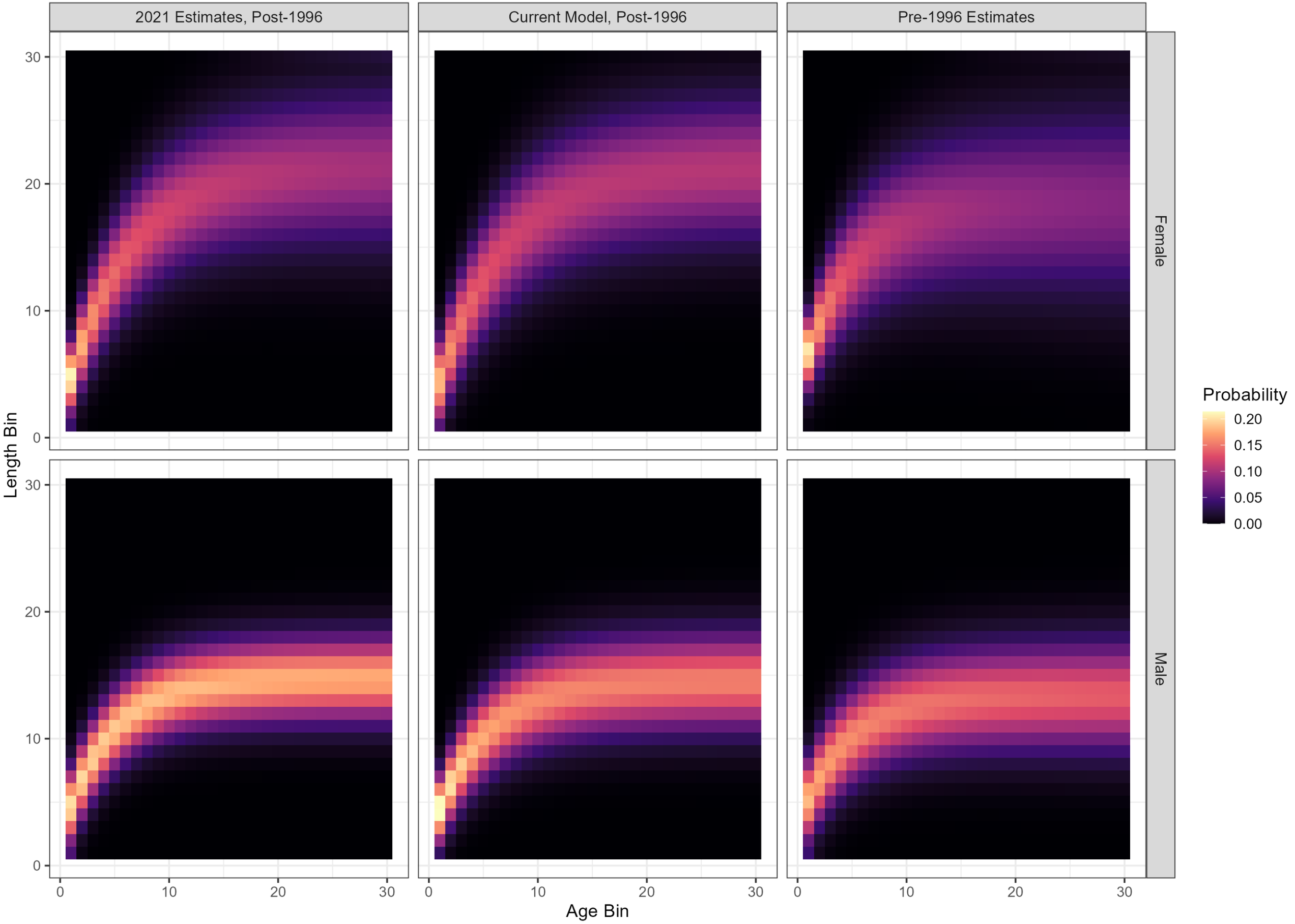

accept_stat
1.00
0.75
0.50
0.25
0.00
divergent
1.00
0.75
0.50
0.25
0.00
energy
25000
20000
15000
10000
5000
0
value
log_stepsize
2.5
0.0
-2.5
-5.0
-7.5
n_leapfrog
1000
750
500
250
0
0
5000
10000
15000
20000
iteration
chain
1
2
3
4

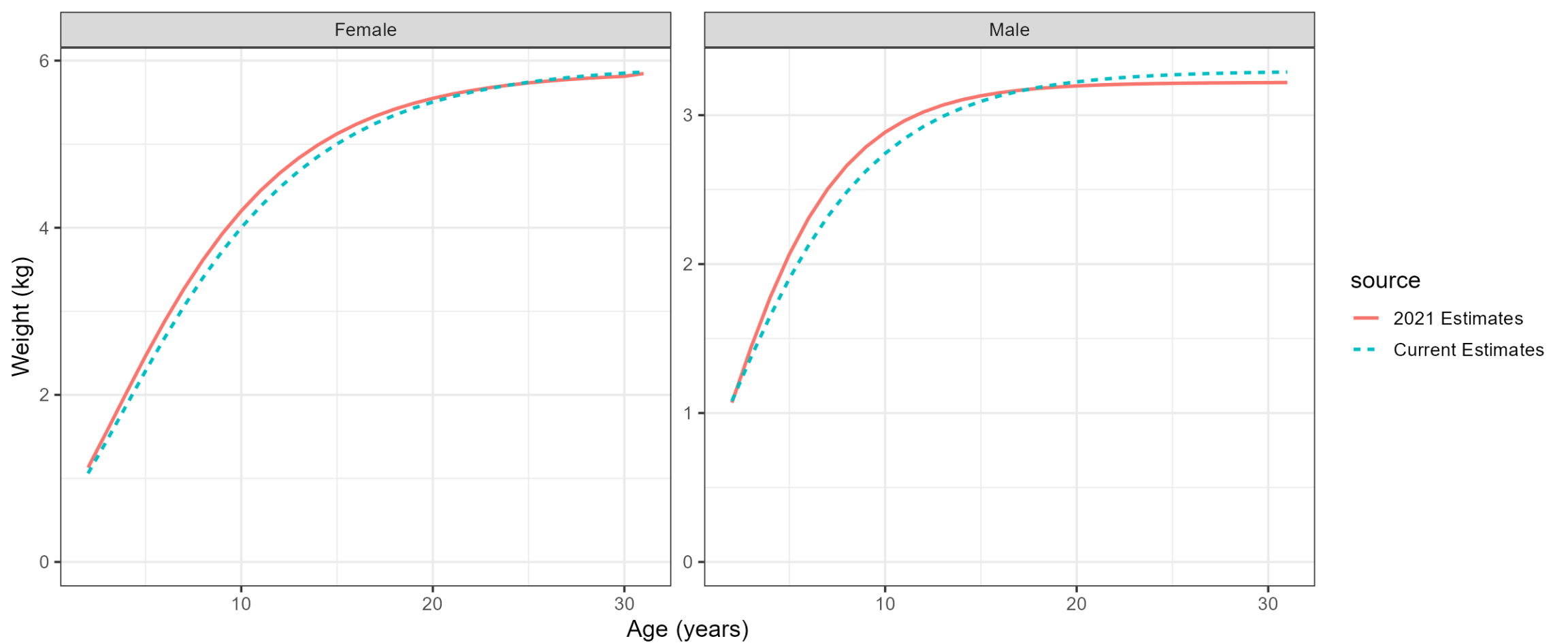

132 divergent transitions after warmup
Standardized value (posterior SDs)
6
4
2
0
-2
-4
fish_fixed_sel_pars[12]
fish_fixed_sel_pars[8]
fish_fixed_sel_pars[2]
fish_fixed_sel_pars[10]
fish_fixed_sel_pars[6]
ln_RecDevs[5]
ln_RecDevs[6]
ln_F_devs[51]
ln_F_devs[39]
fish_fixed_sel_pars[7]
Divergent
Non-divergent

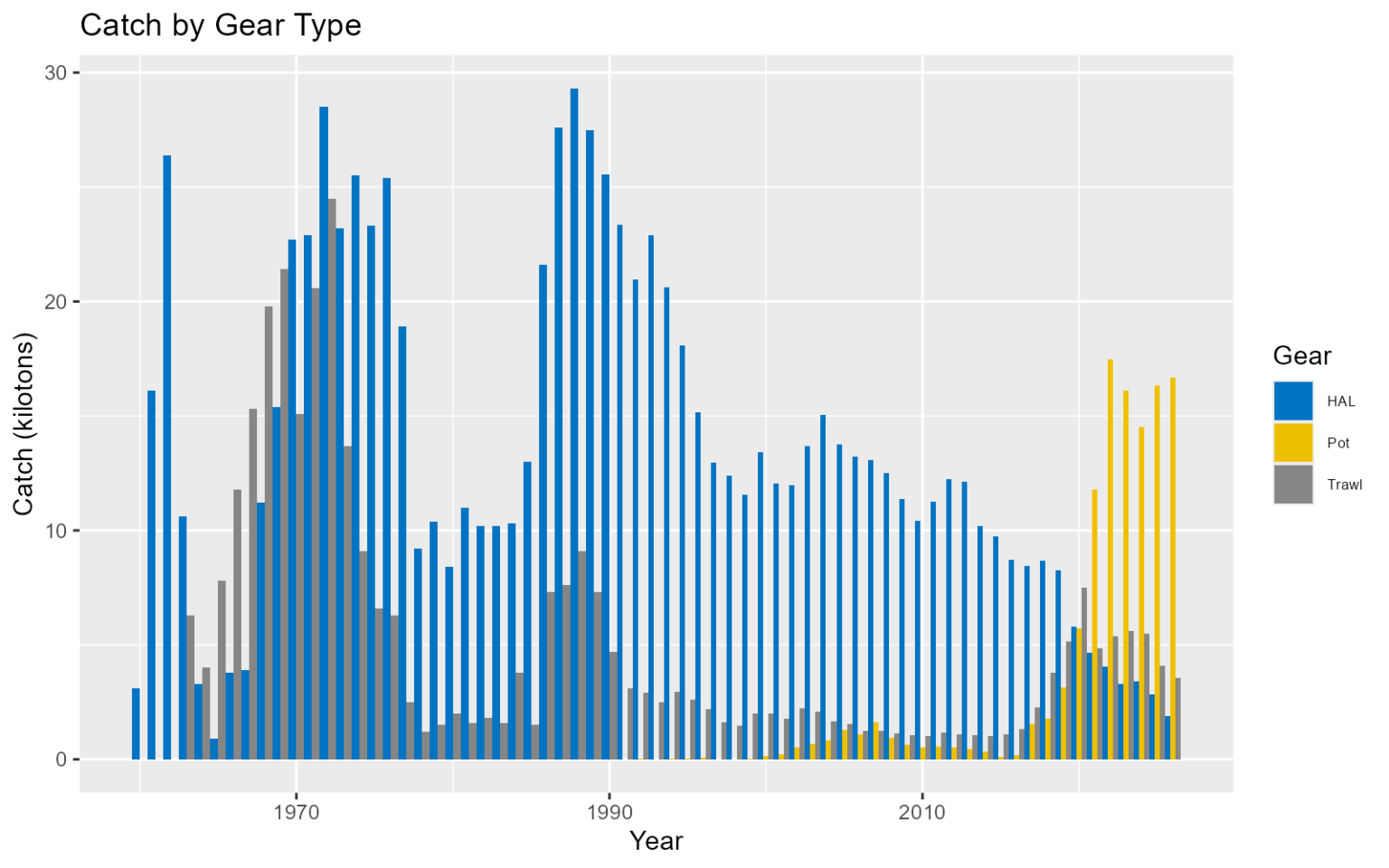

300
200
100
0
SSB (kt)
B40%
B35%
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

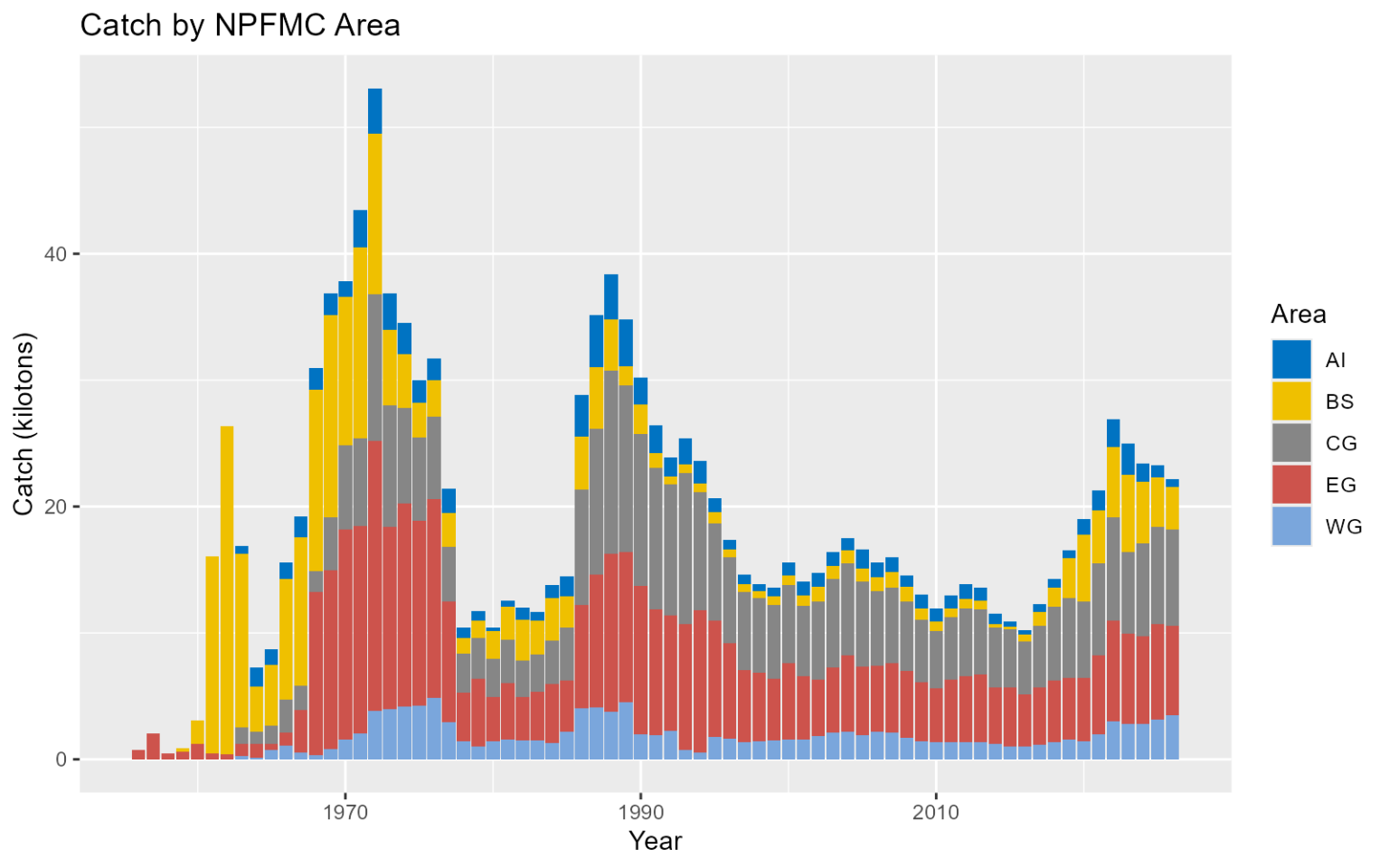

Total Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

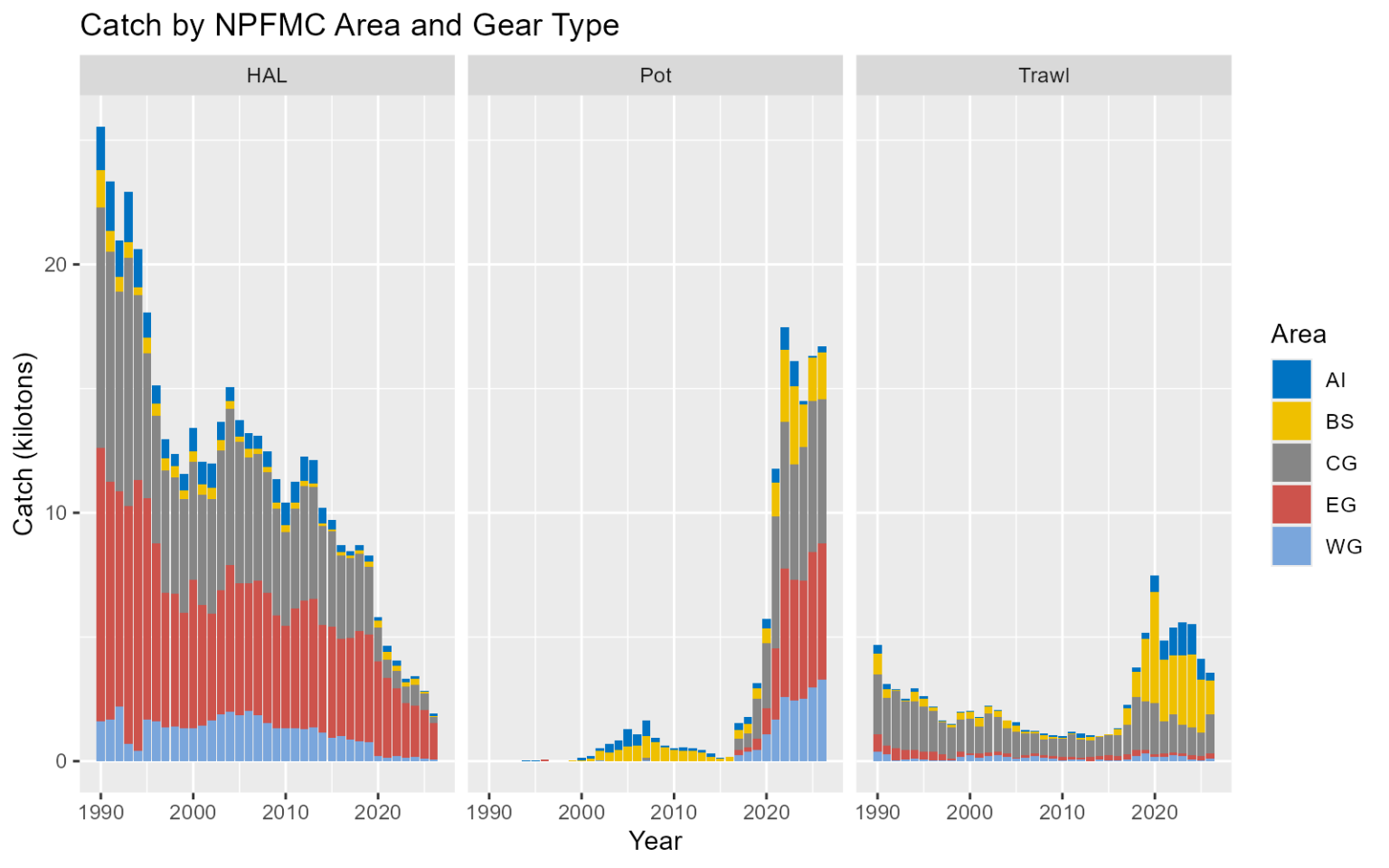

Age-2 Recruitment (millions of individuals)
150
100
50
0
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

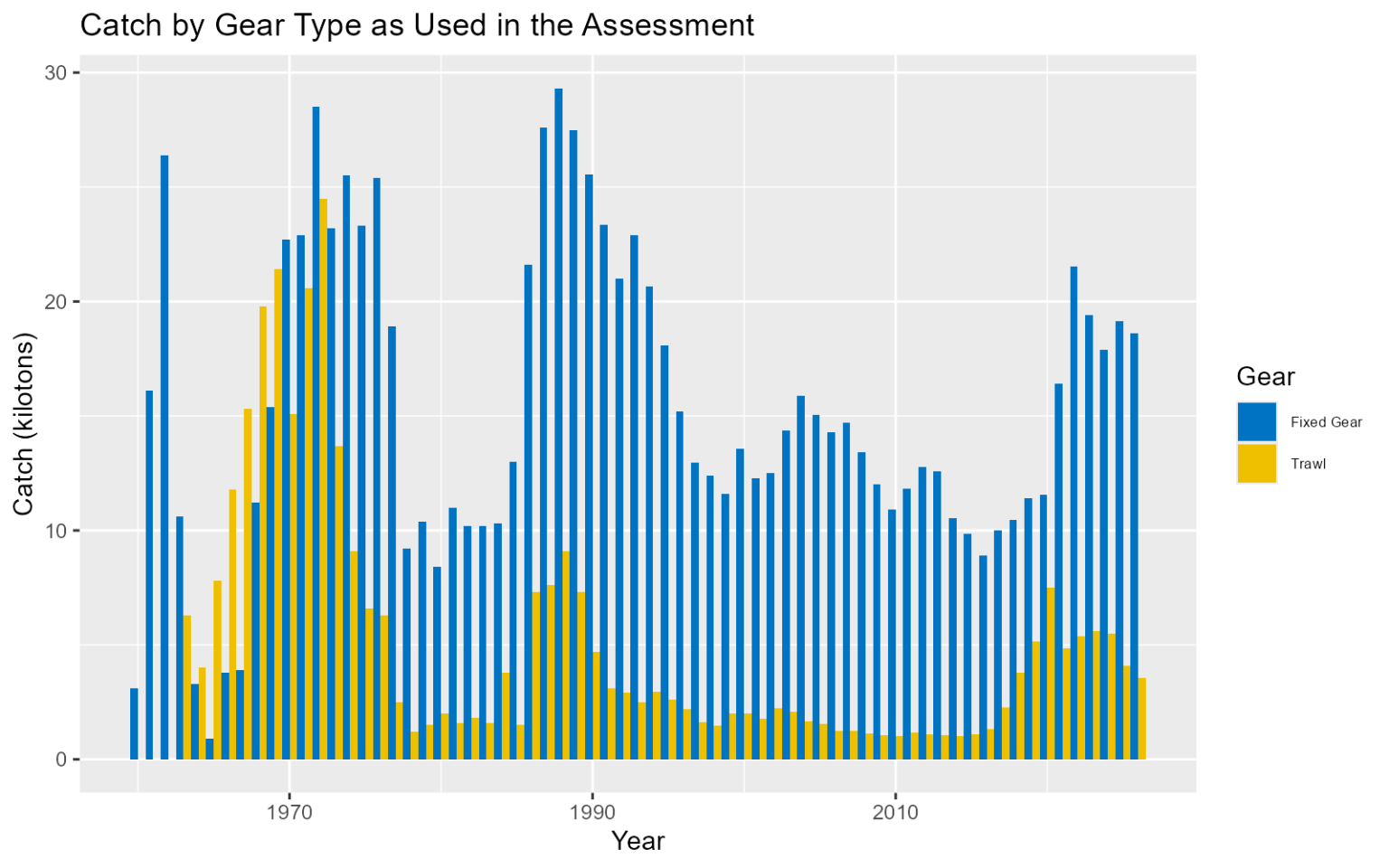

2021 Estimates, Post-1996
Current Model, Post-1996
Pre-1996 Estimates
Female
Male
Length Bin
Age Bin
0
10
20
30
Probability
0.20
0.15
0.10
0.05
0.00

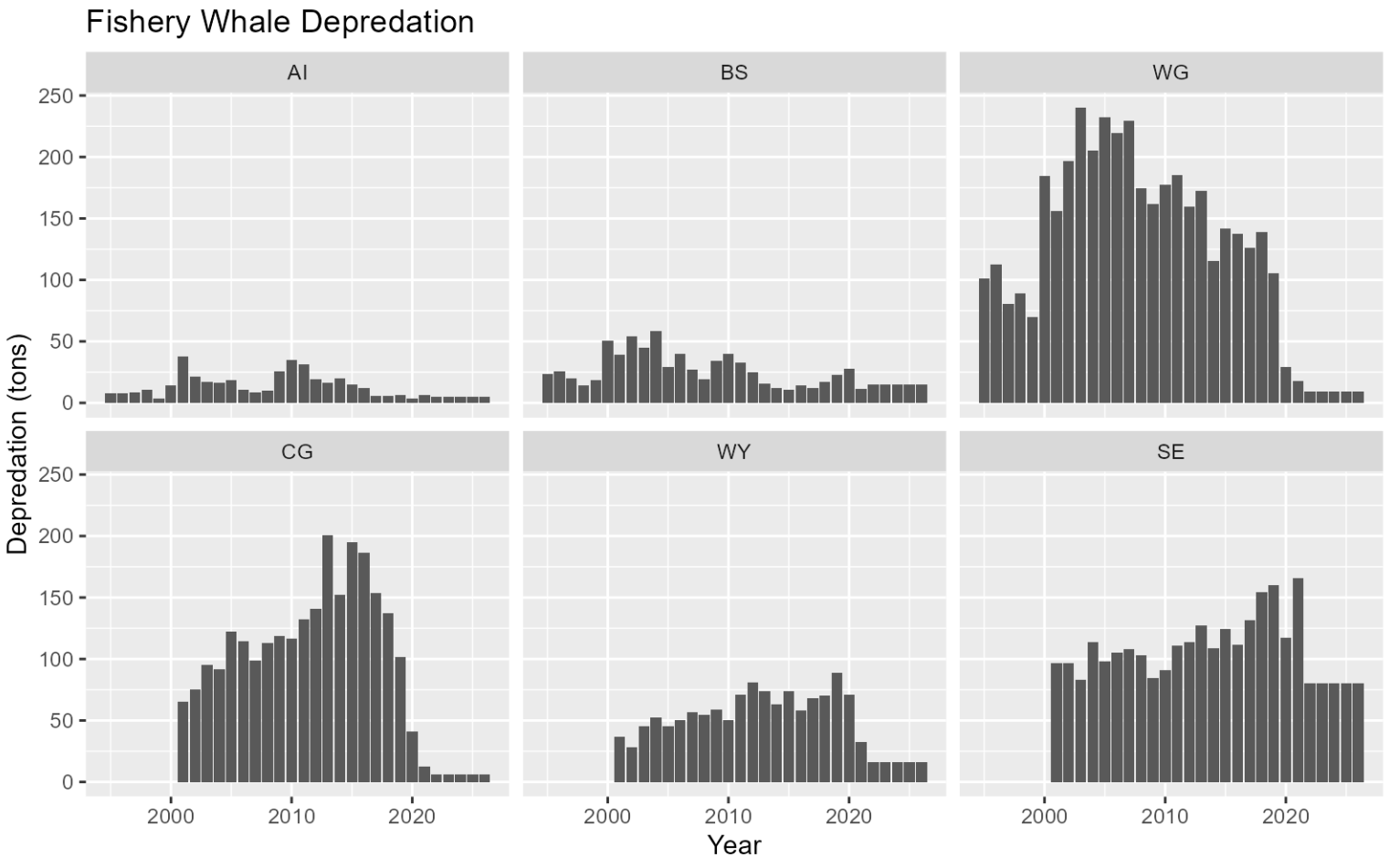

Female
Male
Weight (kg)
Age (years)
0
2
4
6
1
3
10
20
30
source
2021 Estimates
Current Estimates

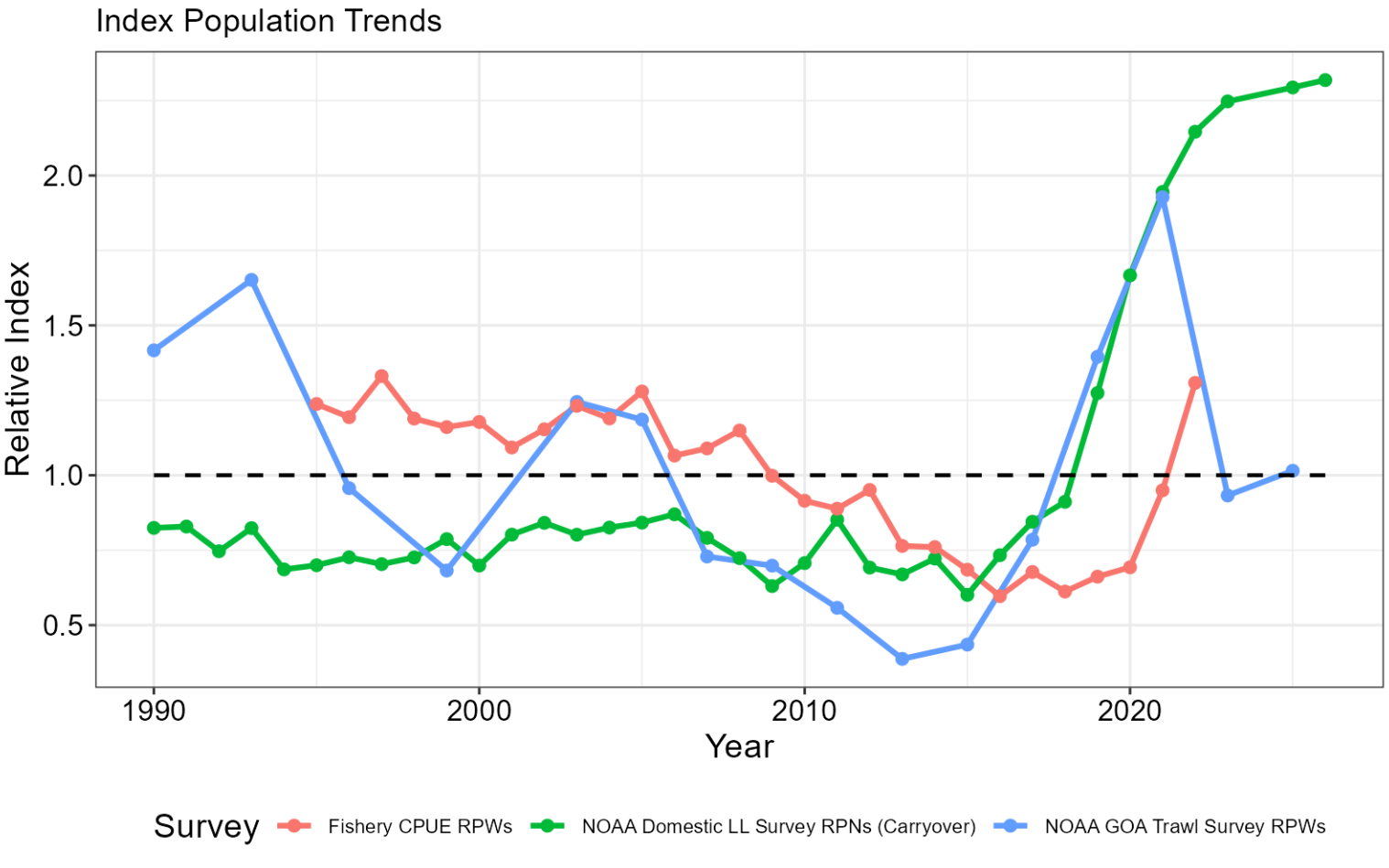

Catch by Gear Type
Catch (kilotons)
30
20
10
0
1970
1990
2010
Year
Gear
HAL
Pot
Trawl

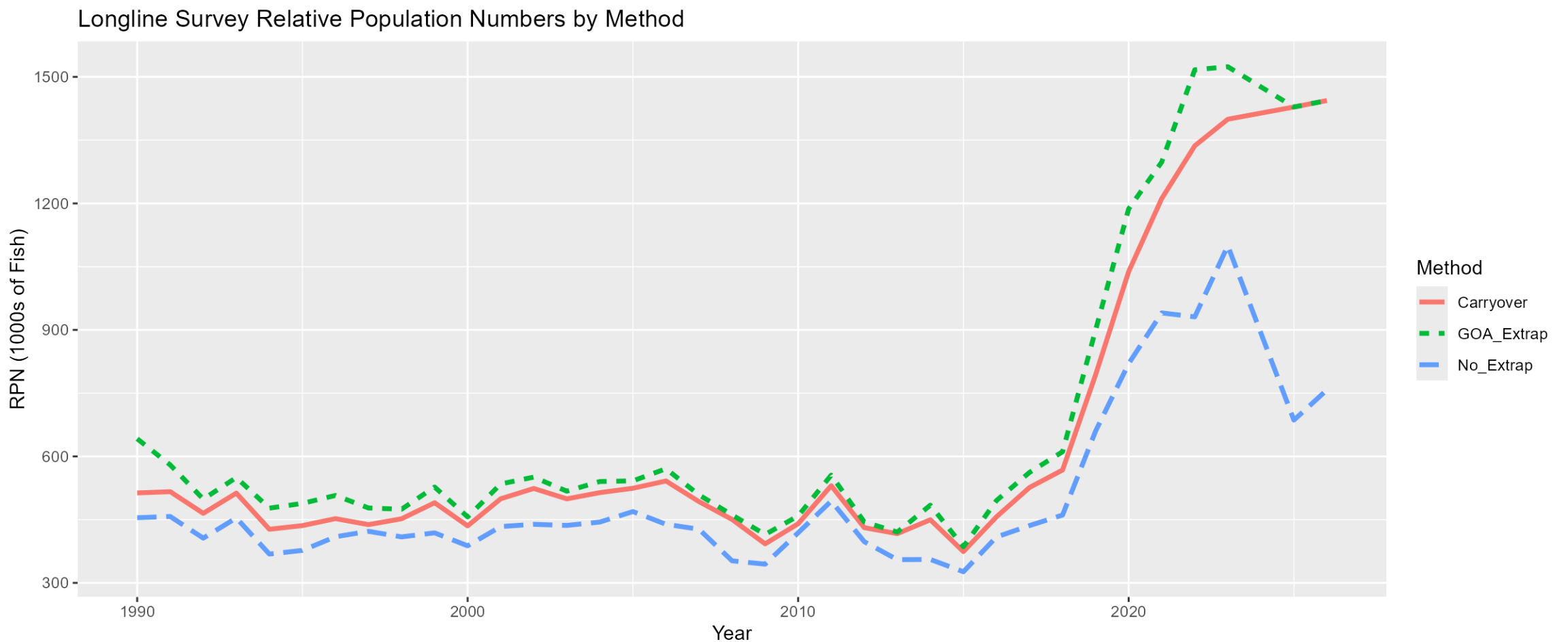

Catch by NPFMC Area
Catch (kilotons)
0
20
40
1970
1990
2010
Year
Area
AI
BS
CG
EG
WG

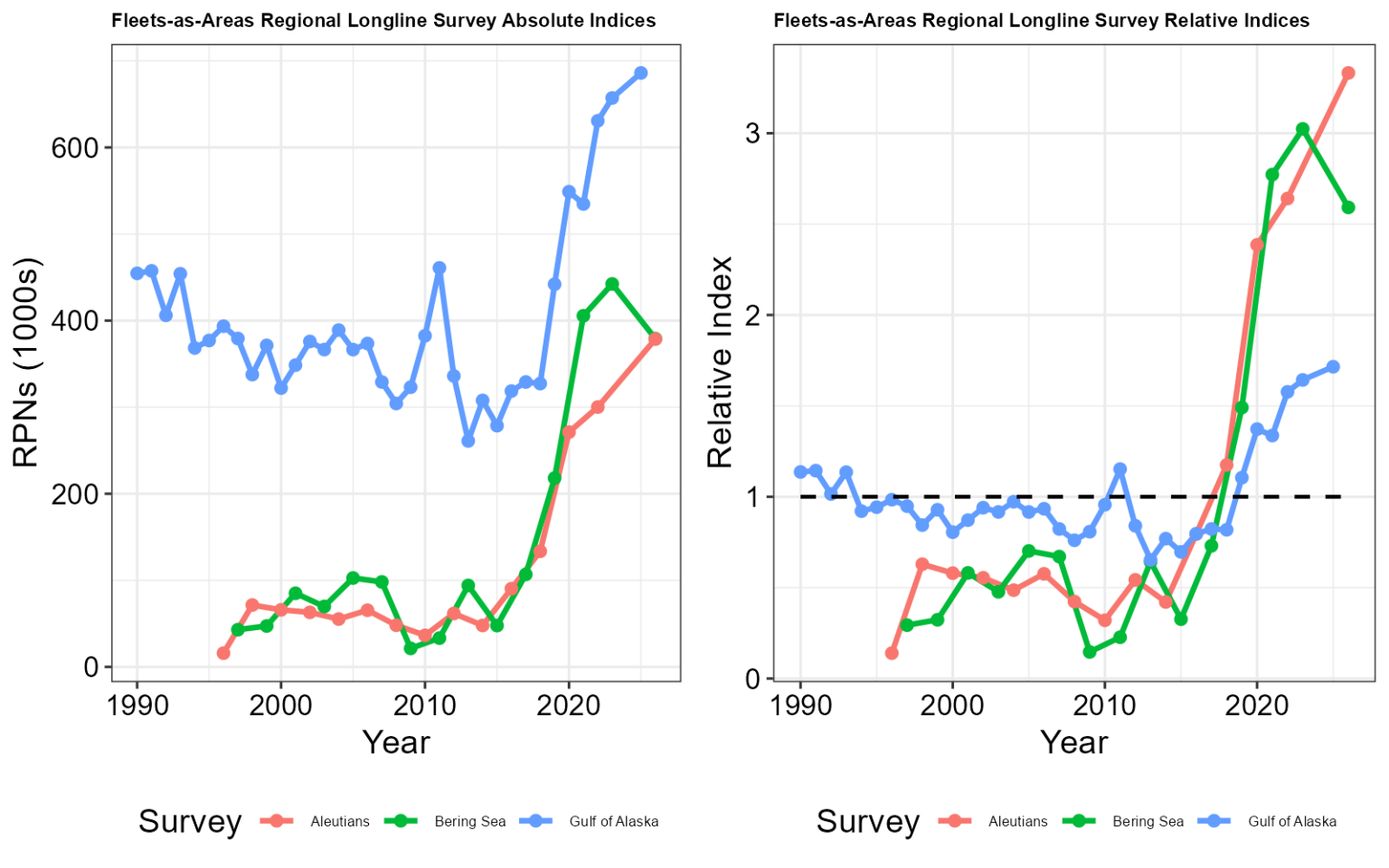

Catch by NPFMC Area and Gear Type
HAL
Pot
Trawl
Catch (kilotons)
0
10
20
1990
2000
2010
2020
Year
Area
AI
BS
CG
EG
WG

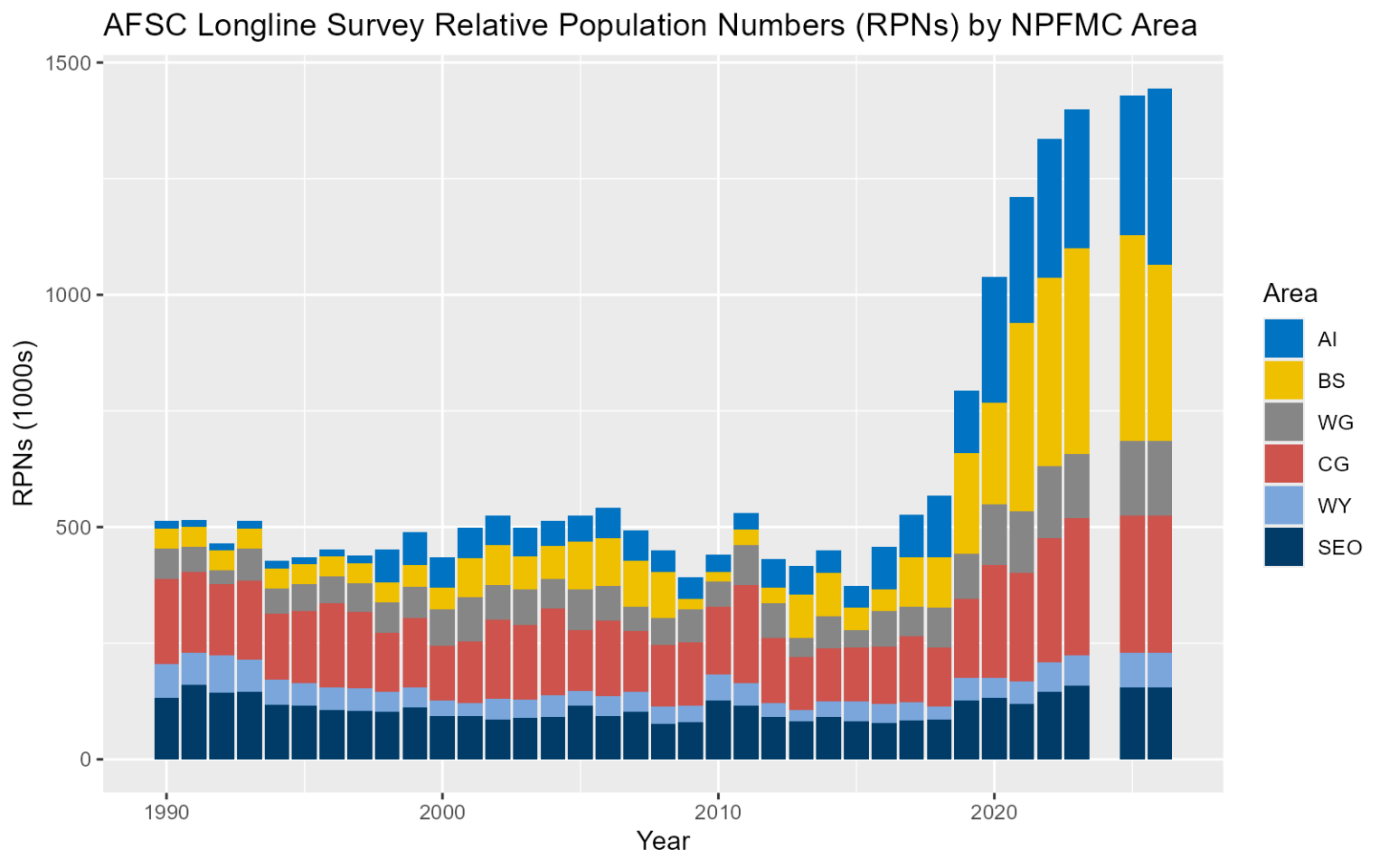

Catch by Gear Type as Used in the Assessment
Catch (kilotons)
30
20
10
0
1970
1990
2010
Year
Gear
Fixed Gear
Trawl

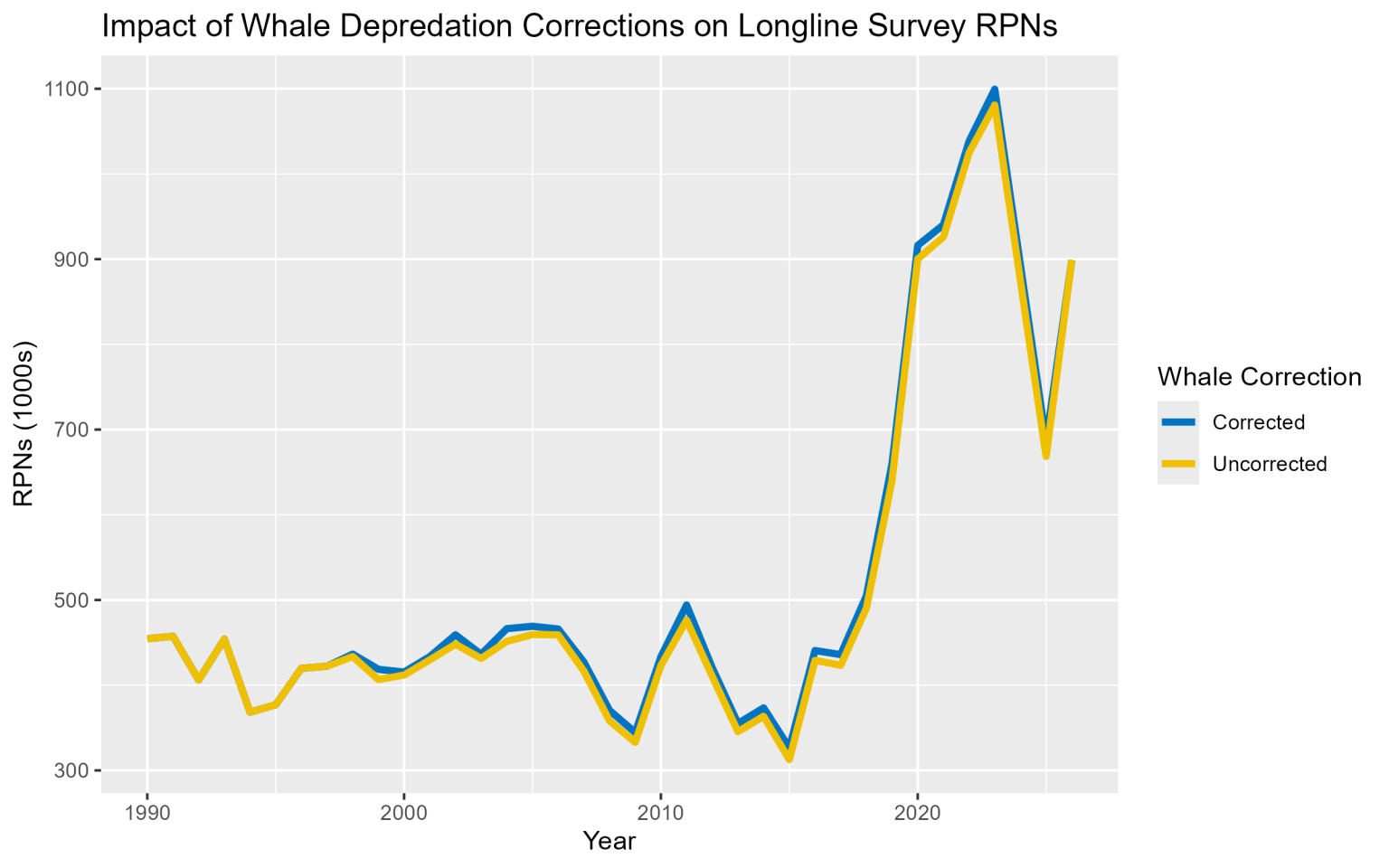

Fishery Whale Depredation
AI
BS
WG
CG
WY
SE
Depredation (tons)
0
50
100
150
200
250
2000
2010
2020
Year

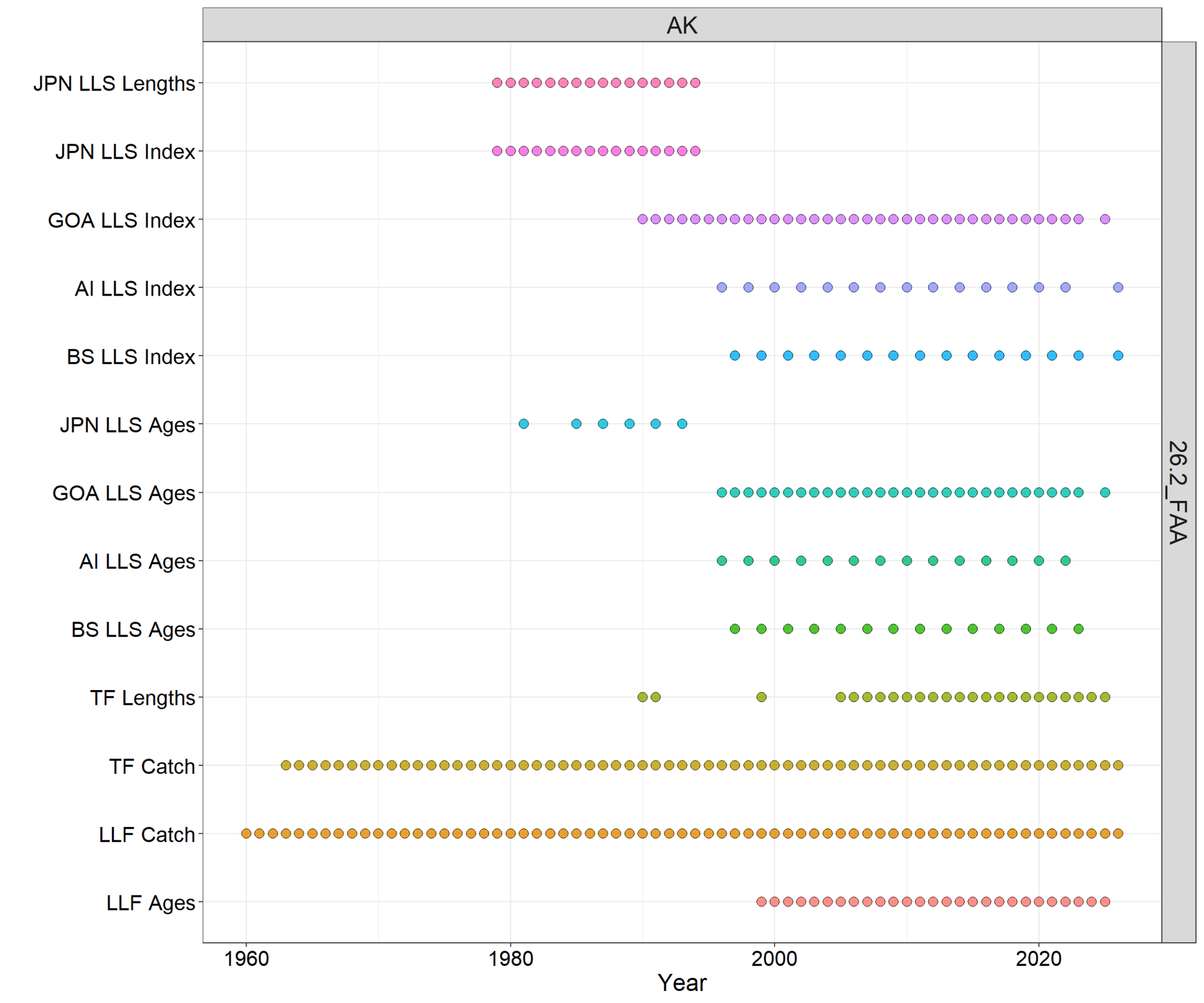

Index Population Trends
Relative Index
2.0
1.5
1.0
0.5
1990
2000
2010
2020
Year
Survey
Fishery CPUE RPWs
NOAA Domestic LL Survey RPNs (Carryover)
NOAA GOA Trawl Survey RPWs

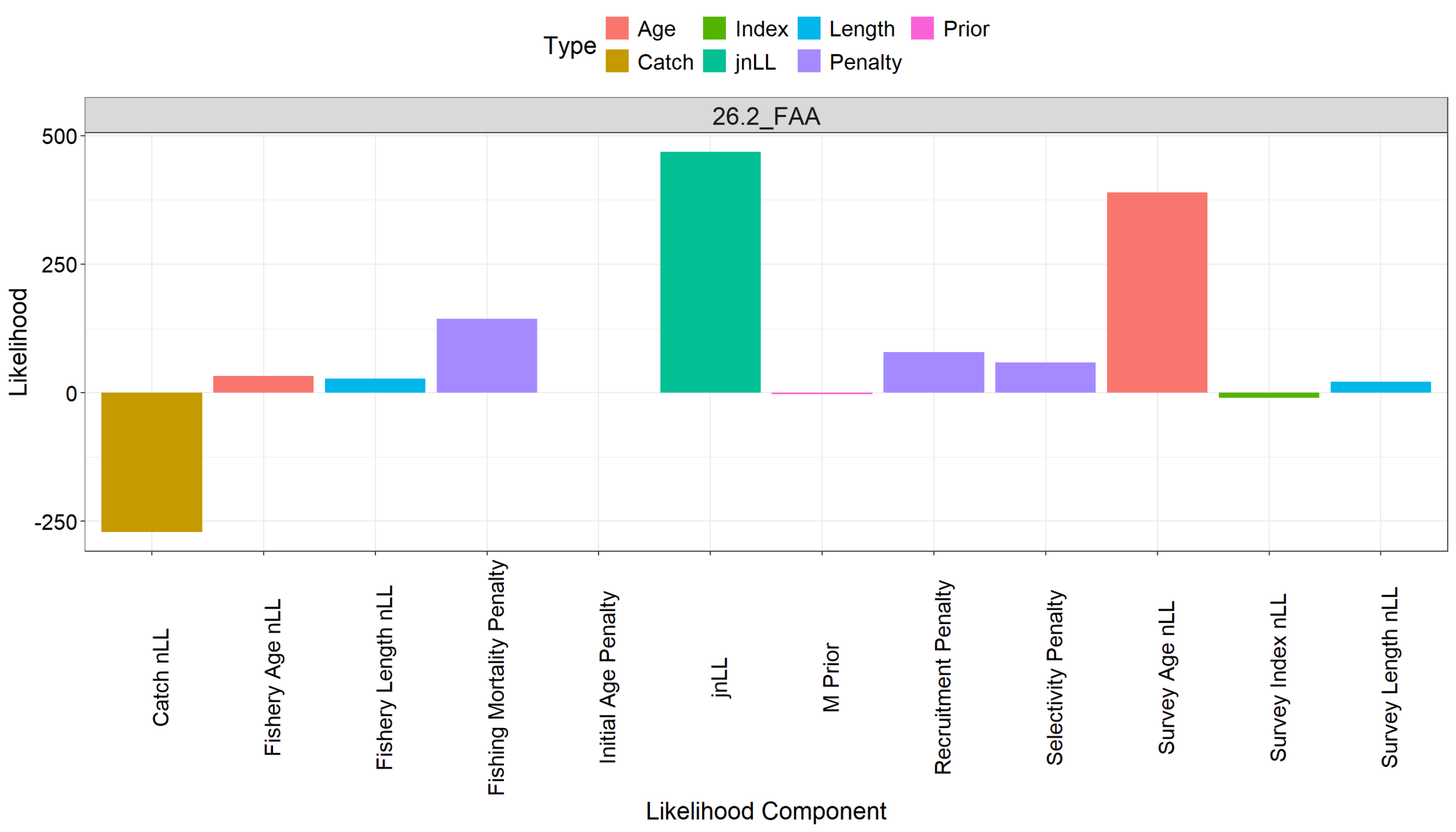

Longline Survey Relative Population Numbers by Method
RPN (1000s of Fish)
1500
1200
900
600
300
1990
2000
2010
2020
Year
Method
Carryover
GOA_Extrap
No_Extrap


Fleets-as-Areas Regional Longline Survey Absolute Indices
RPNs (1000s)
0
200
400
600
1990
2000
2010
2020
Year
Survey
Aleutians
Bering Sea
Gulf of Alaska
Fleets-as-Areas Regional Longline Survey Relative Indices
Relative Index
0
1
2
3
1990
2000
2010
2020
Year
Survey
Aleutians
Bering Sea
Gulf of Alaska


AFSC Longline Survey Relative Population Numbers (RPNs) by NPFMC Area
RPNs (1000s)
1500
1000
500
0
1990
2000
2010
2020
Year
Area
AI
BS
WG
CG
WY
SEO


Impact of Whale Depredation Corrections on Longline Survey RPNs
RPNs (1000s)
1100
900
700
500
300
1990
2000
2010
2020
Year
Whale Correction
Corrected
Uncorrected


AK
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.2_FAA
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.2_FAA
500
250
0
-250
Likelihood
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

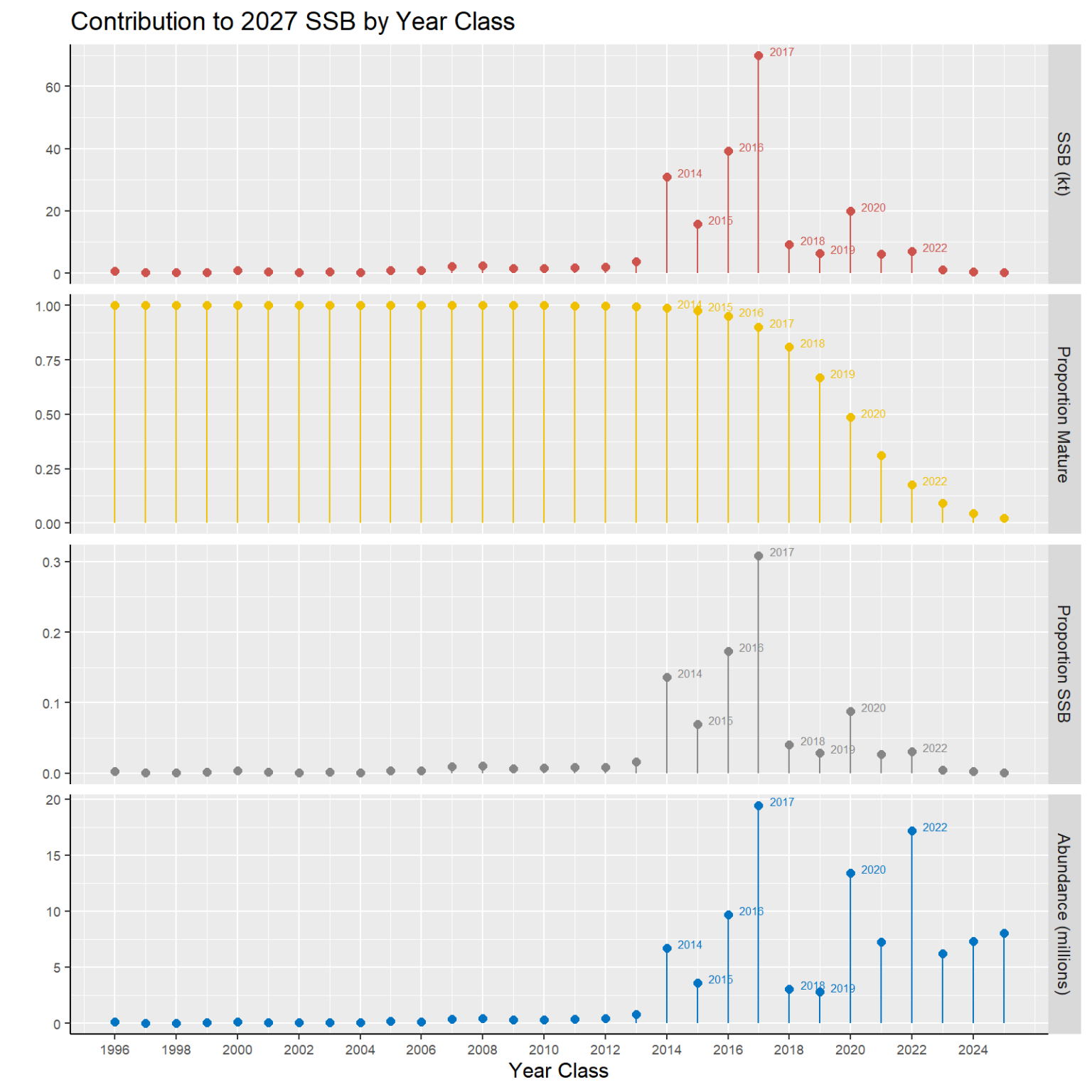
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

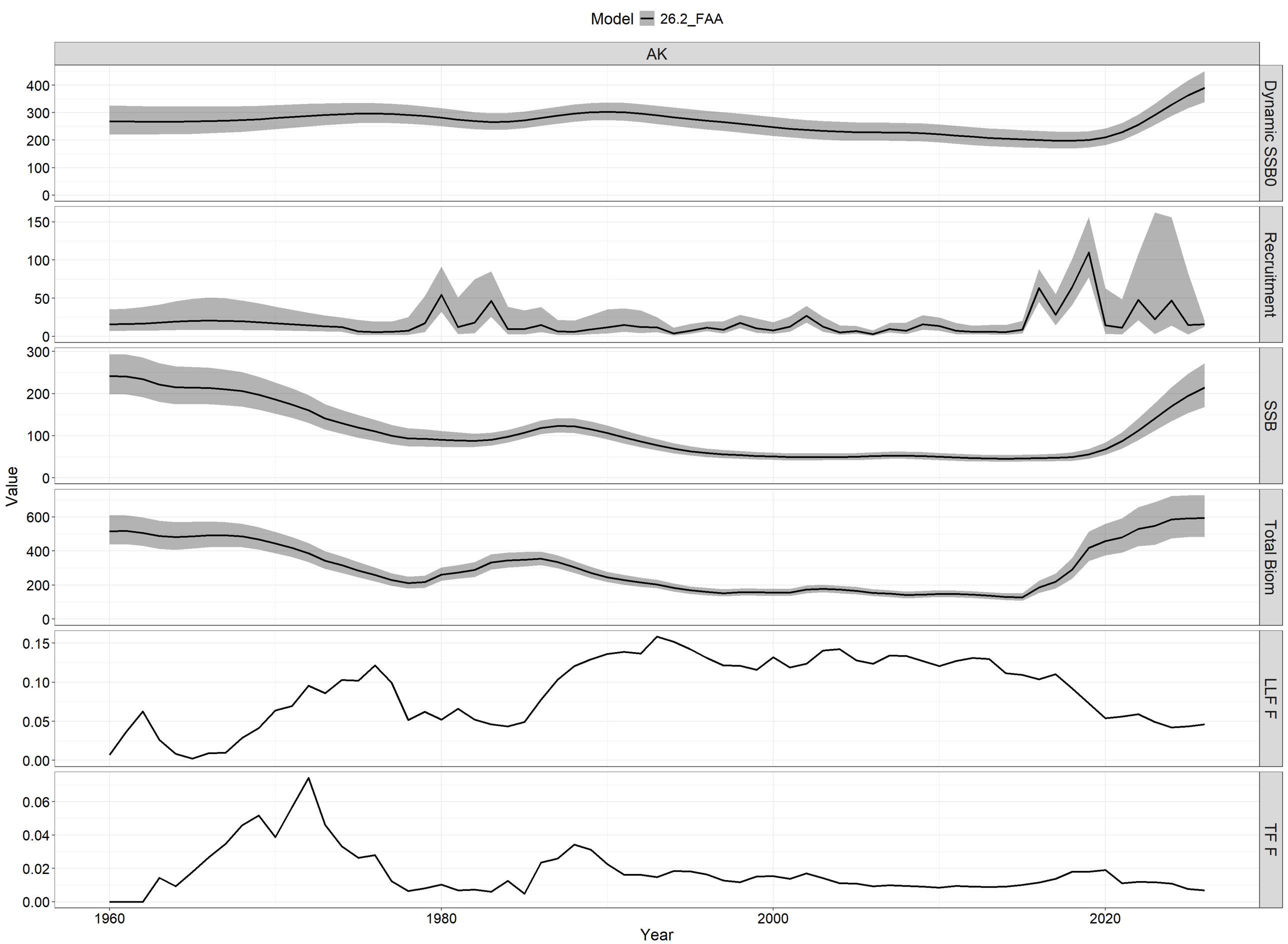
Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

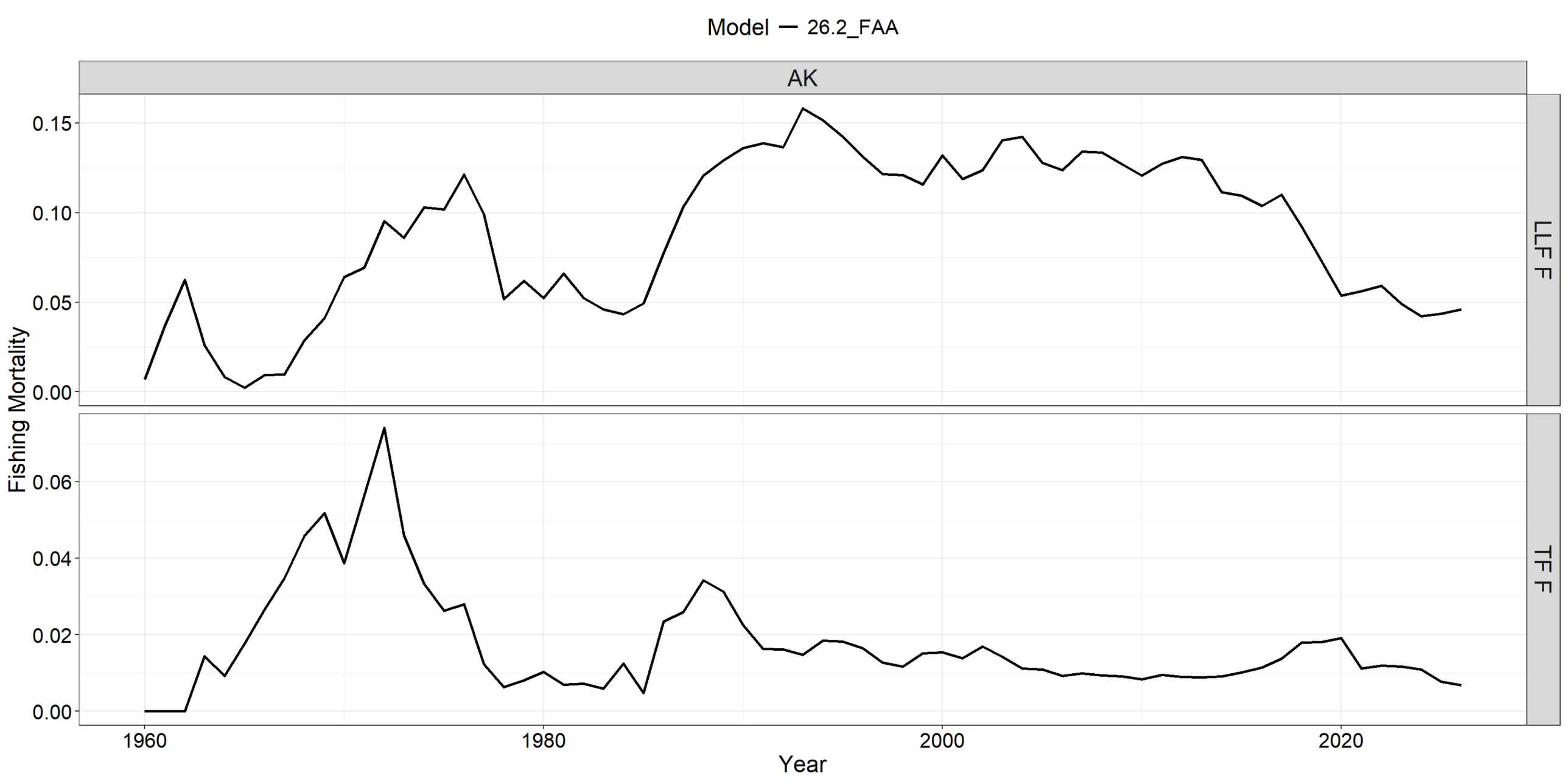
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

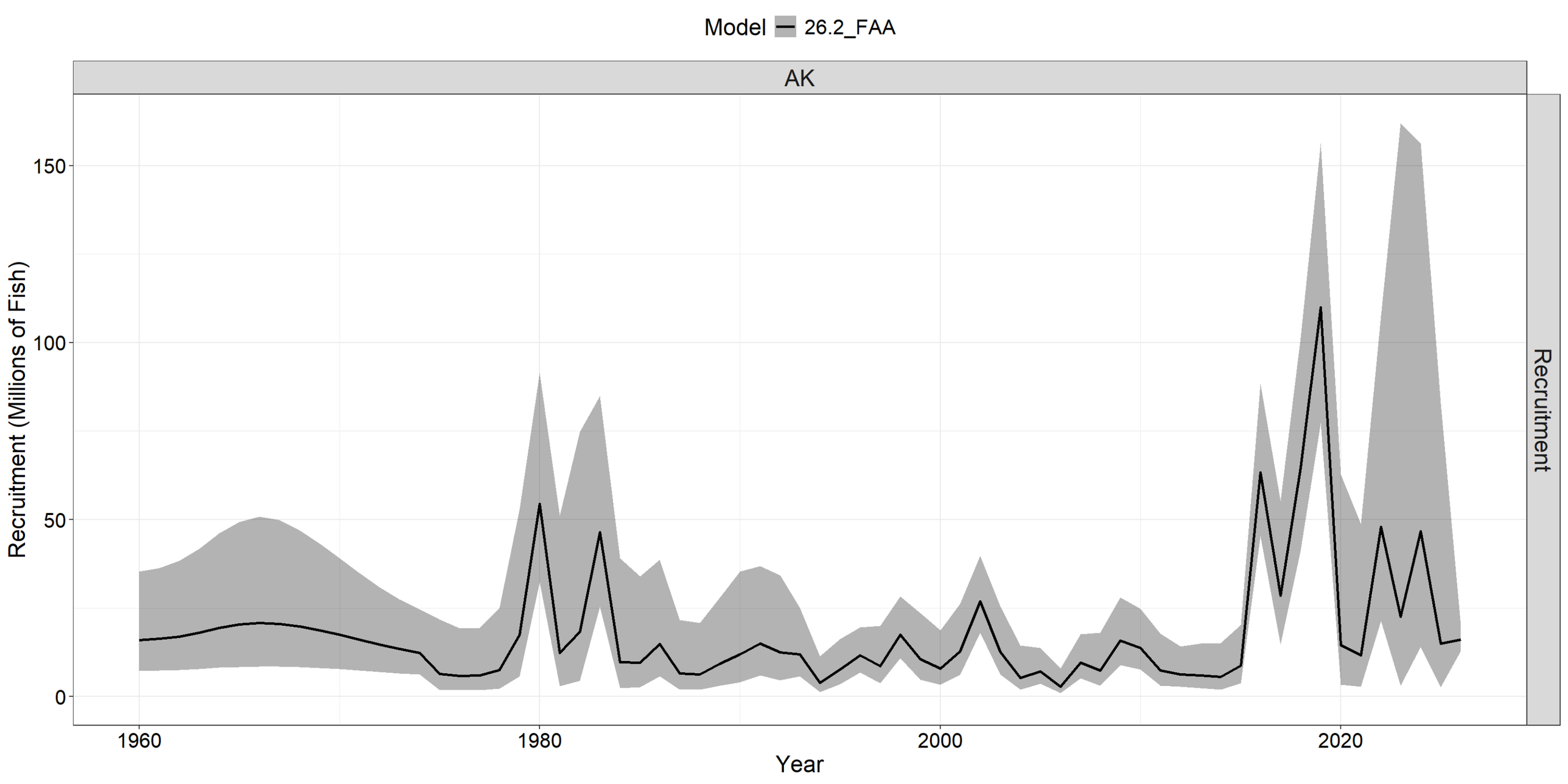
Numbers-at-Age (millions)

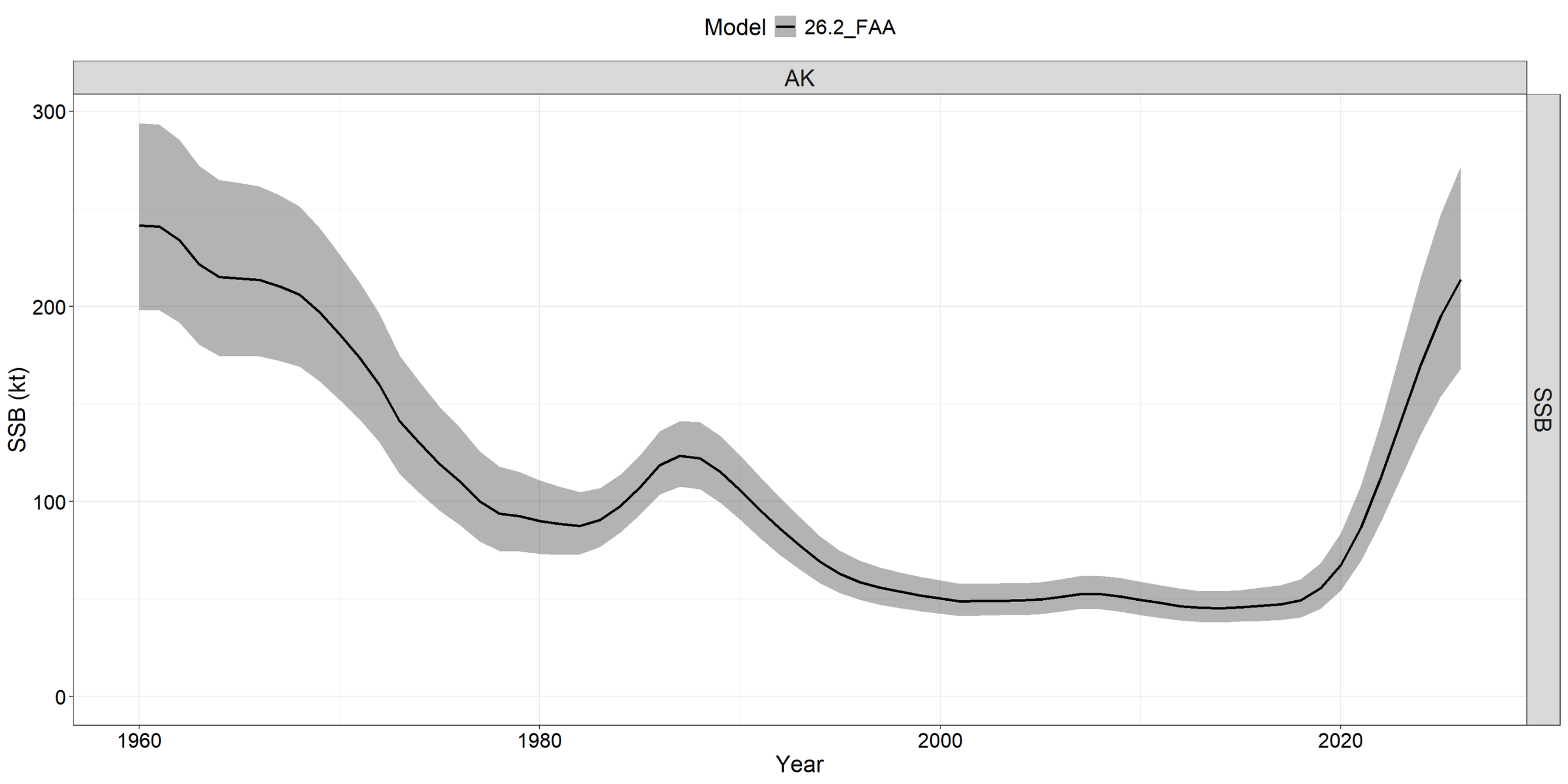
Year

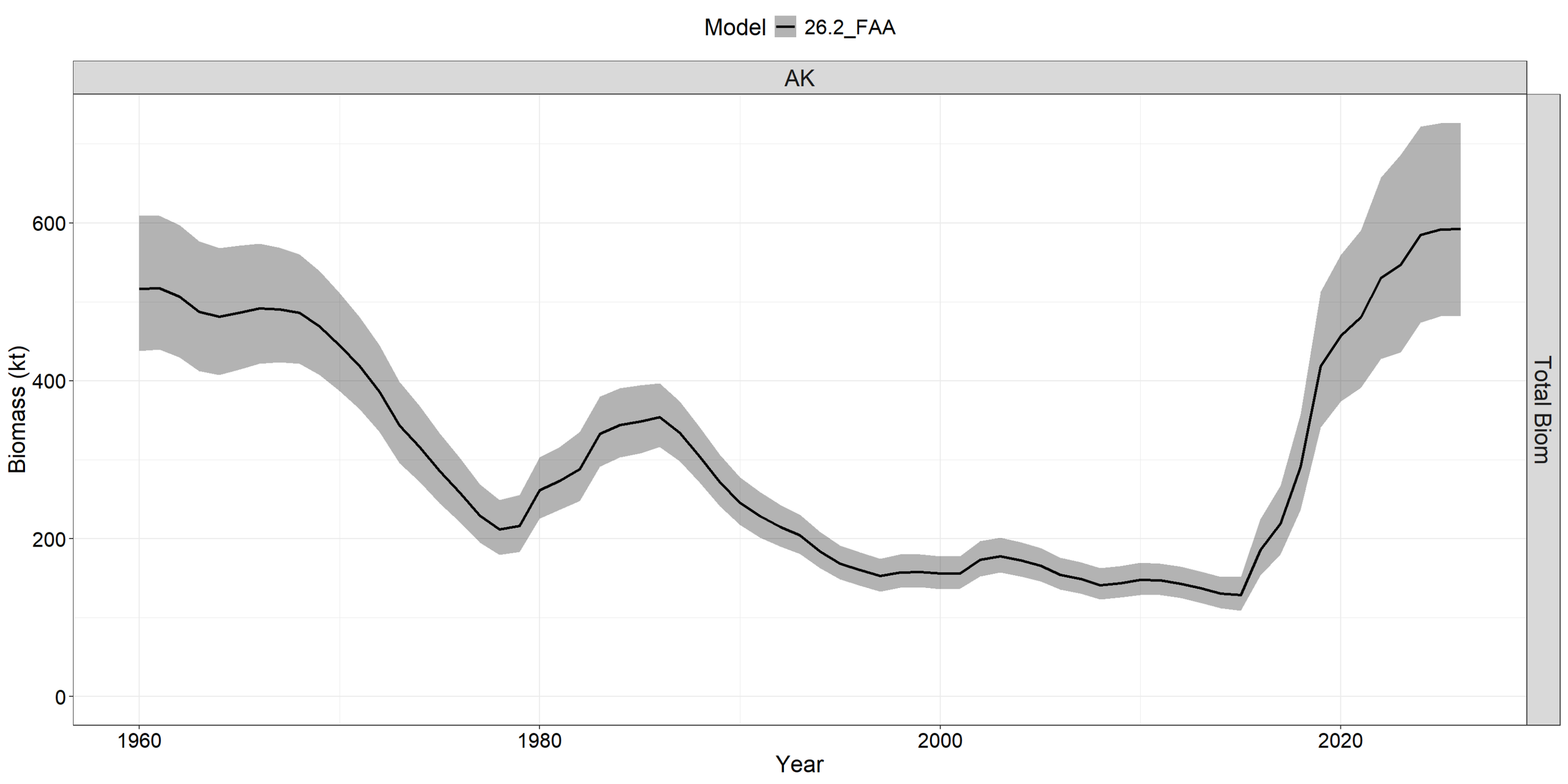

Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
2014
2015
2016
2017
2018
2019
2020
2022

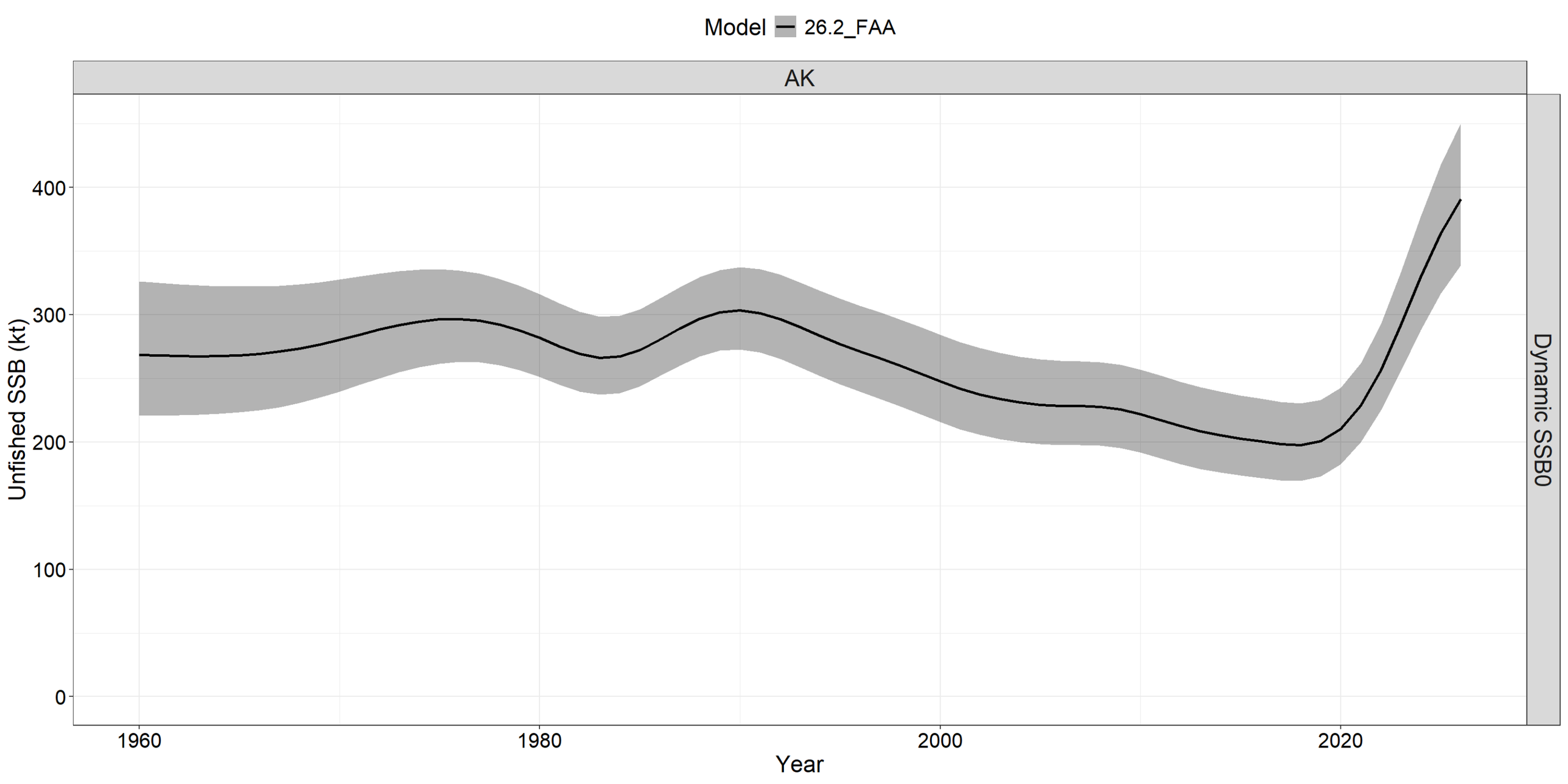

Model
26.2_FAA
AK
Dynamic SSB0
Recruitment
SSB
Total Biom
LLF F
TF F
Value
Year
0
100
200
300
400
50
150
600
0.00
0.02
0.04
0.06
0.10
0.15
1960
1980
2000
2020

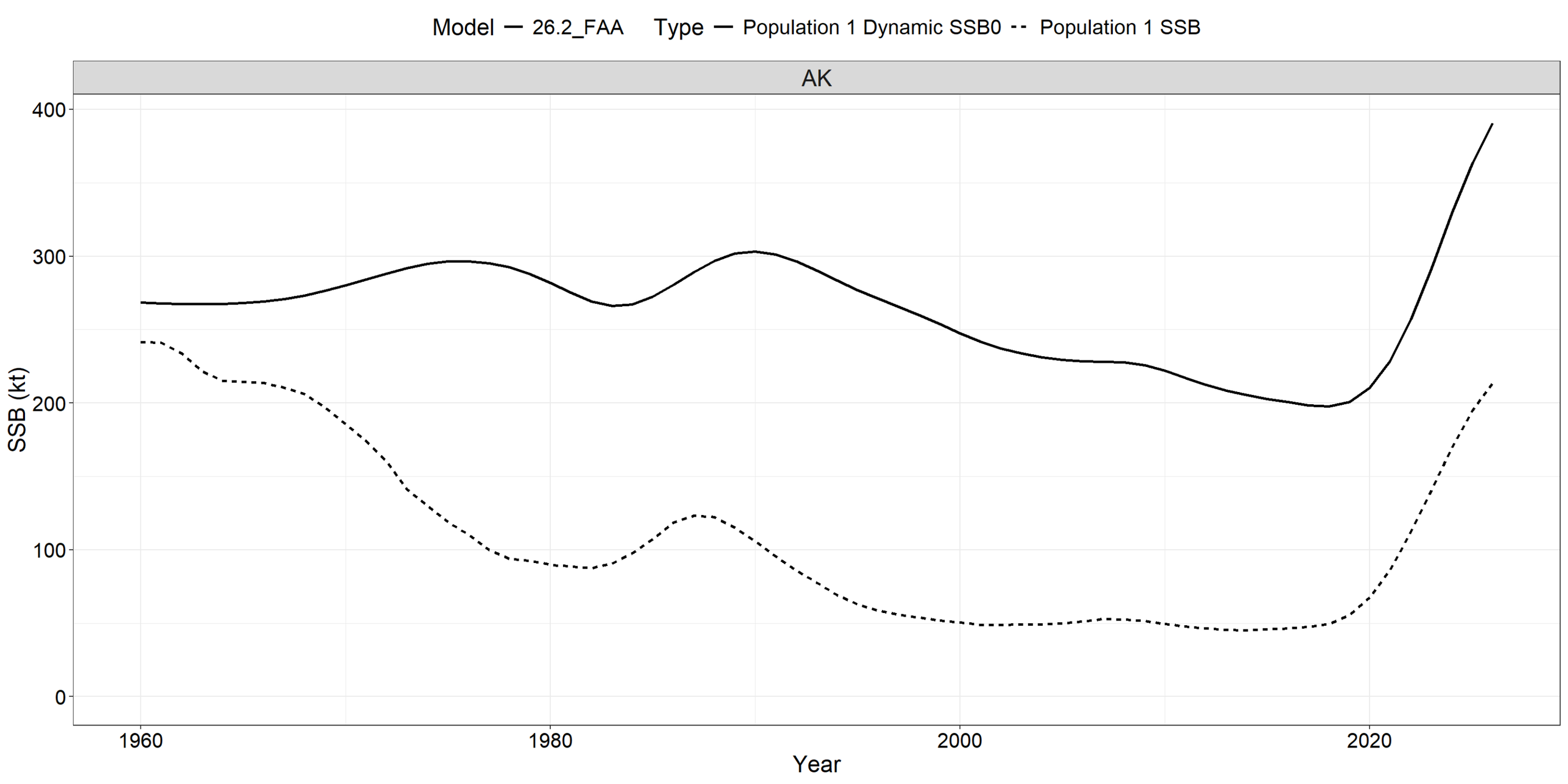

Model — 26.2_FAA
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year

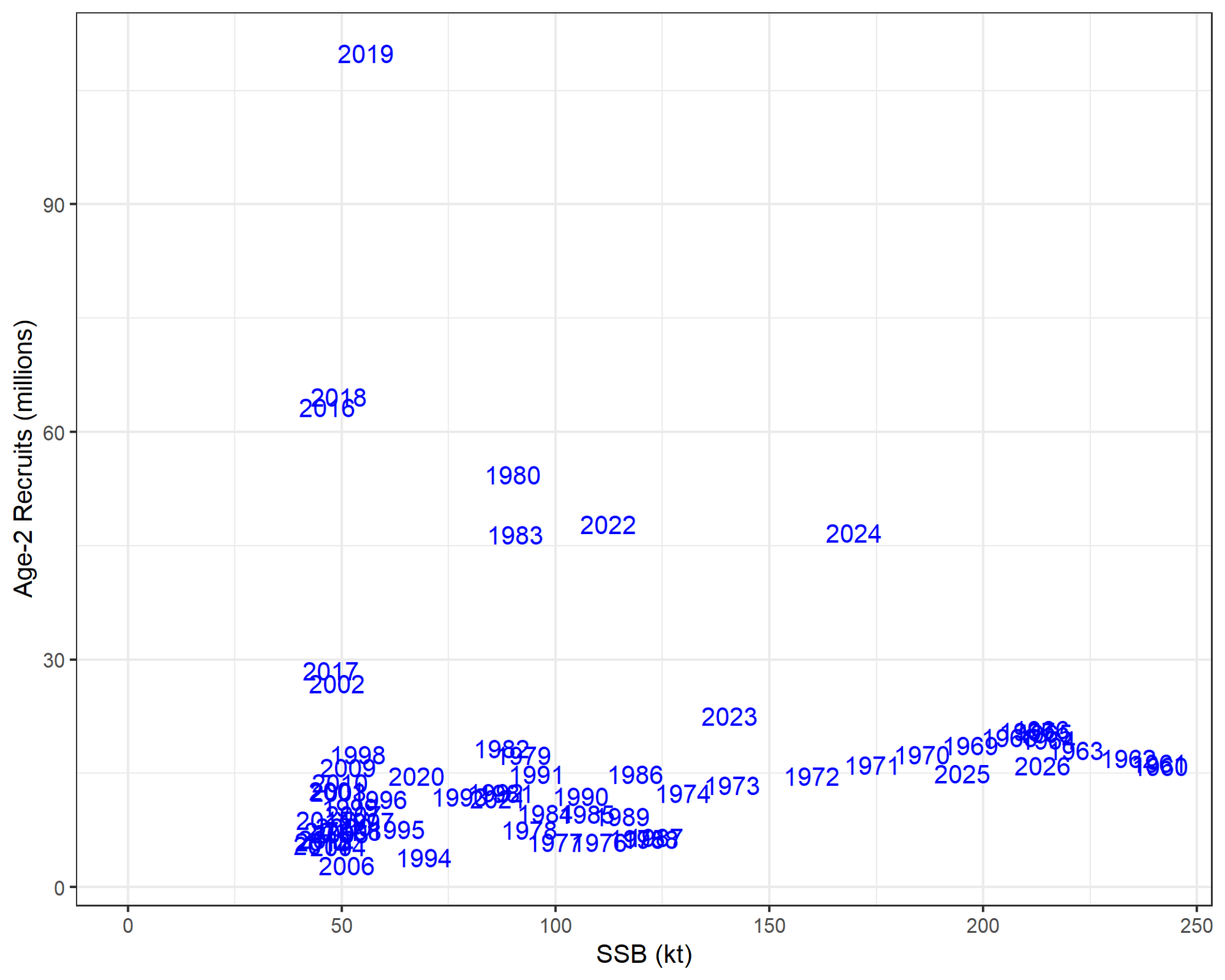

Model
26.2_FAA
AK
Recruitment (Millions of Fish)
150
100
50
0
1960
1980
2000
2020
Year
Recruitment

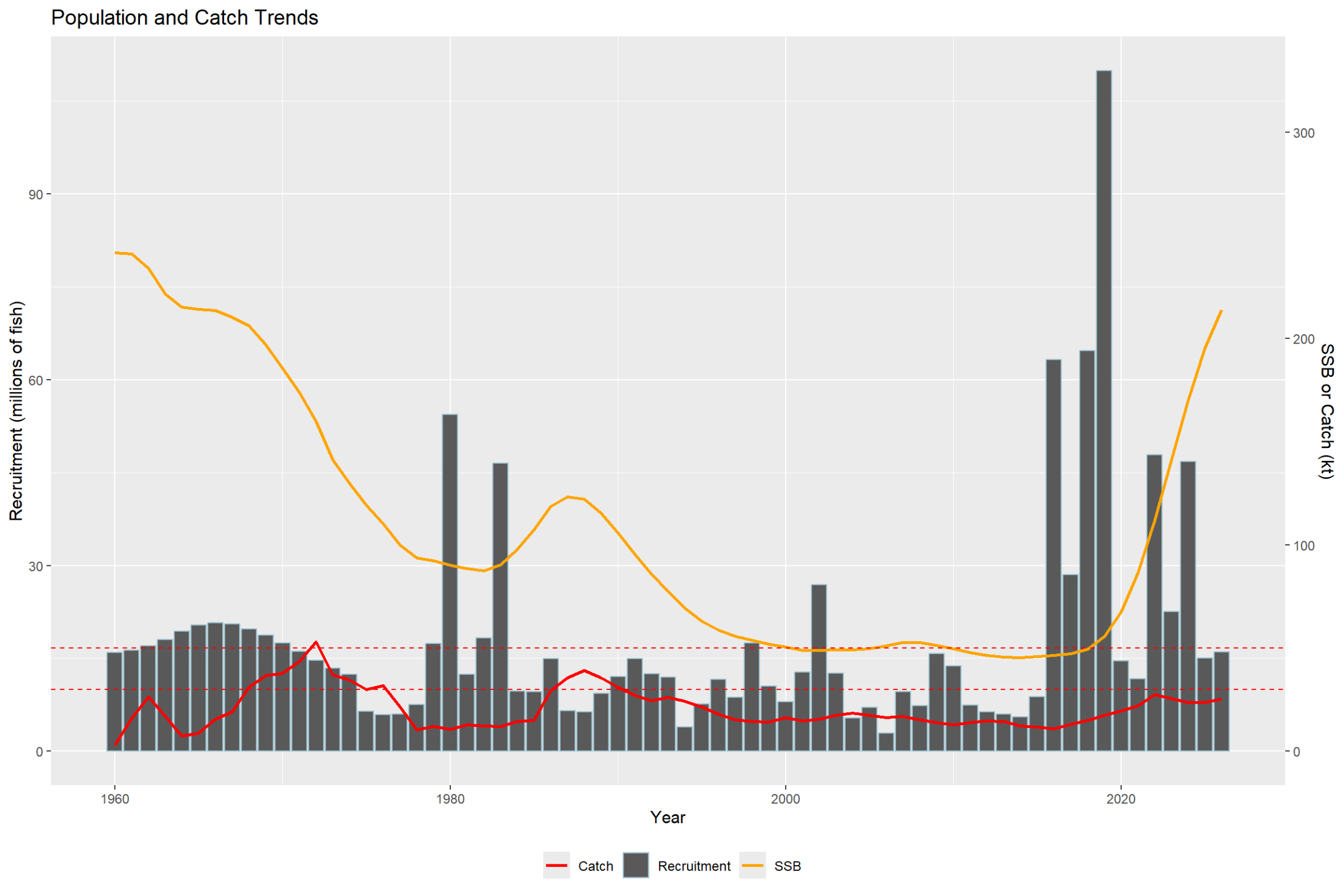

Model
26.2_FAA
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

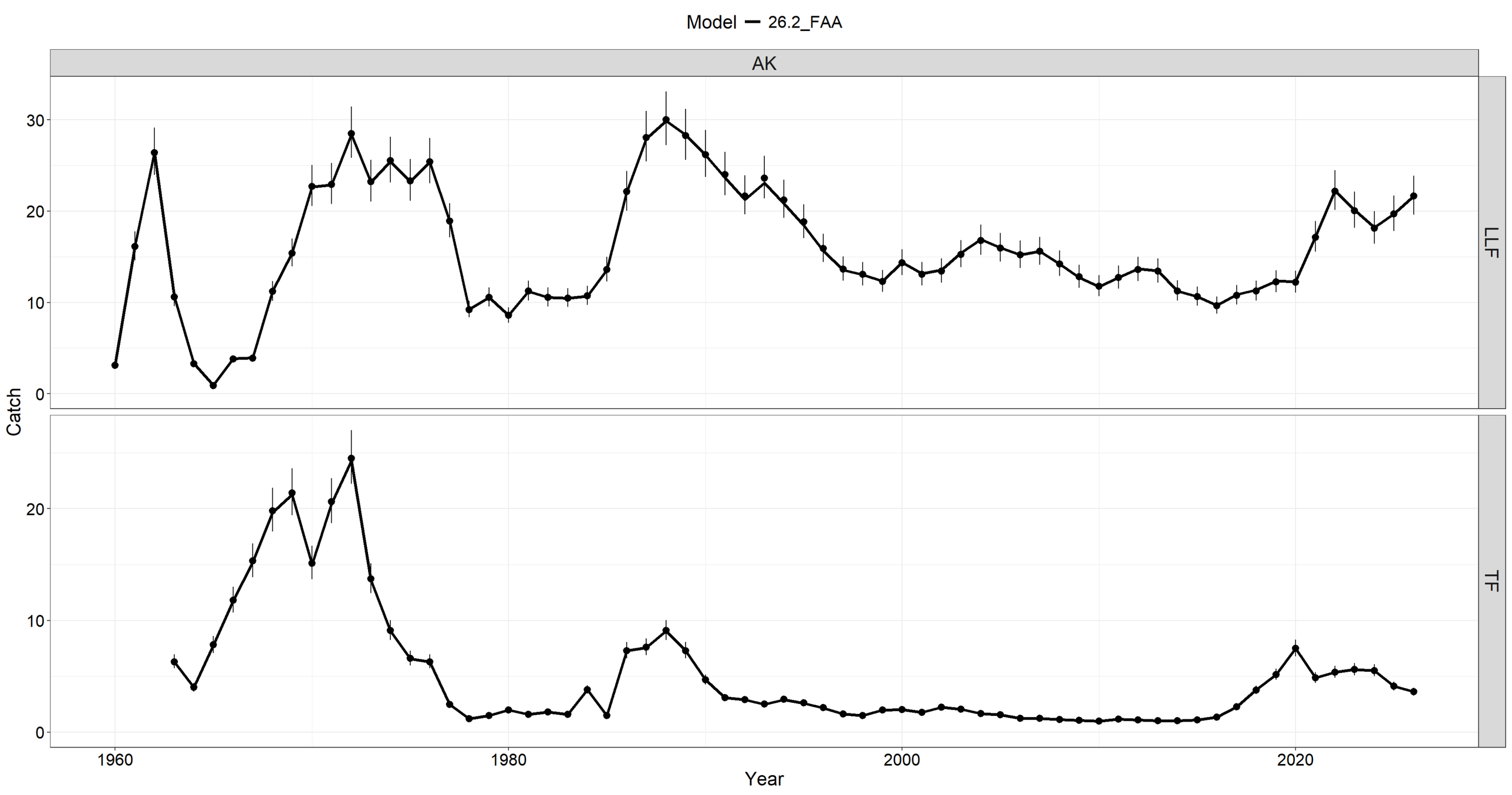

Model
26.2_FAA
AK
Total Biom
Biomass (kt)
0
200
400
600
1960
1980
2000
2020
Year


Model
26.2_FAA
AK
Dynamic SSB0
Unfished SSB (kt)
0
100
200
300
400
1960
1980
2000
2020
Year


Model — 26.2_FAA Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year


Age-2 Recruits (millions)
SSB (kt)
0
30
60
90
0
50
100
150
200
250
2019
2018
2016
1980
1983
2022
2024
2017
2002
2023
1998
2009
2020
1991
1986
1974
1973
1972
1971
1970
1969
1963
2025
2026
1994
2006


Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
0
30
60
90
0
100
200
300
1960
1980
2000
2020
Year
Catch
Recruitment
SSB


Model — 26.2_FAA
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA

BS LLS

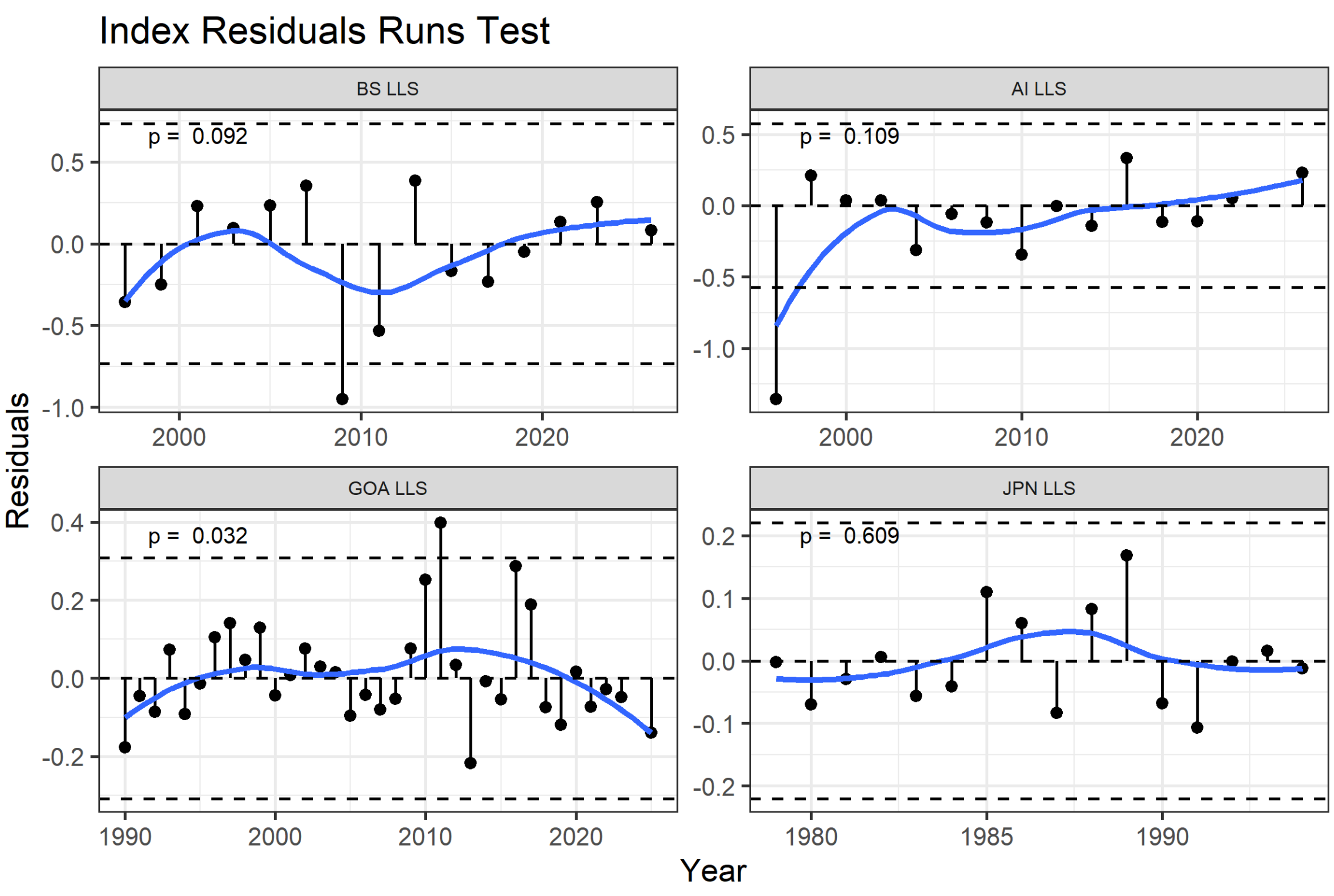
AI LLS

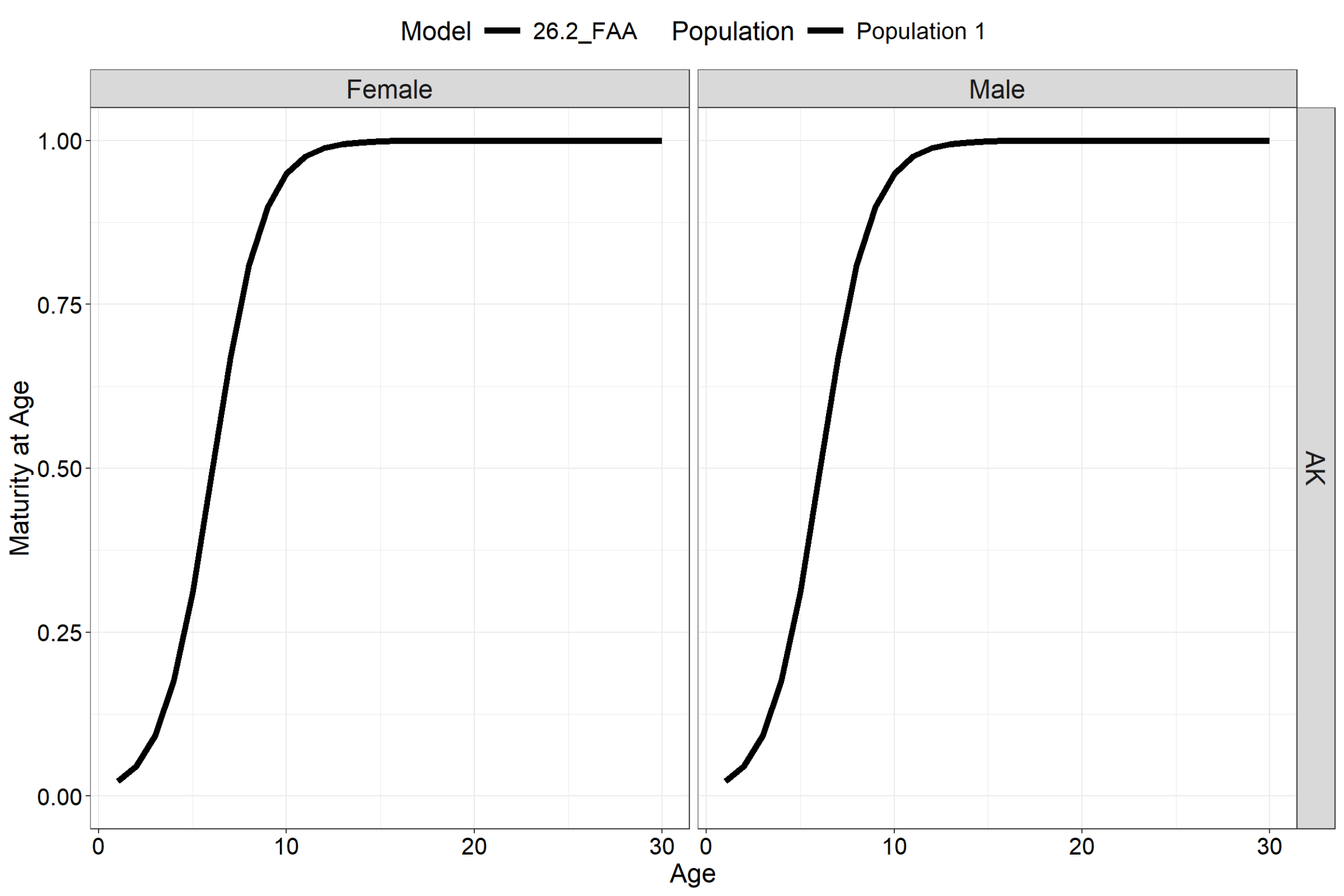
GOA LLS

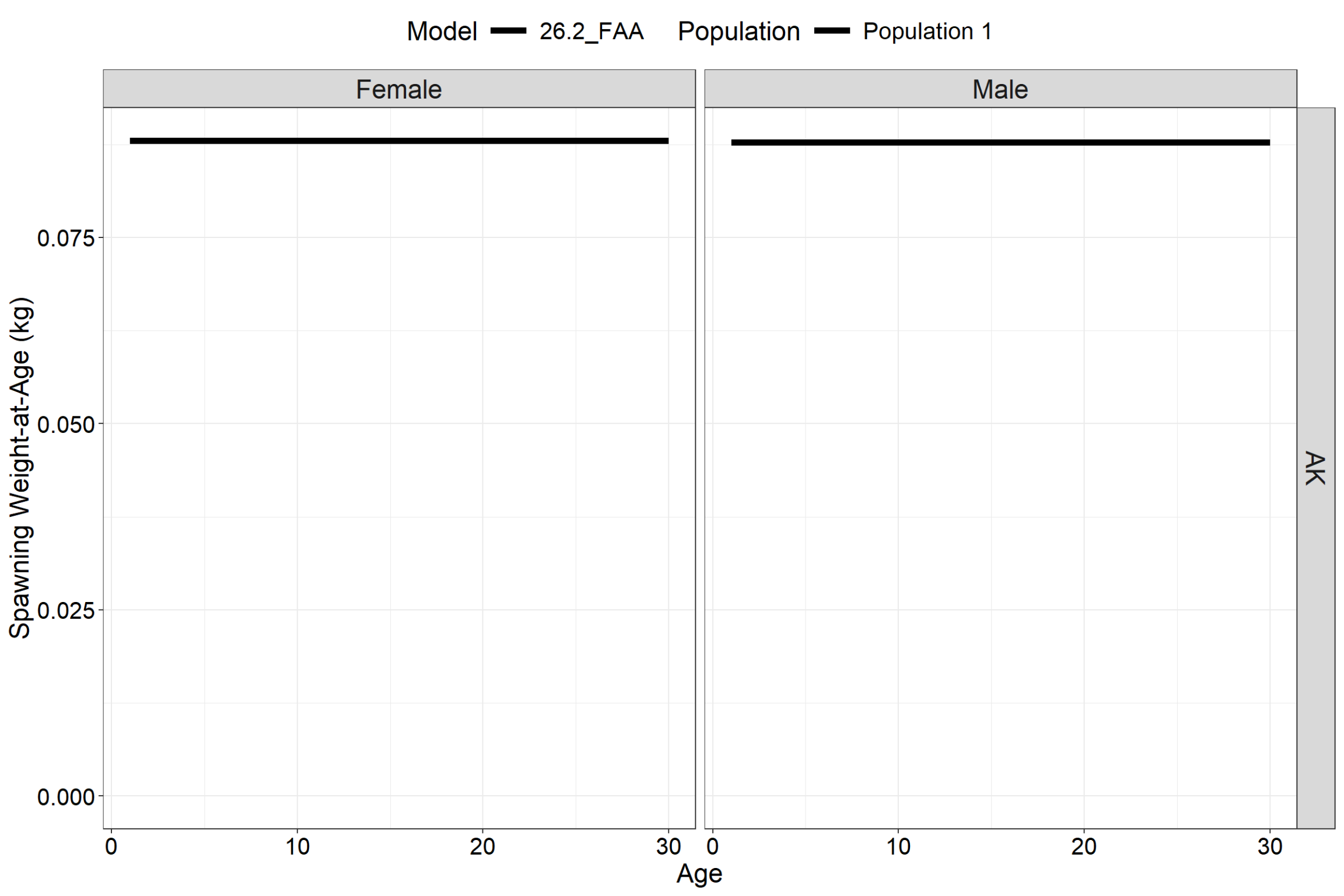
JPN LLS

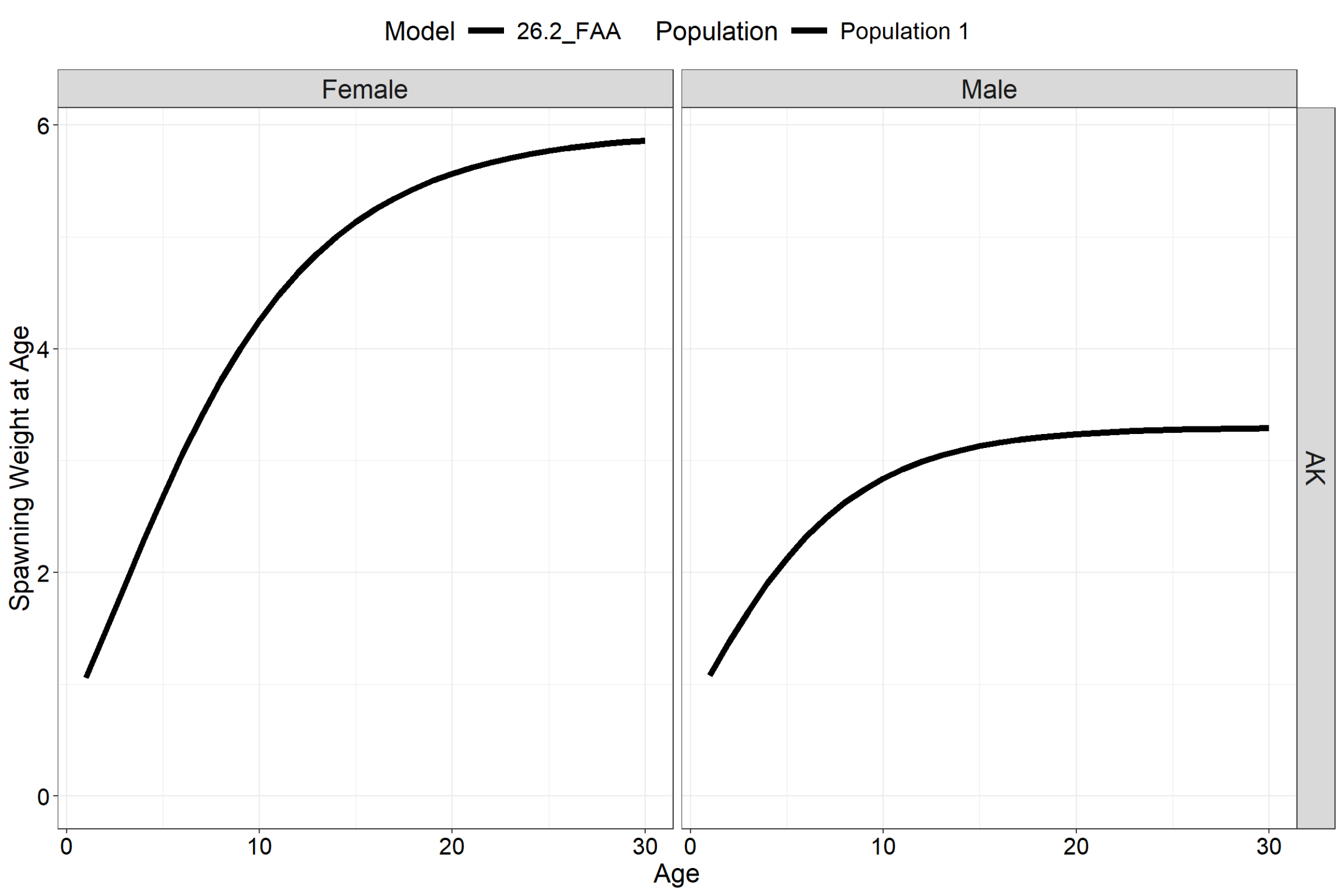
RPNs (1000s)

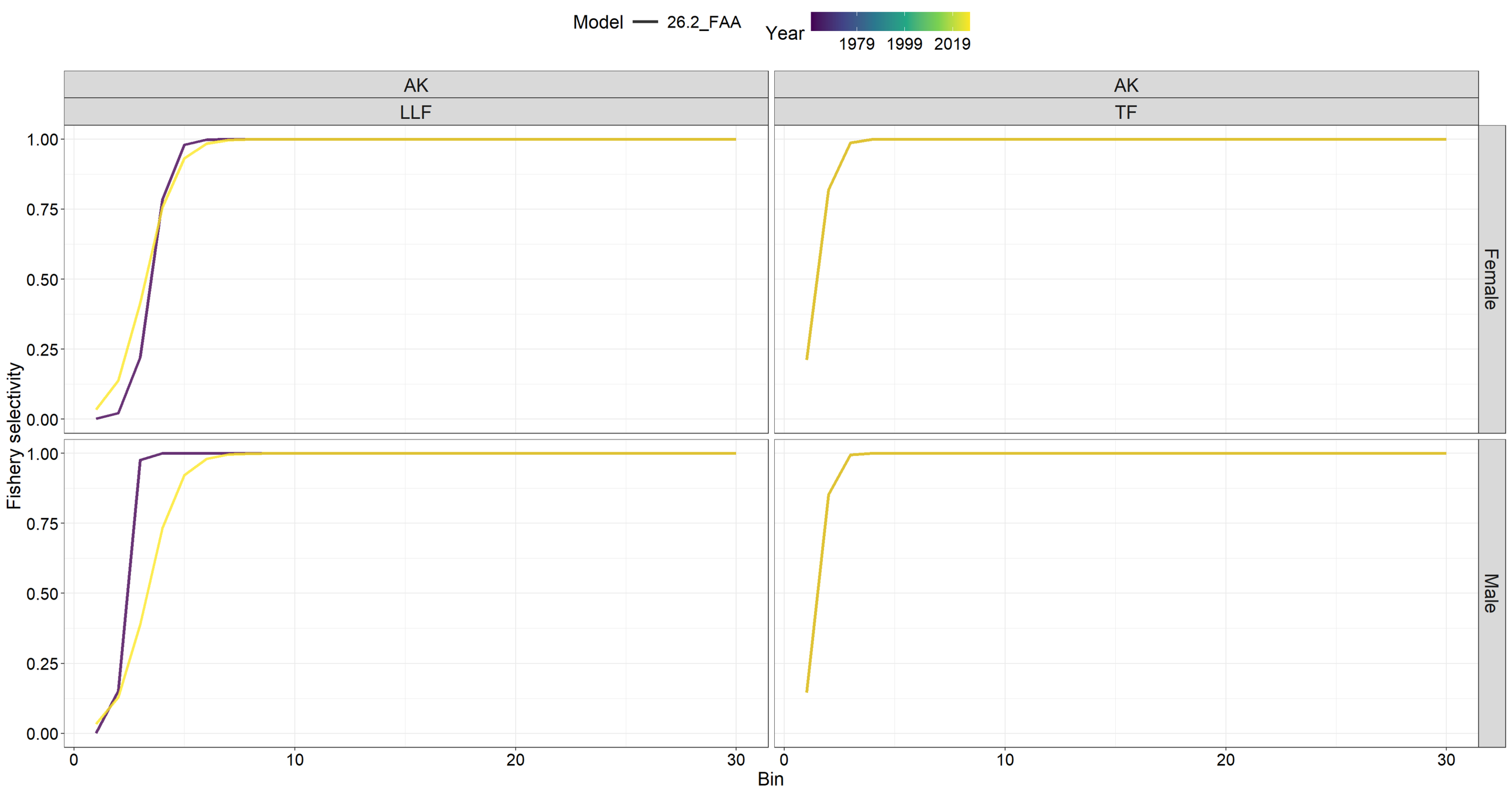
Year

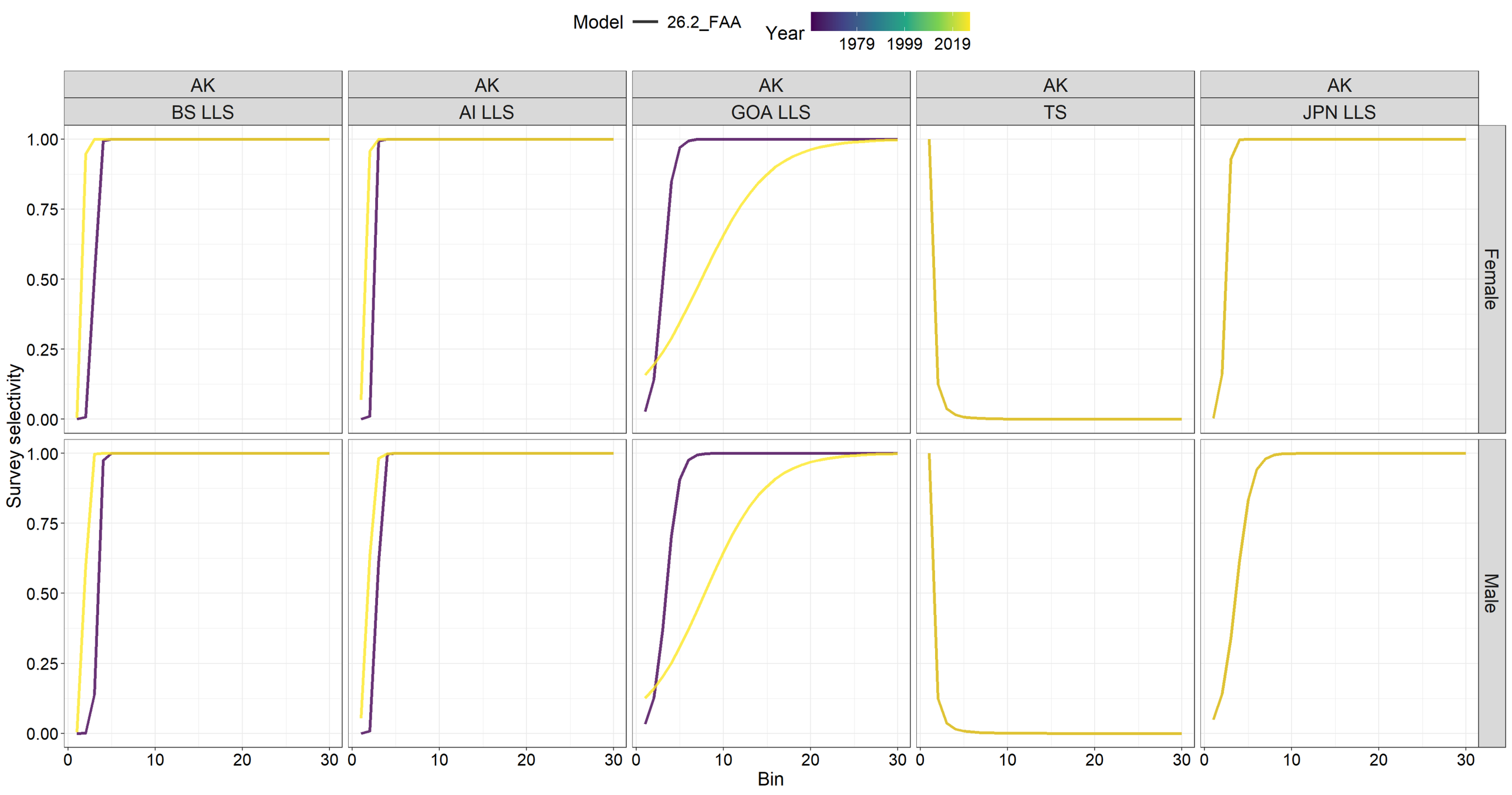

Index Residuals Runs Test
BS LLS
p = 0.092
AI LLS
p = 0.109
GOA LLS
p = 0.032
JPN LLS
p = 0.609
Residuals
Year


Model
26.2_FAA
Population
Population 1
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age

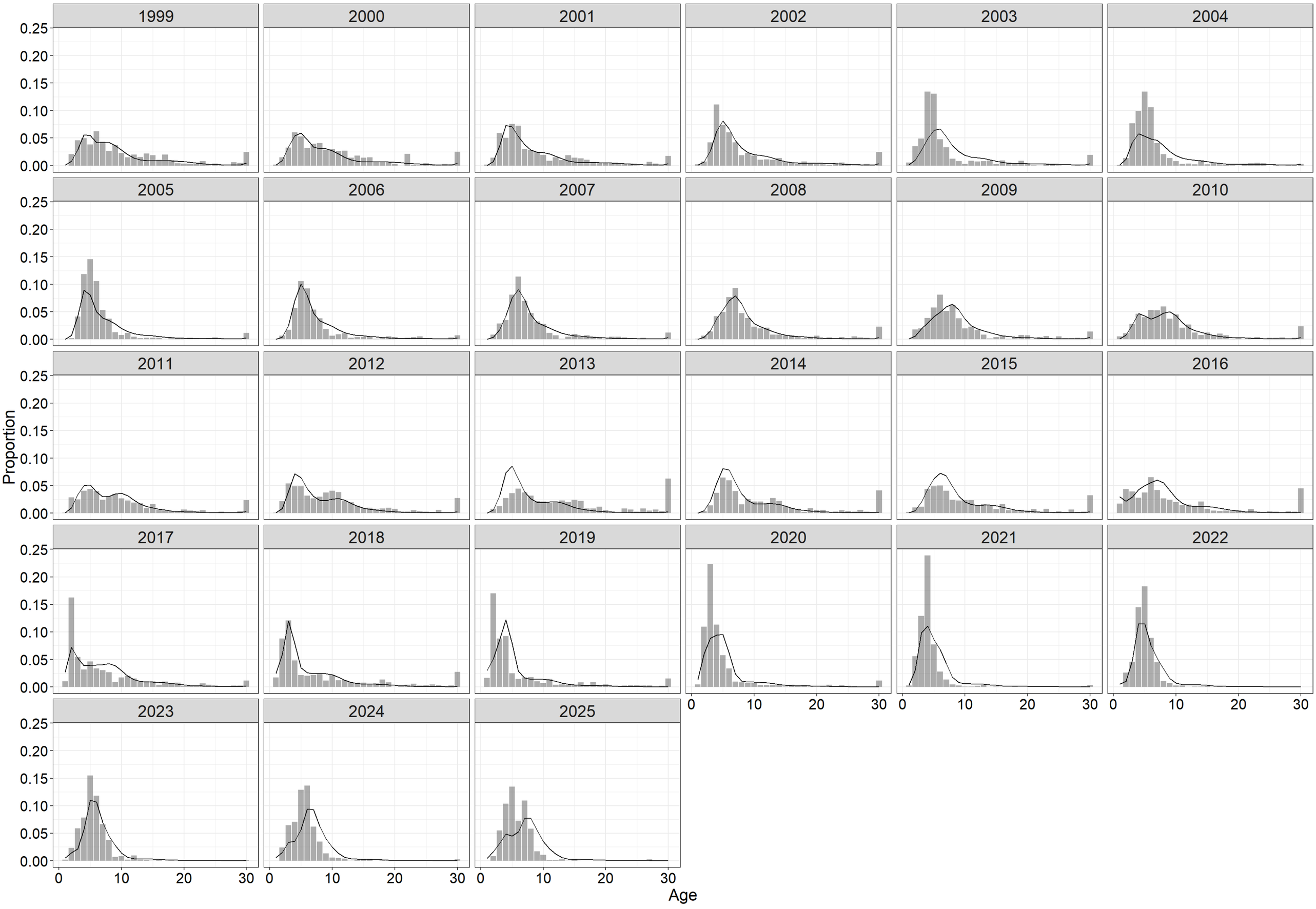

Model 26.2_FAA
Population Population 1
Female
Male
AK
Spawning Weight-at-Age (kg)
0.075
0.050
0.025
0.000
0
10
20
30
Age


Model 26.2_FAA
Population Population 1
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30


Model
26.2_FAA
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

Fixed Gear Fishery Age Comps Female AK



Fixed Gear Fishery Age Comps Male AK

1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

Proportion: 0.00, 0.05, 0.10, 0.15

Age: 0, 10, 20, 30

Fixed Gear Fishery Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=1.00
(0.97-1.03)

Sample quantiles

Theoretical quantiles

Female

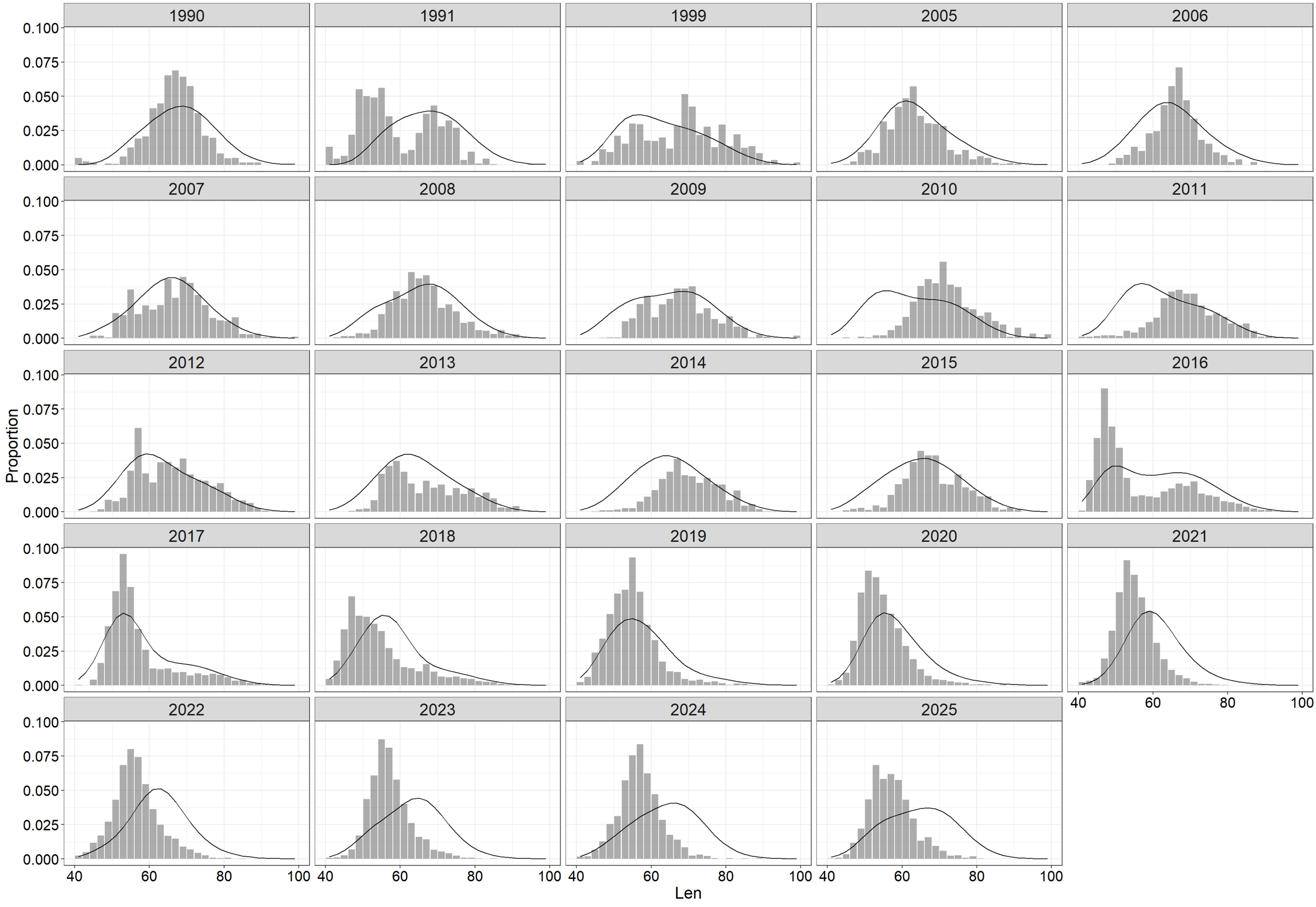
Male

AK

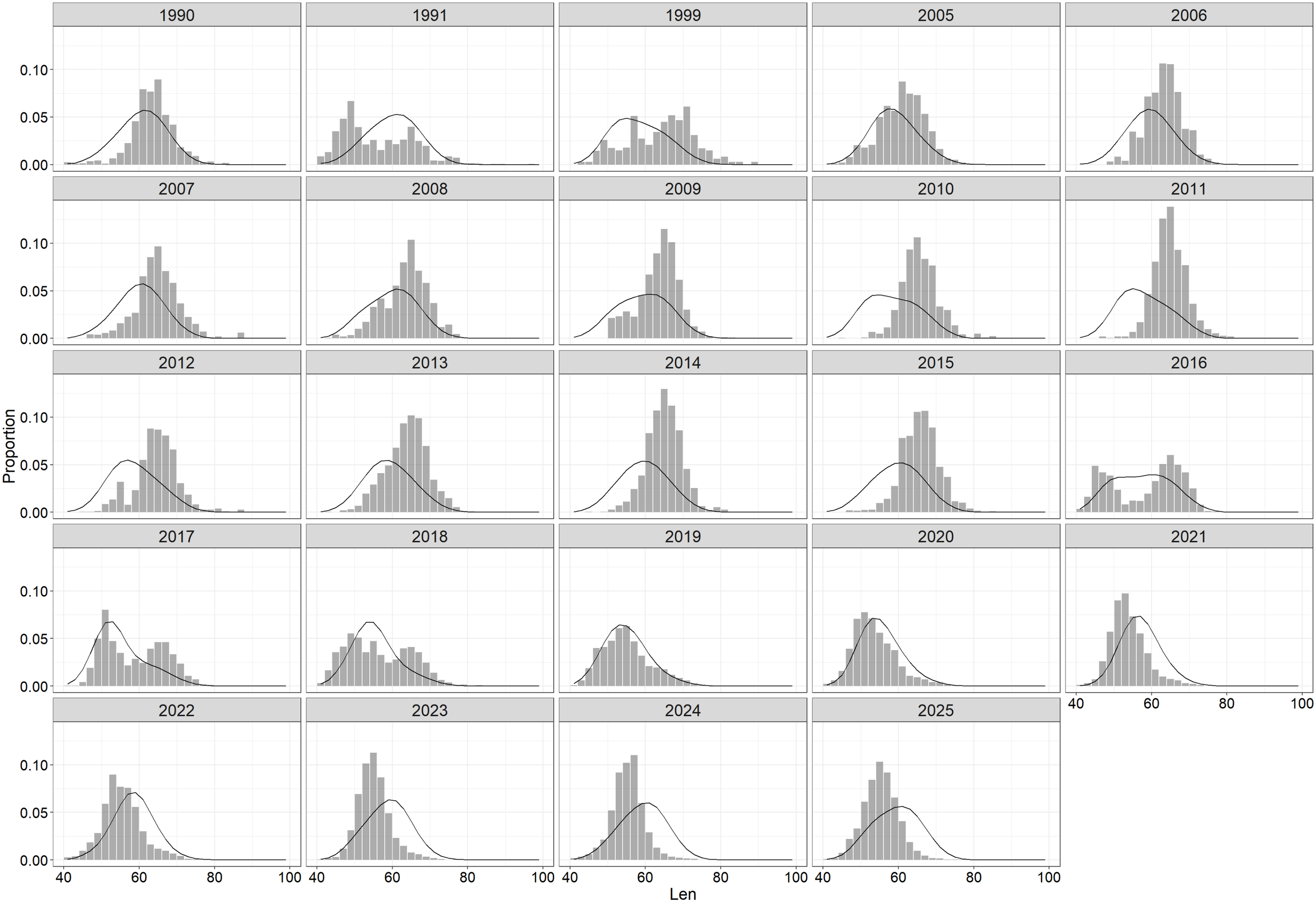
Proportions

Age

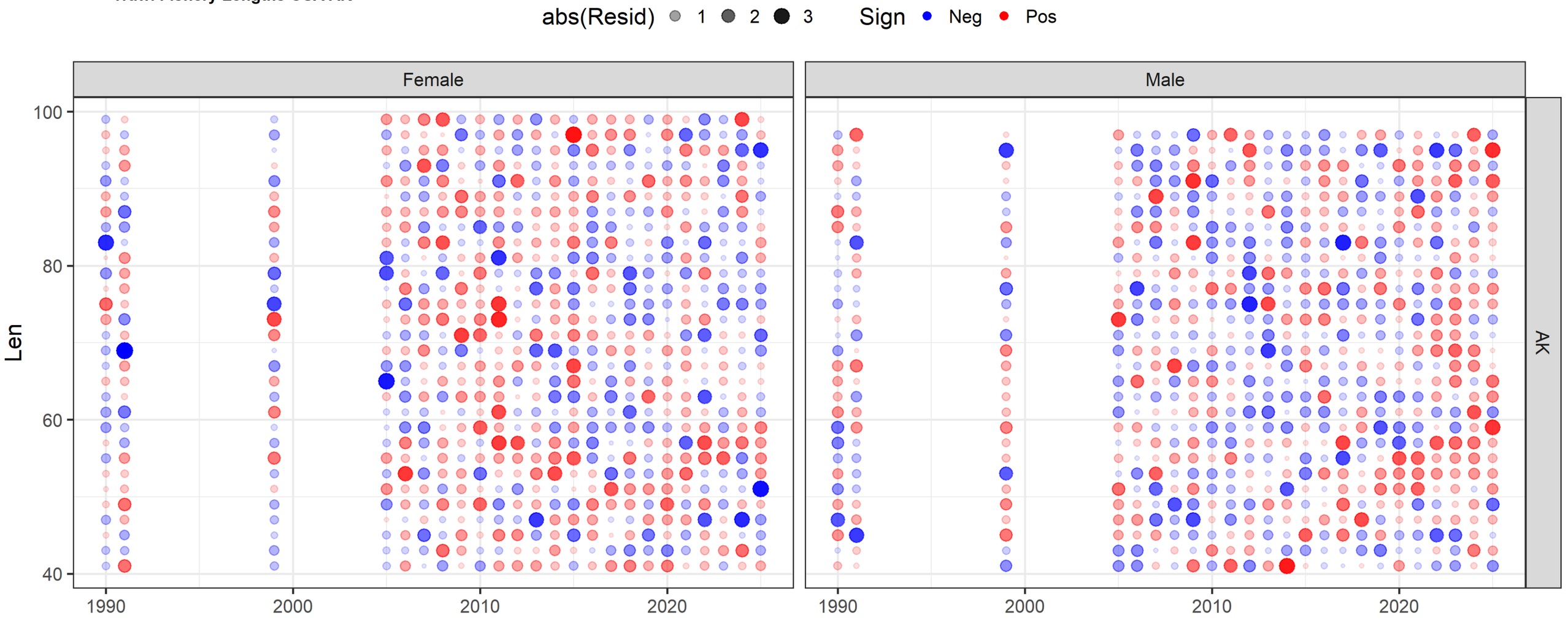
Trawl Fishery Lengths Female AK

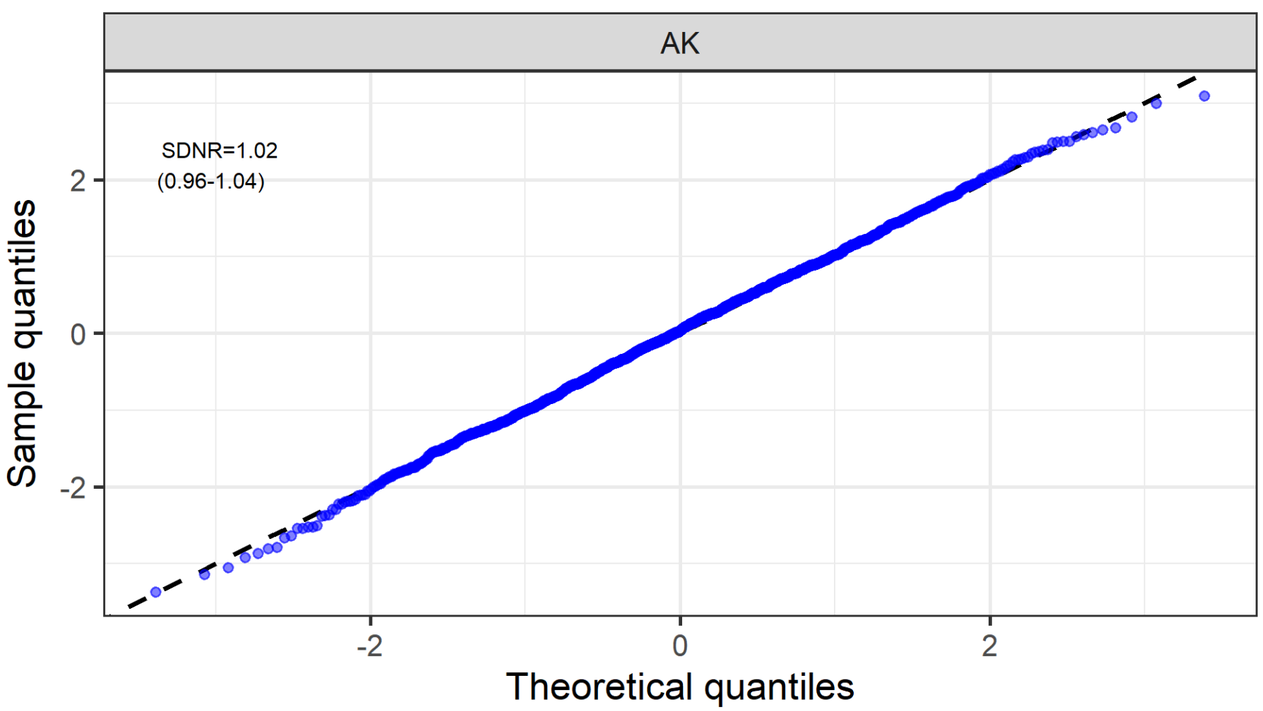


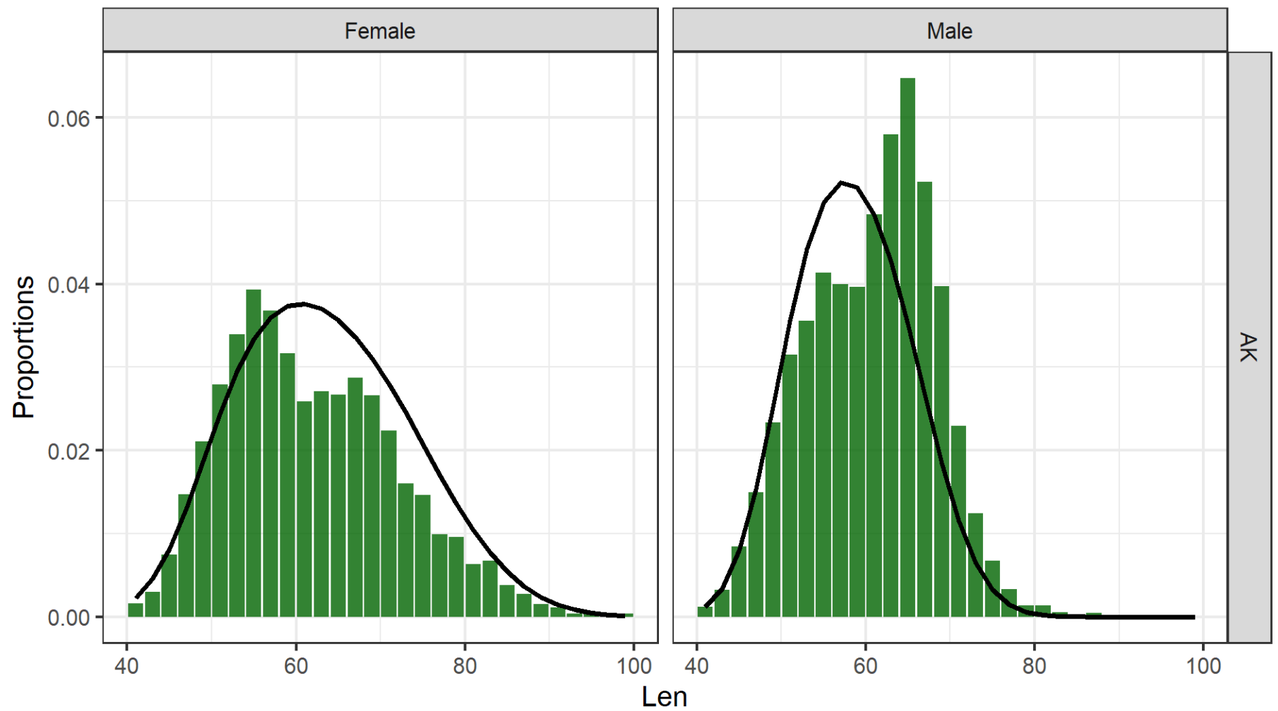
Trawl Fishery Lengths Male AK



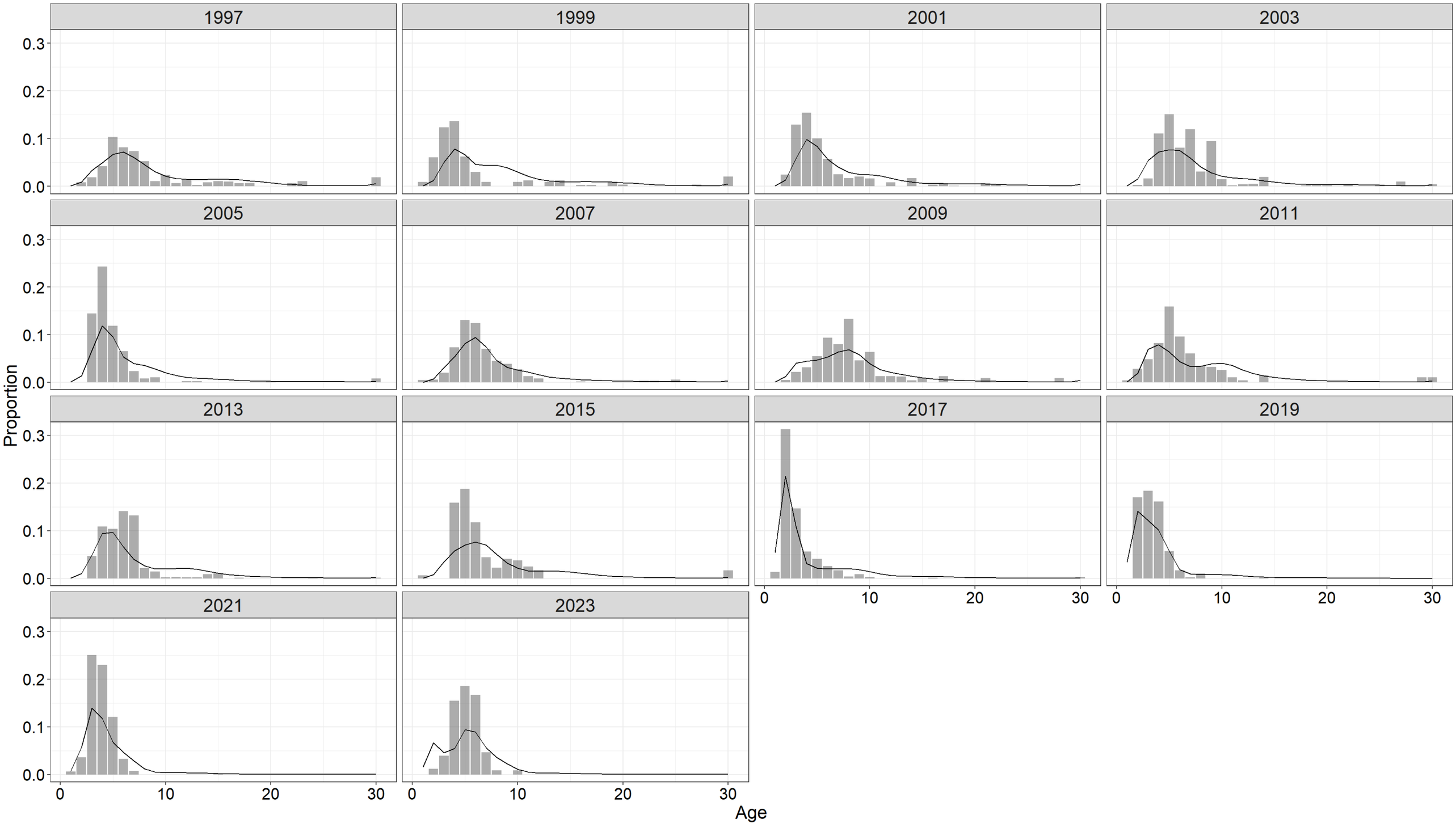
Trawl Fishery Lengths OSA AK







BS LL Survey Age Comps Female AK

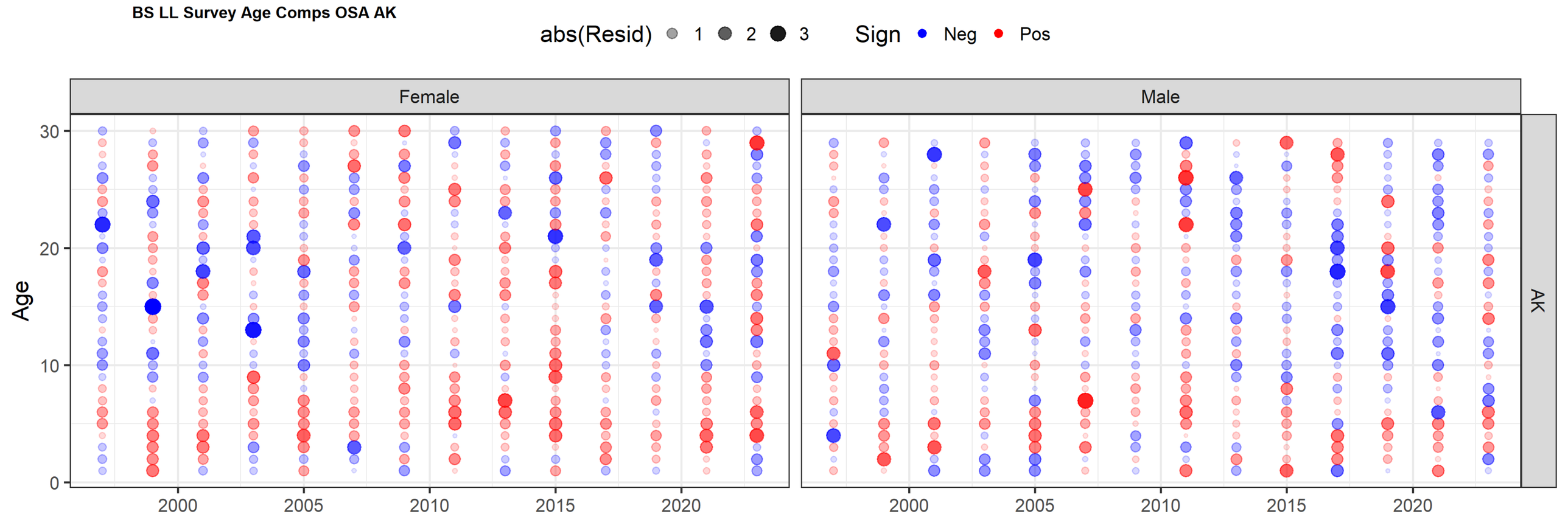


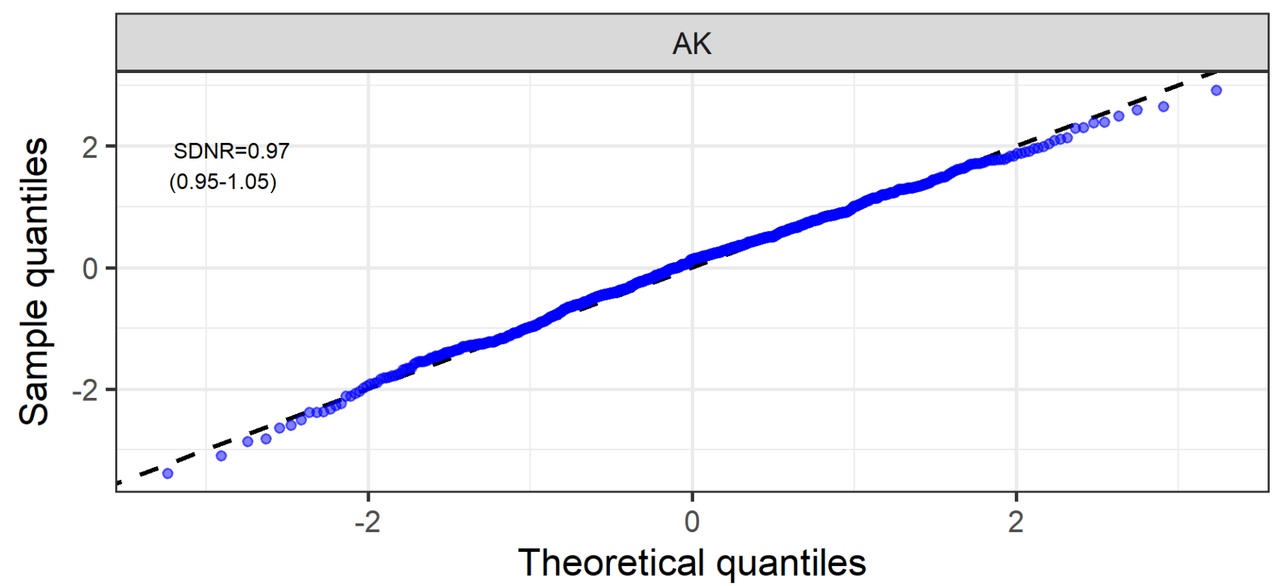
BS LL Survey Age Comps Male AK

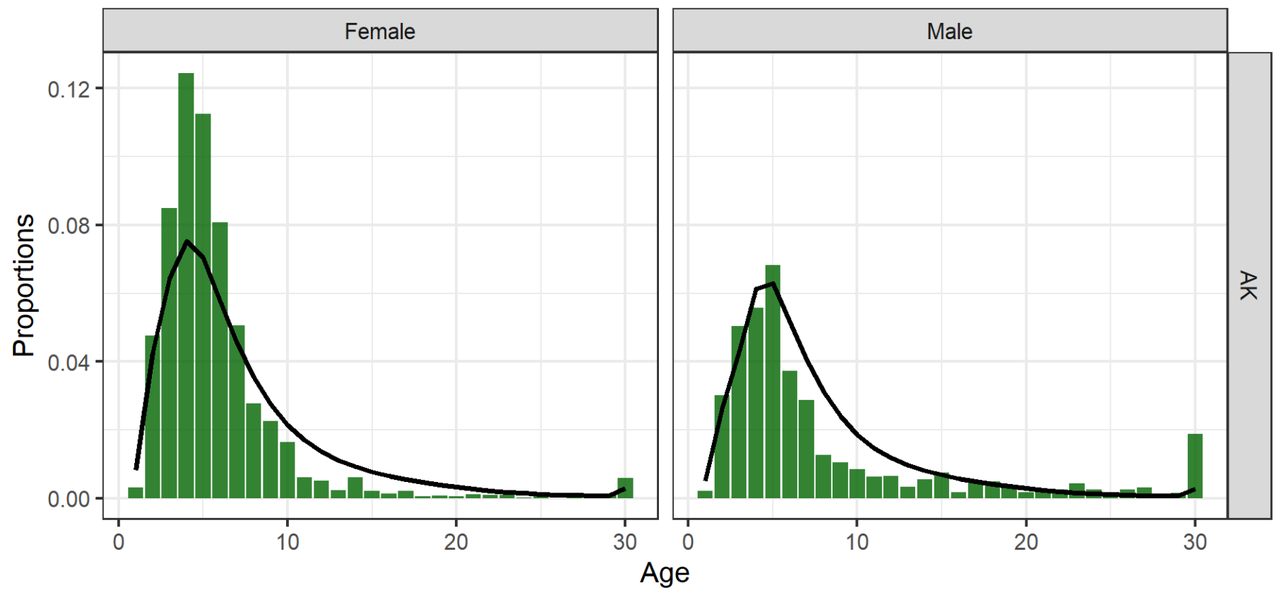
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

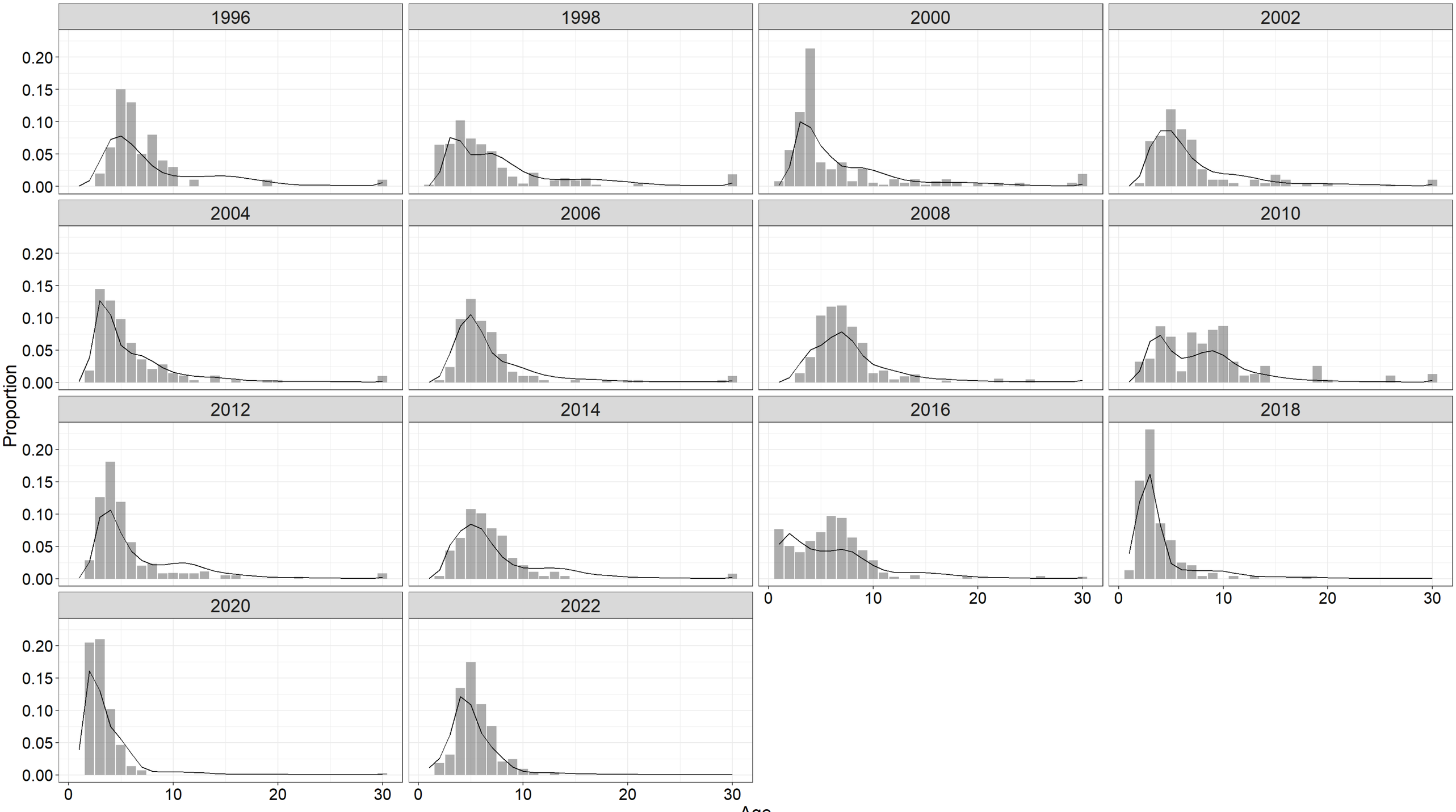
BS LL Survey Age Comps OSA AK







AI LL Survey Age Comps Female AK

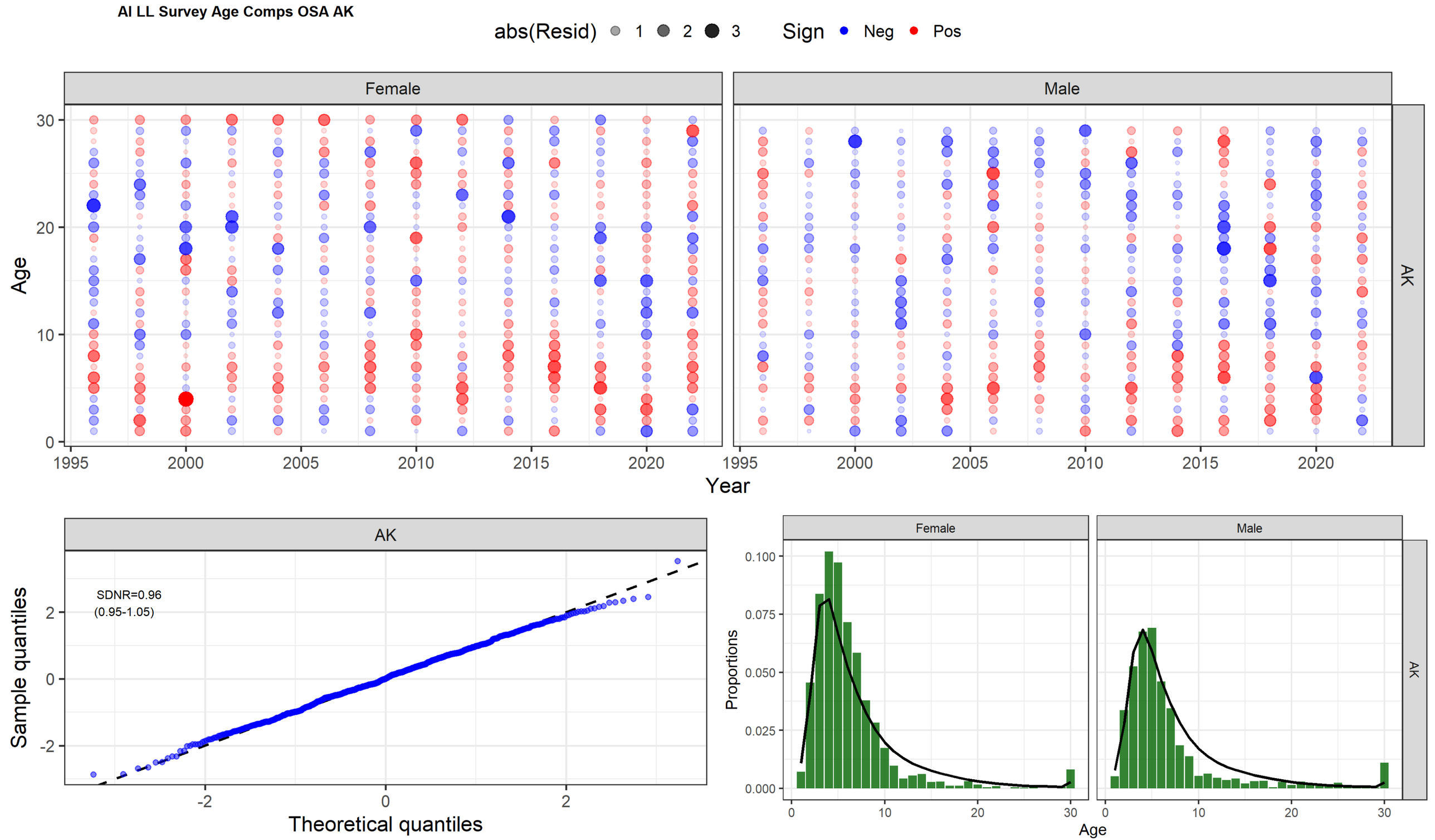


AI LL Survey Age Comps Male AK

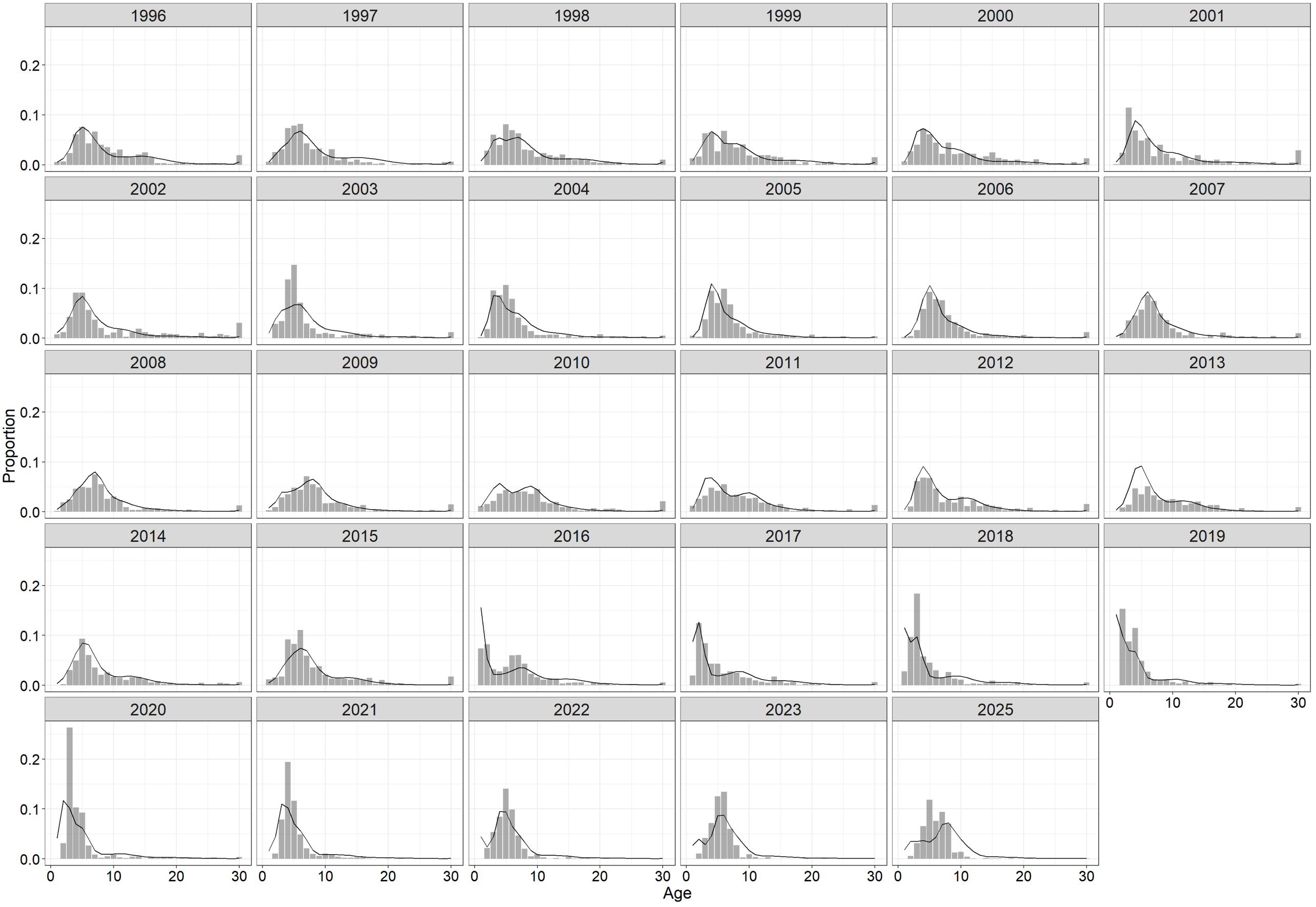
1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

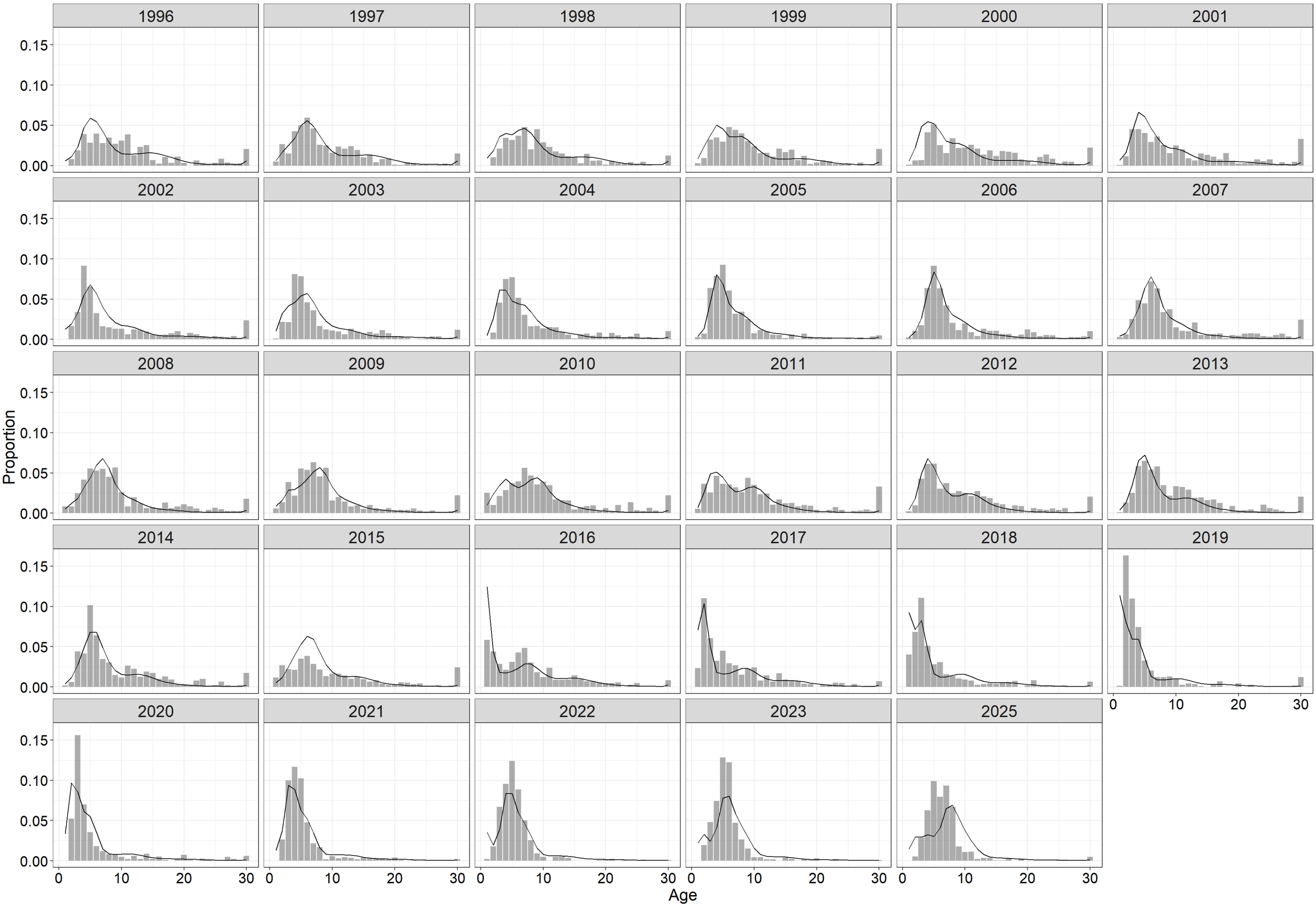
AI LL Survey Age Comps OSA AK



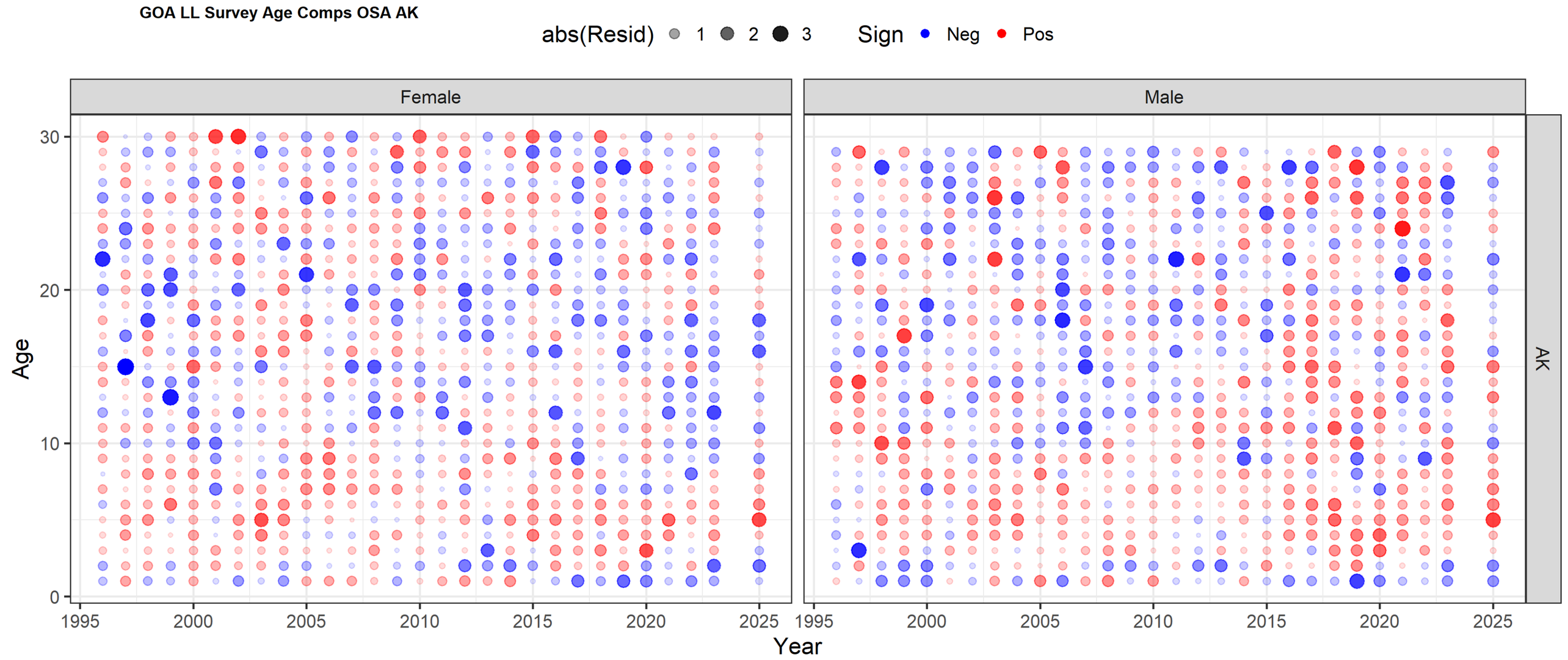
GOA LL Survey Age Comps Female AK

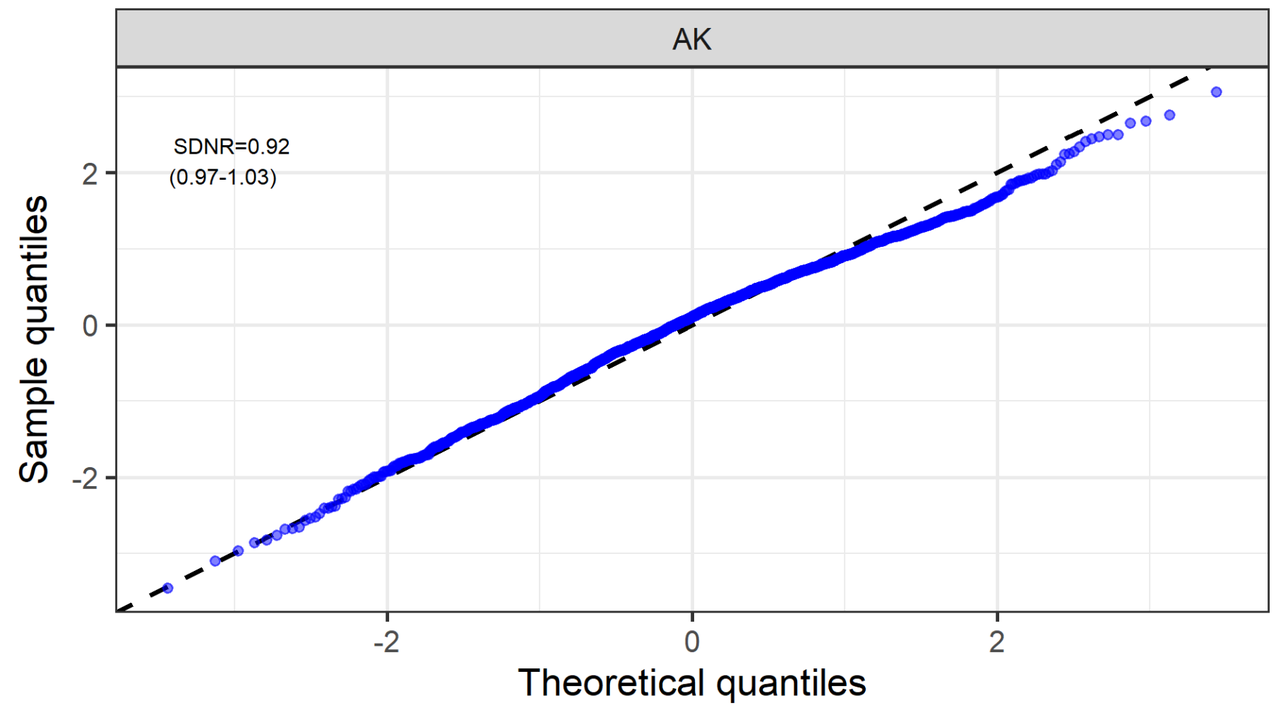


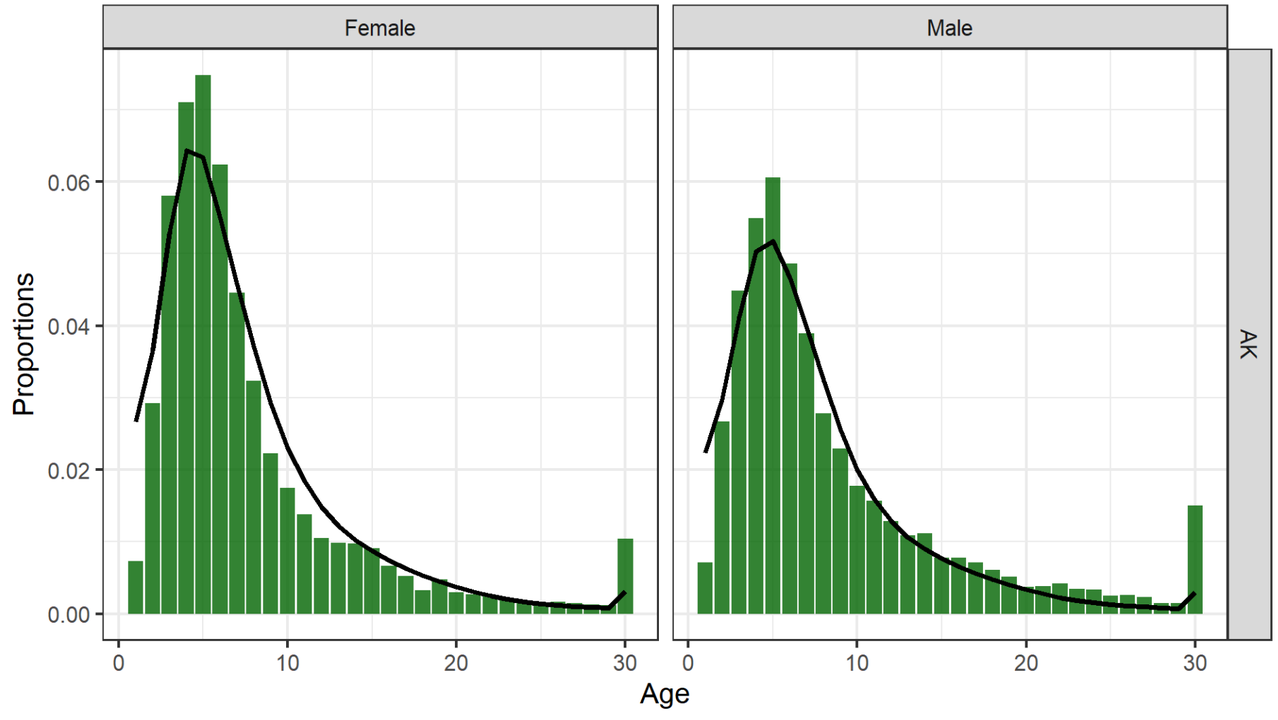
GOA LL Survey Age Comps Male AK

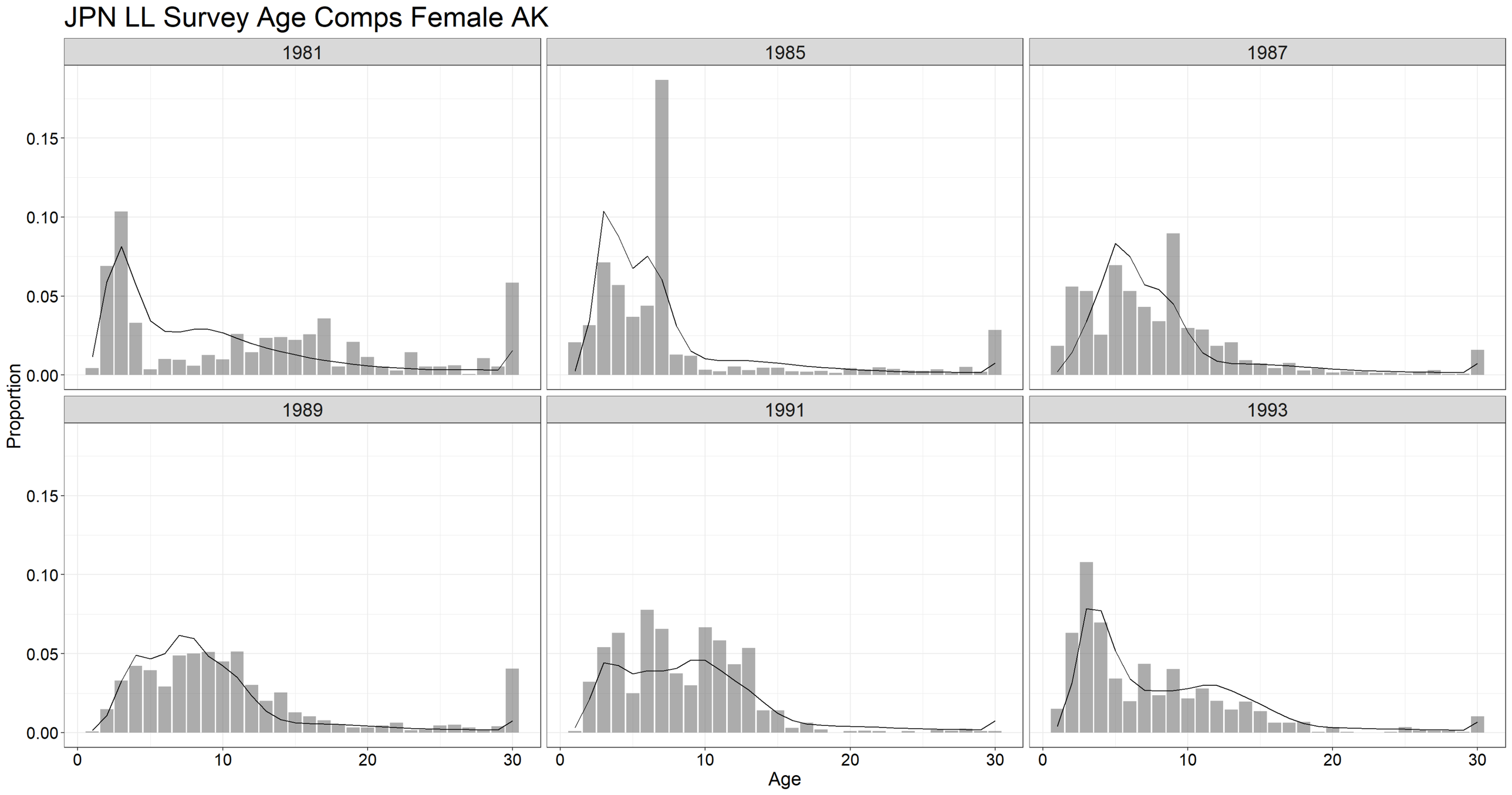


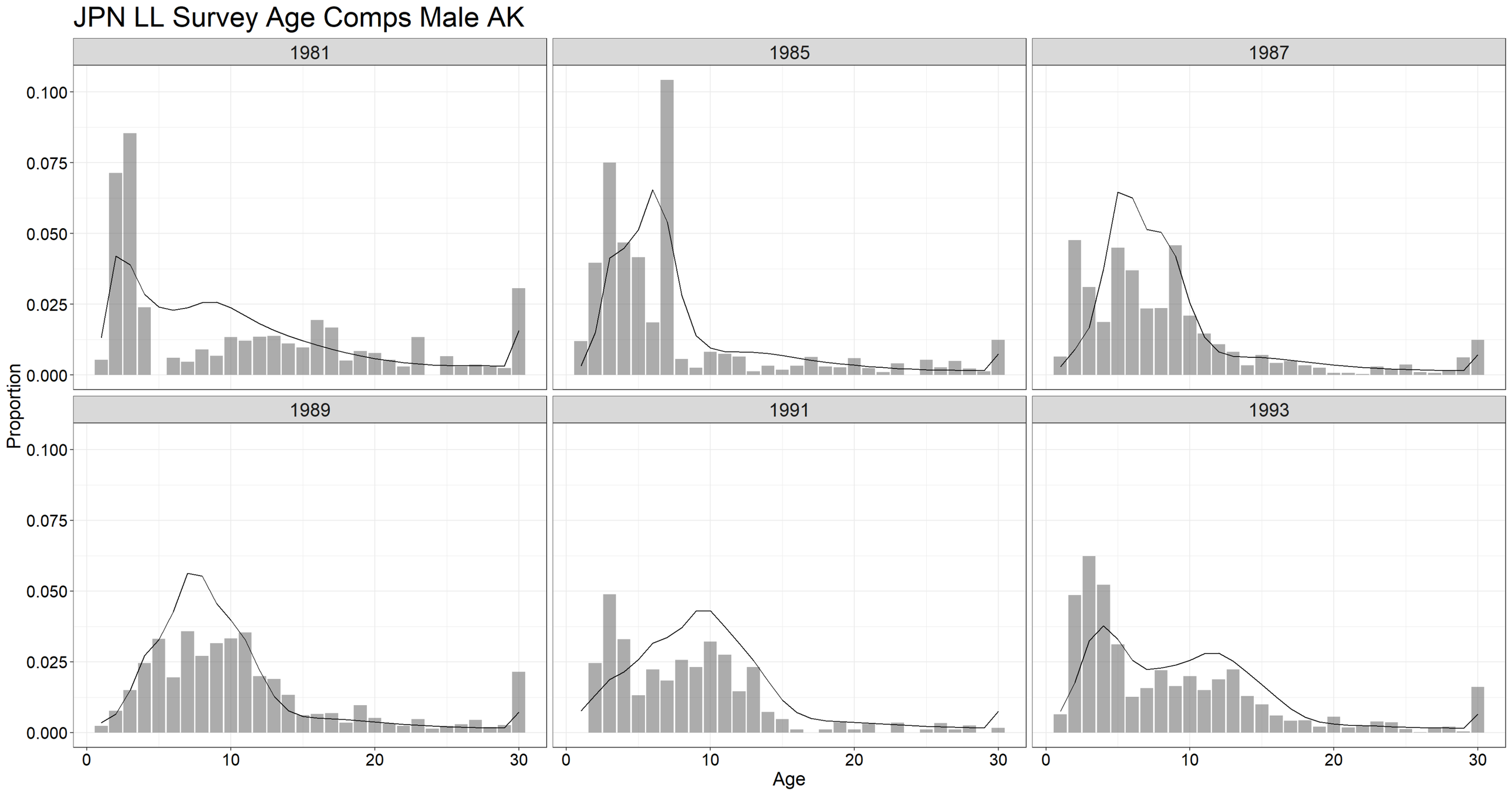
GOA LL Survey Age Comps OSA AK

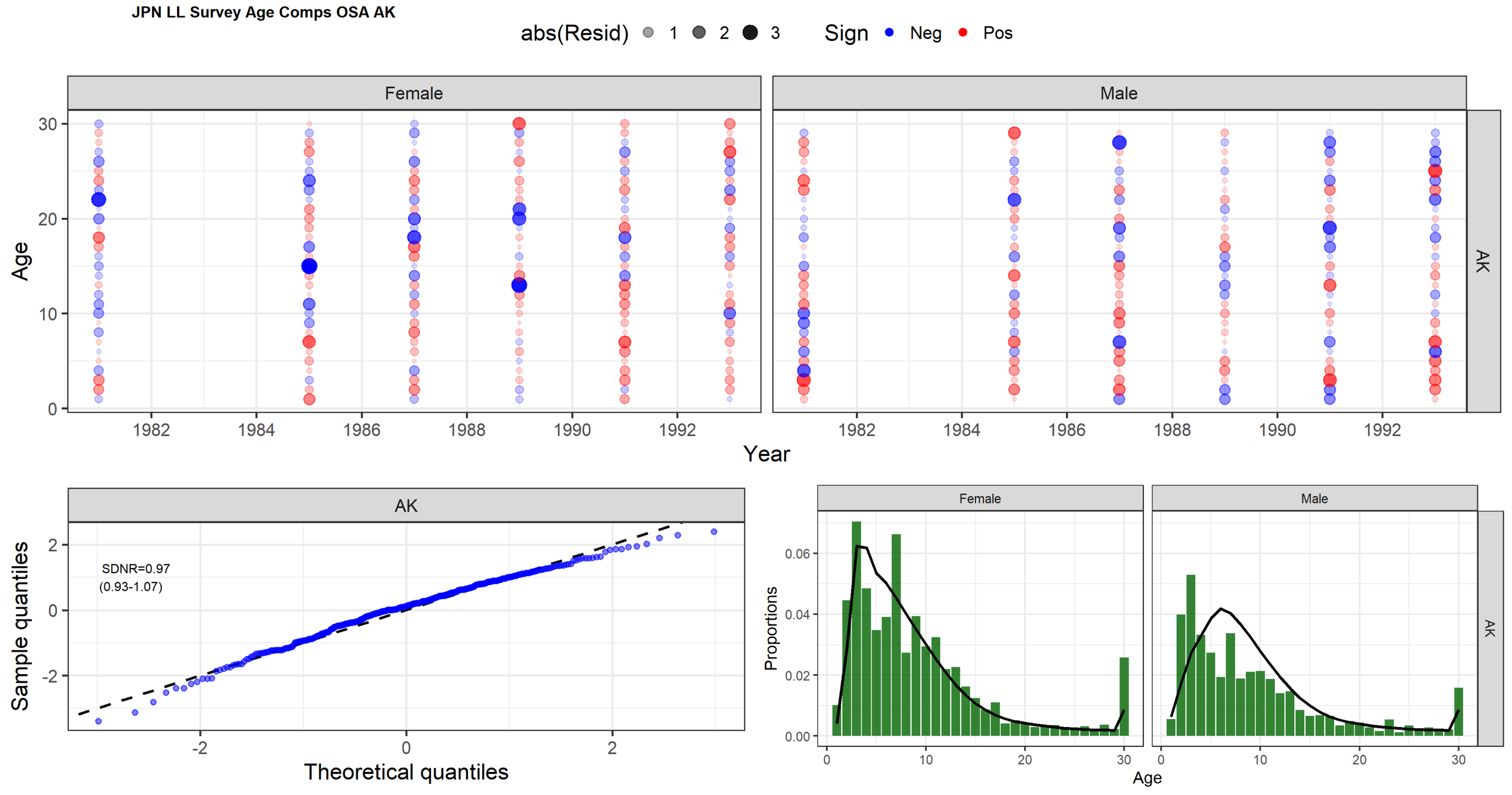







JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
Proportion
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=0.97
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

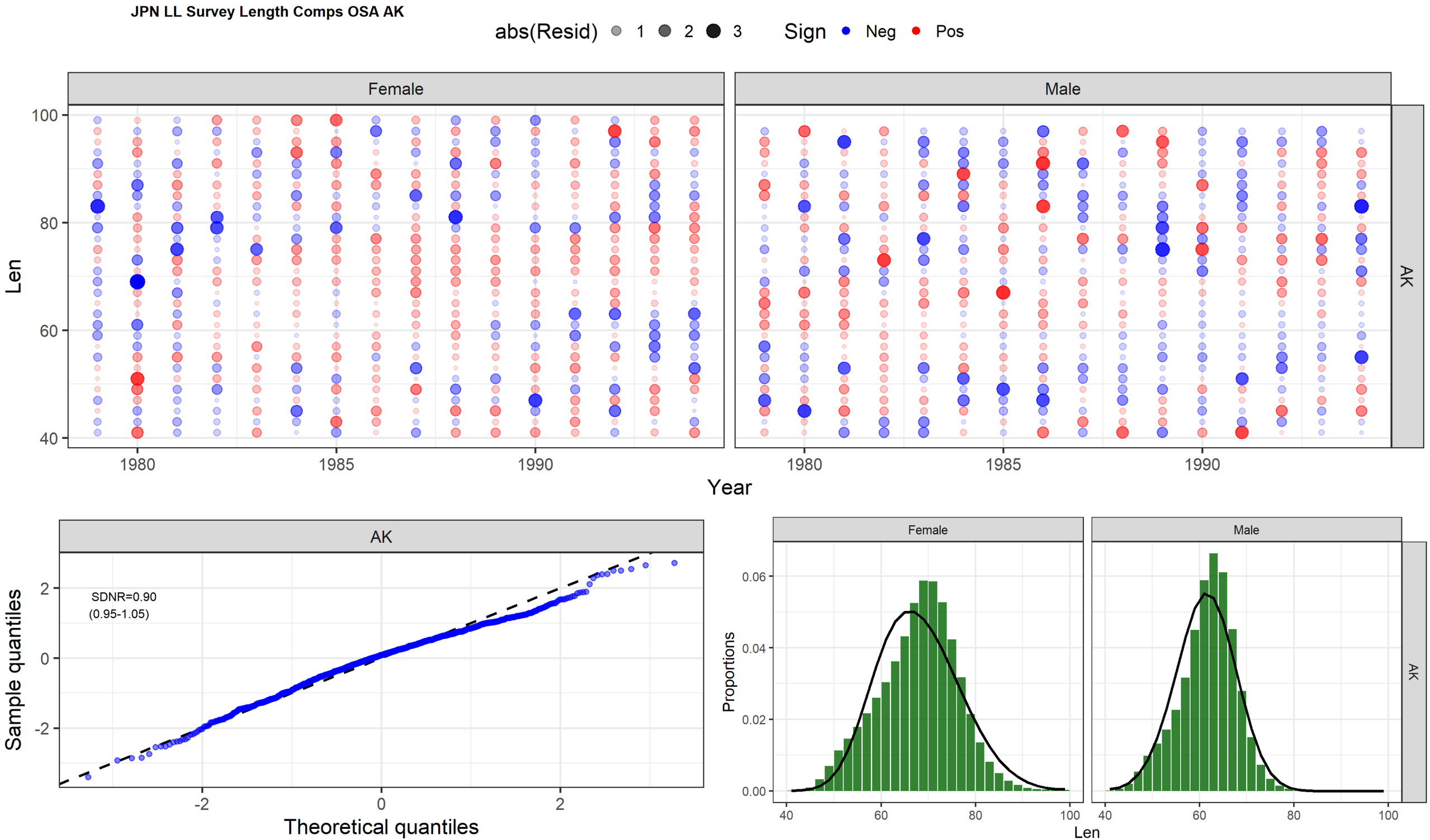
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

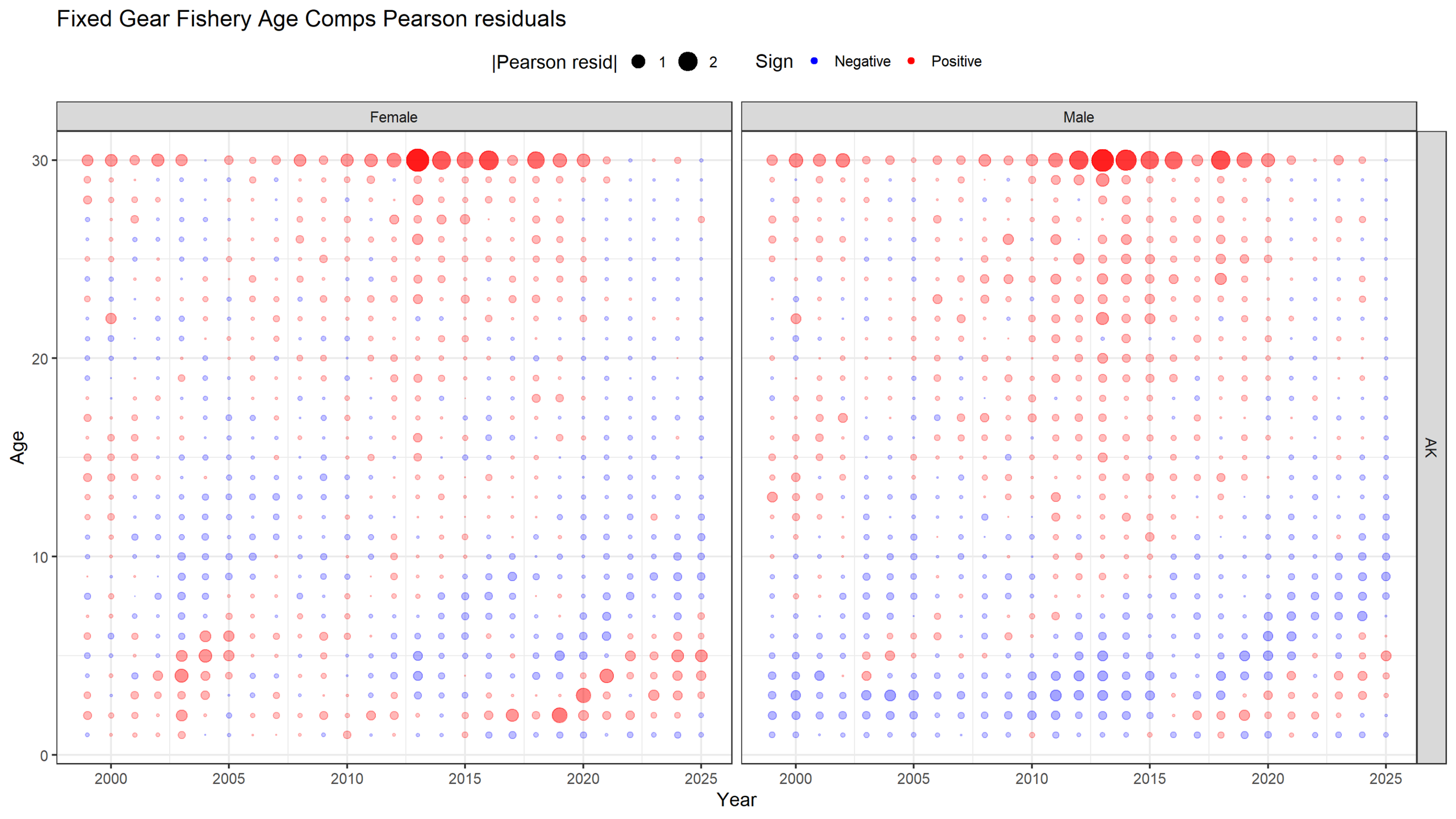
JPN LL Survey Length Comps Male AK

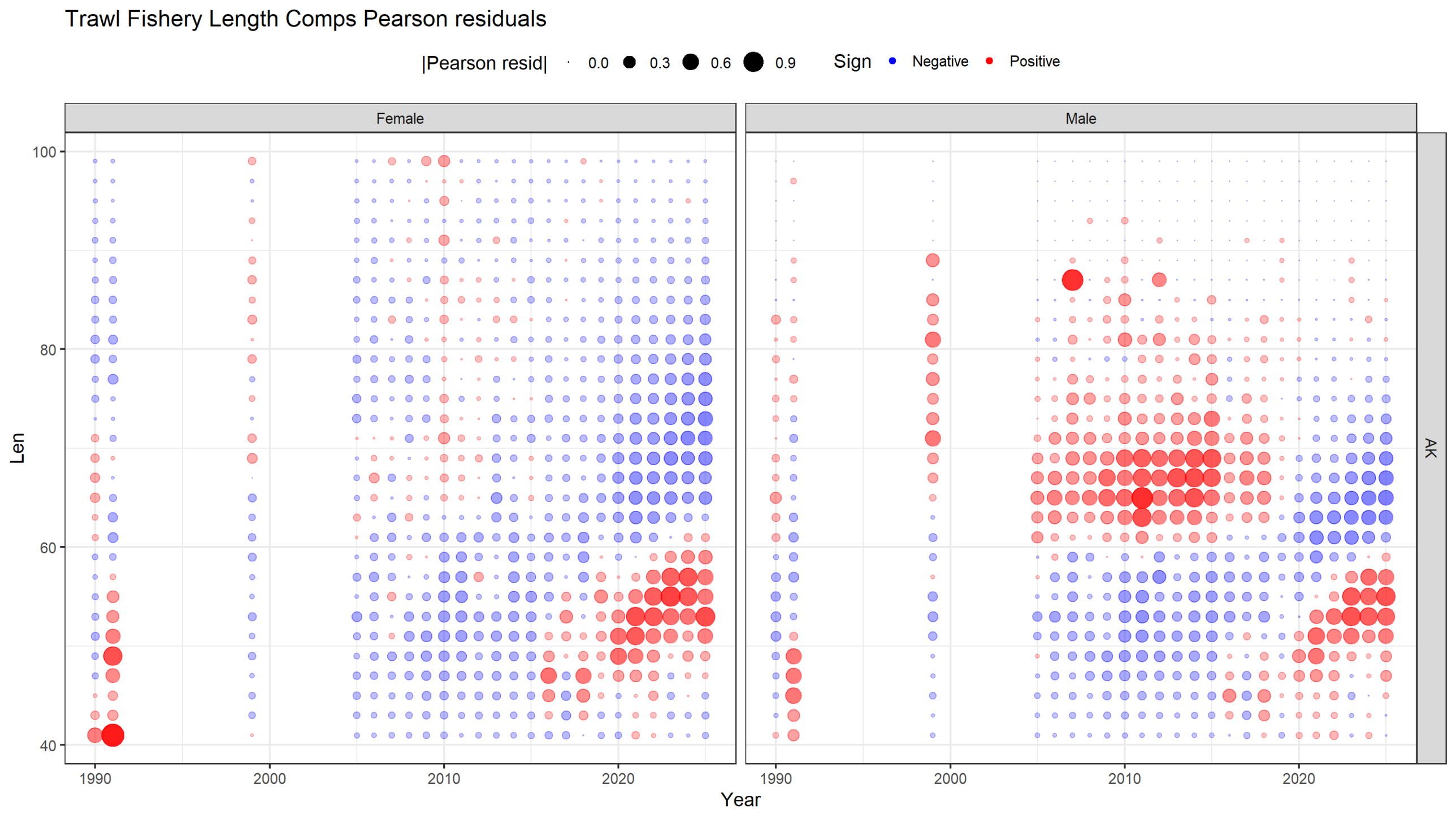
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

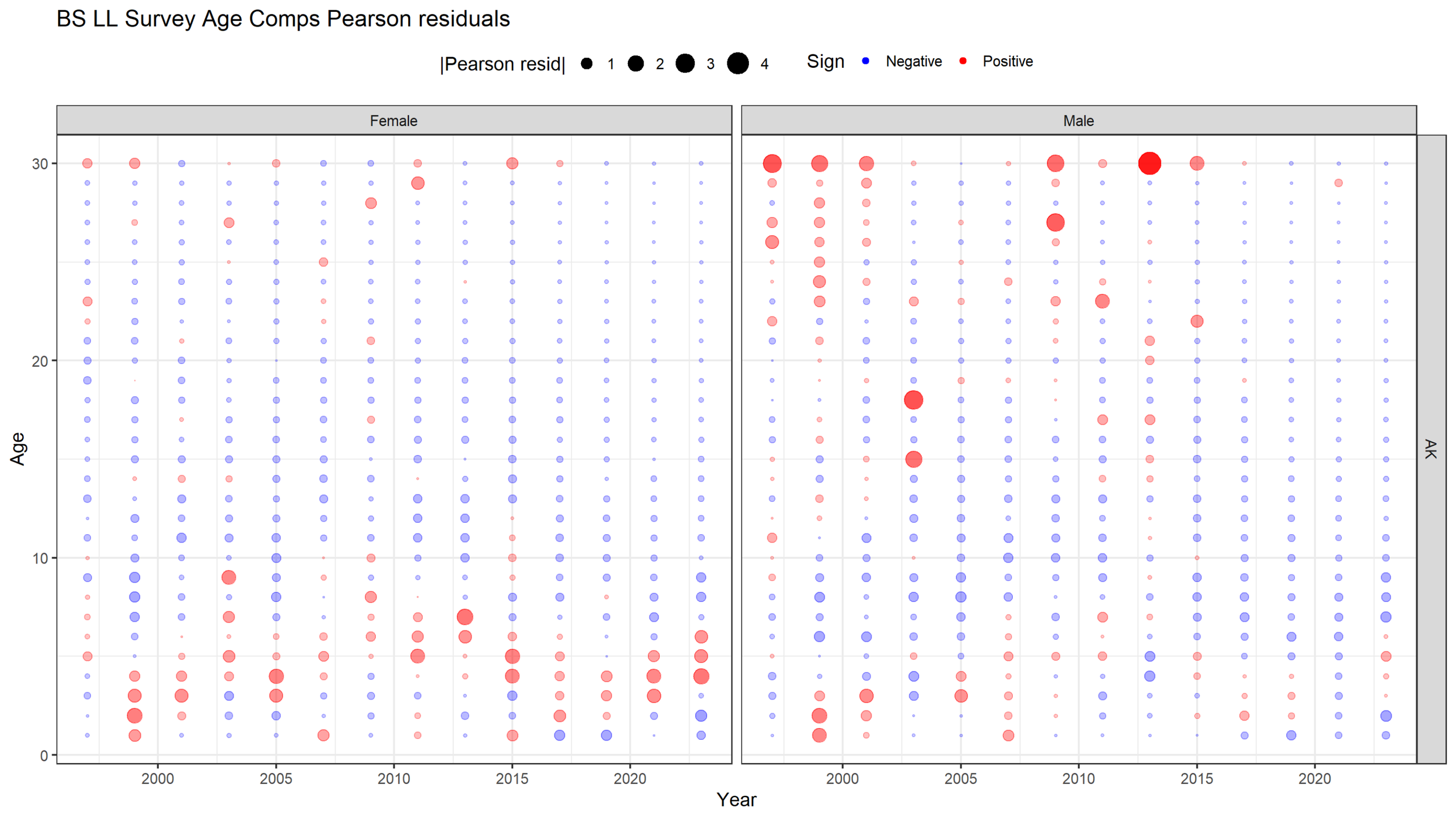
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

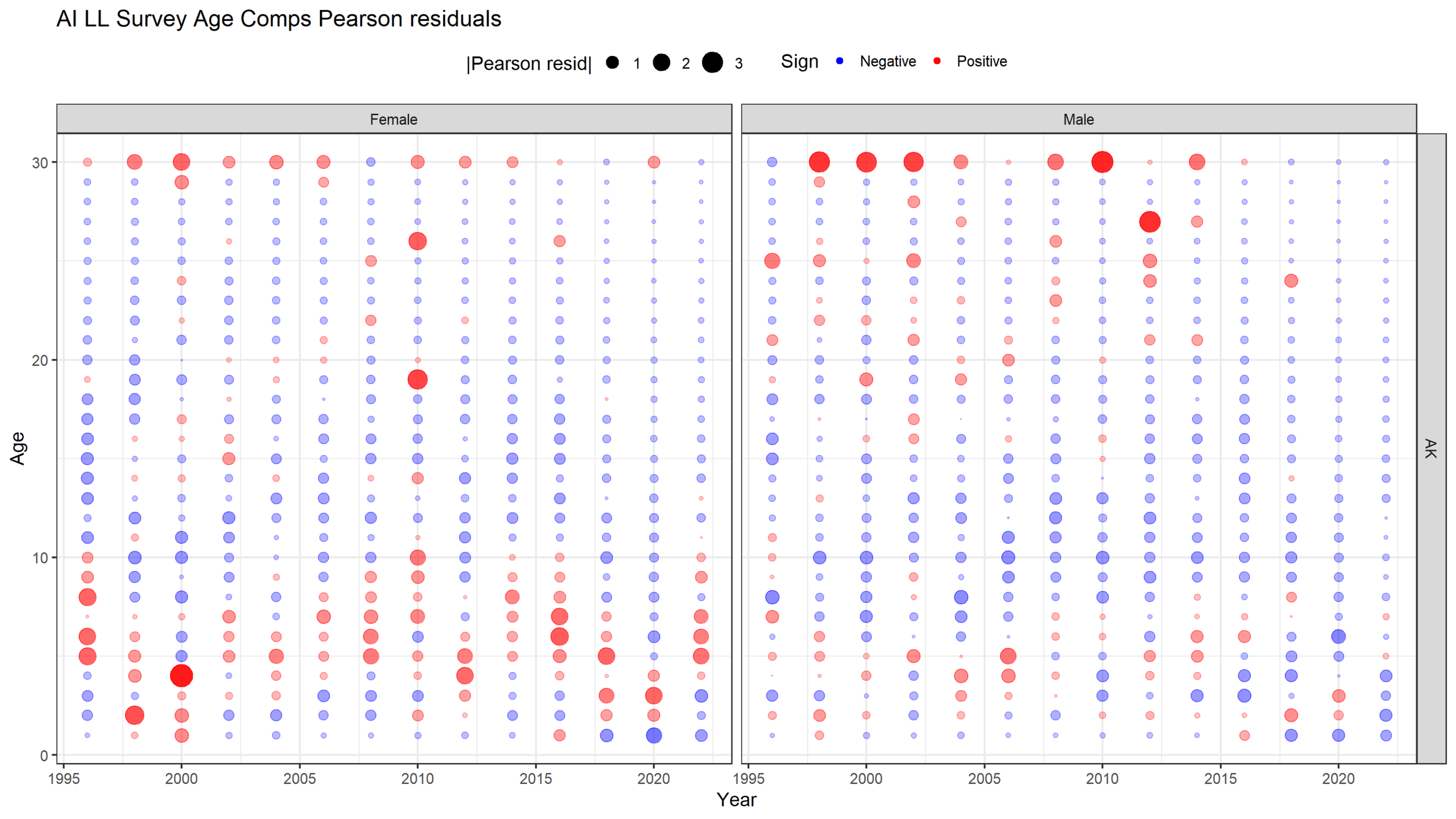
JPN LL Survey Length Comps OSA AK

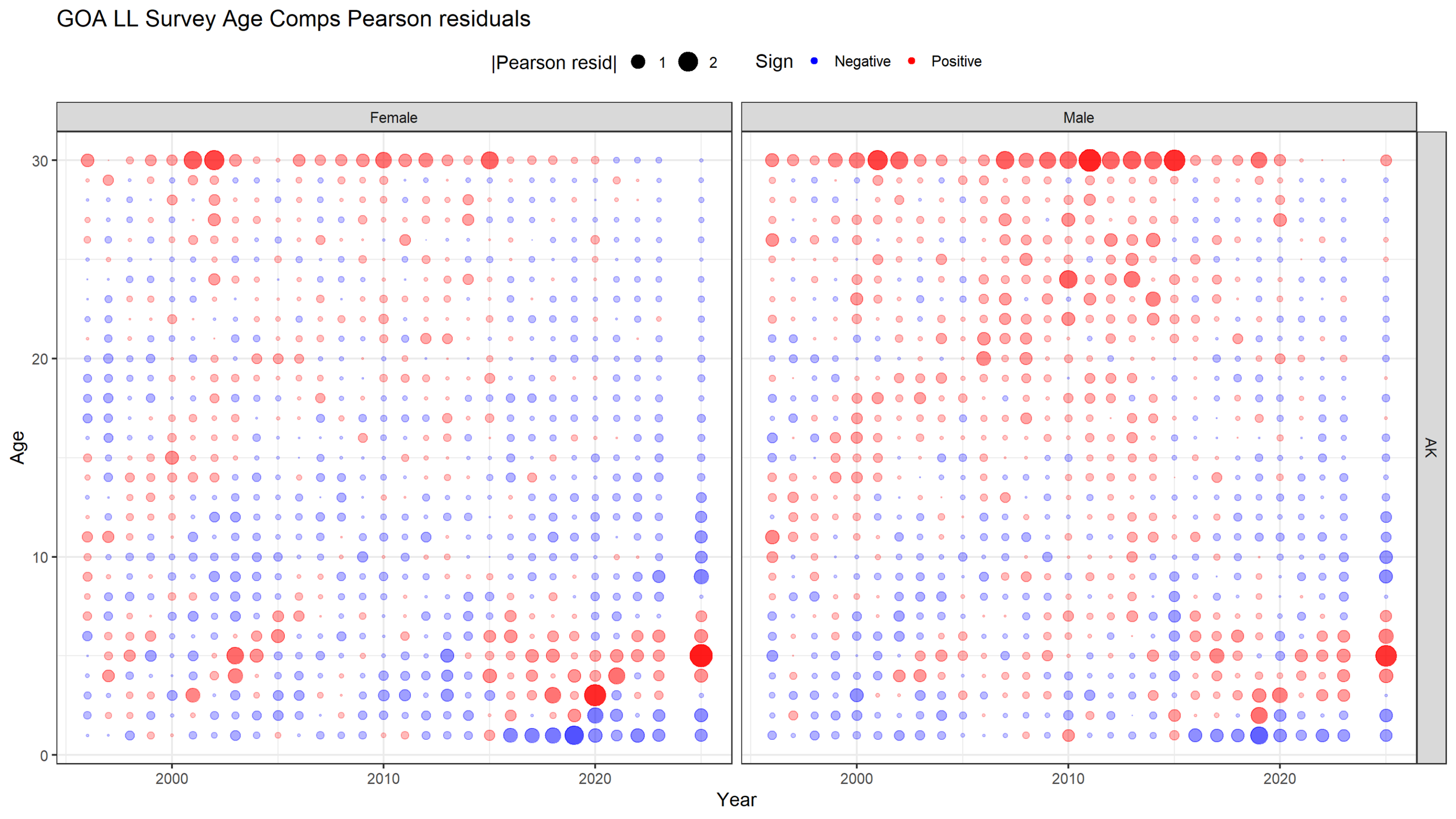


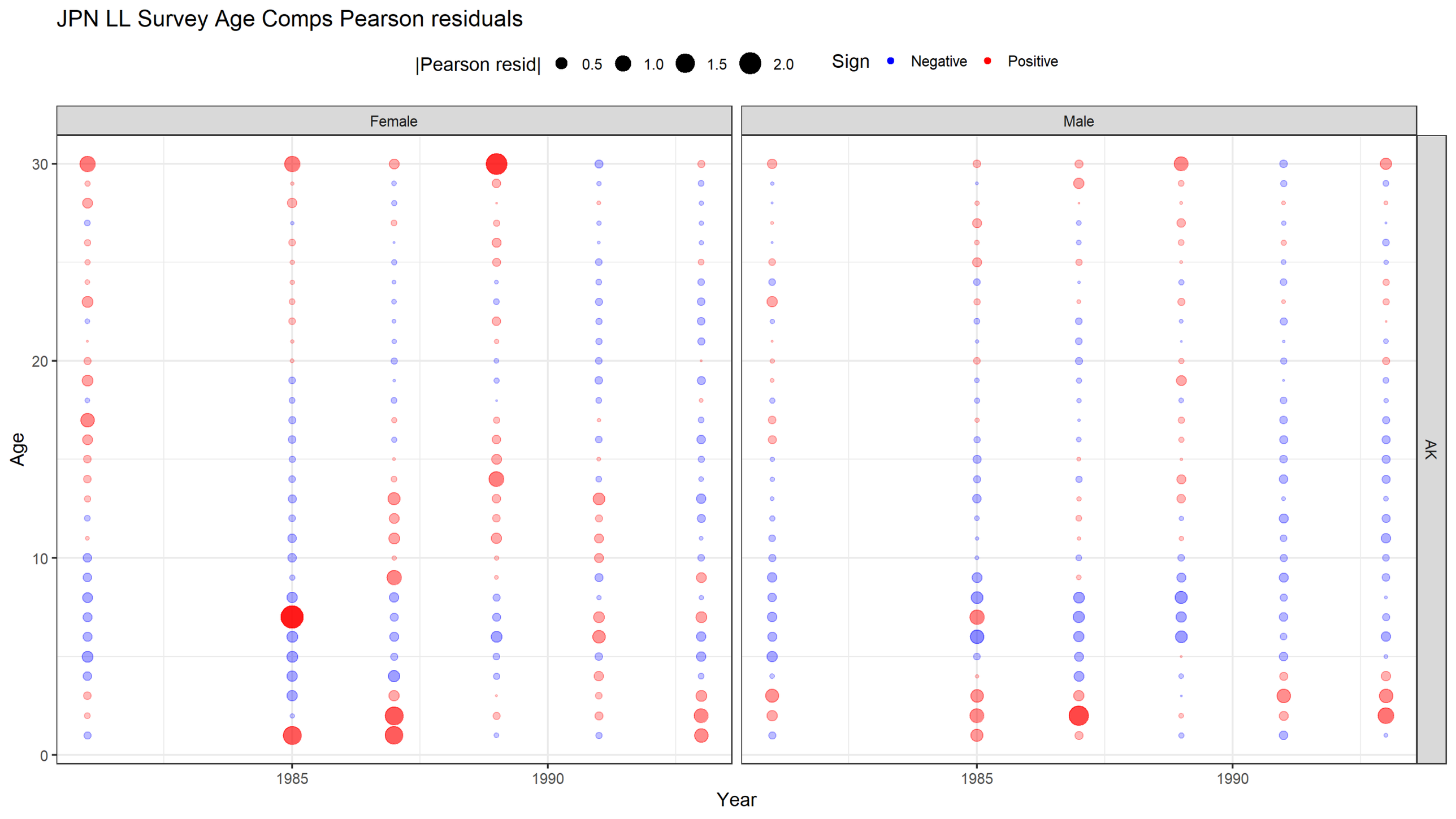

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

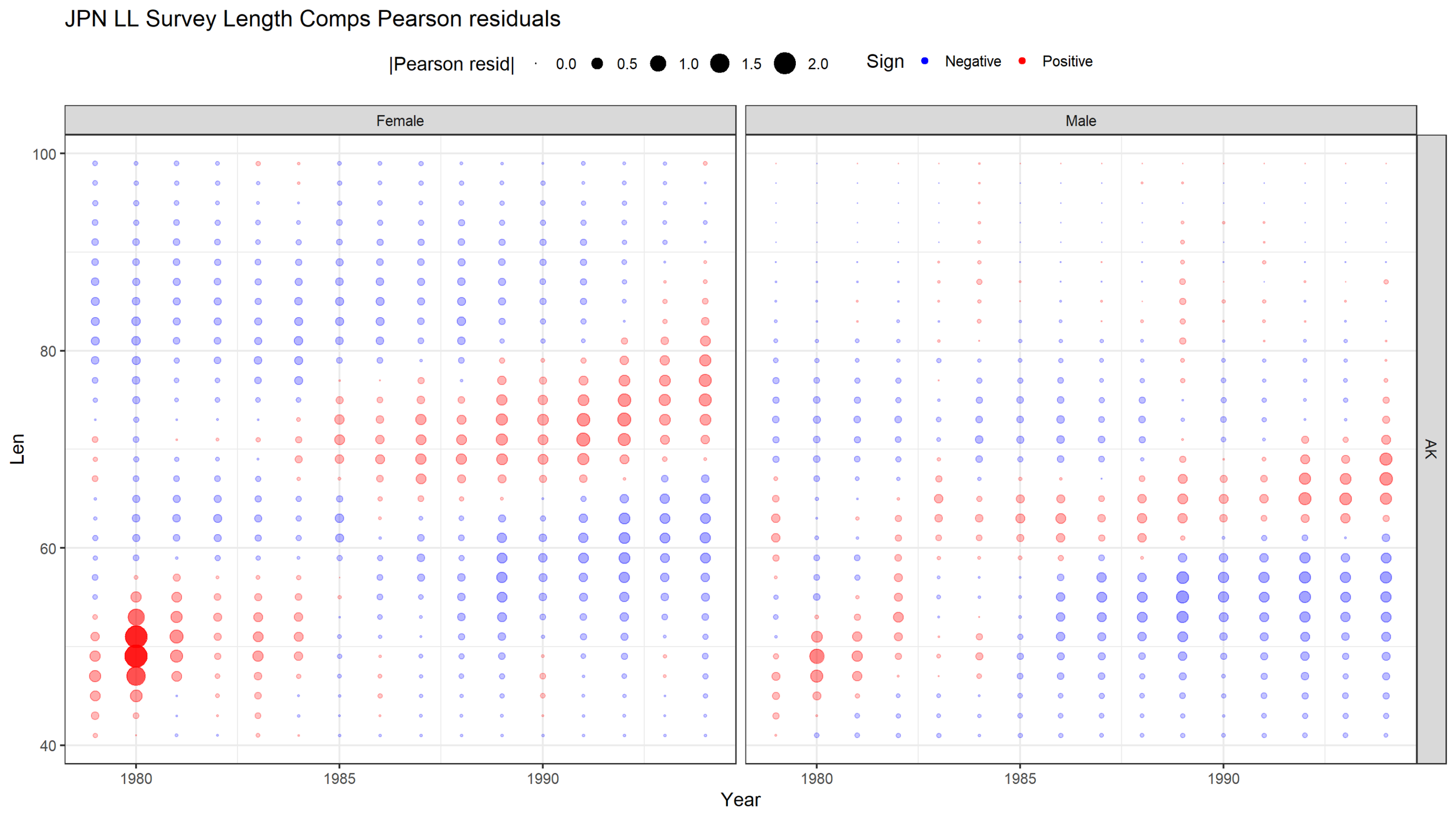

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.3
0.6
0.9
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year


BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

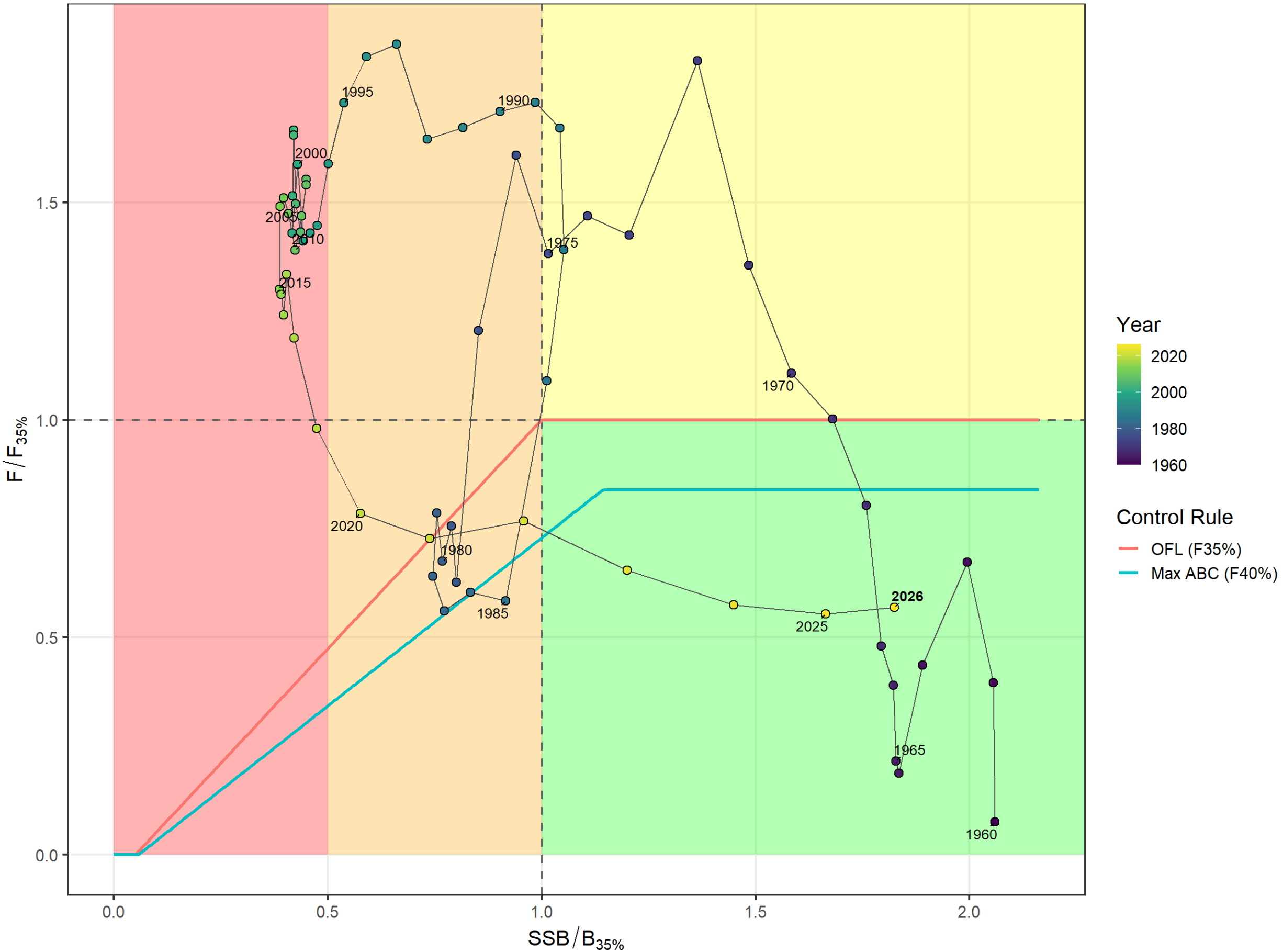

AI LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

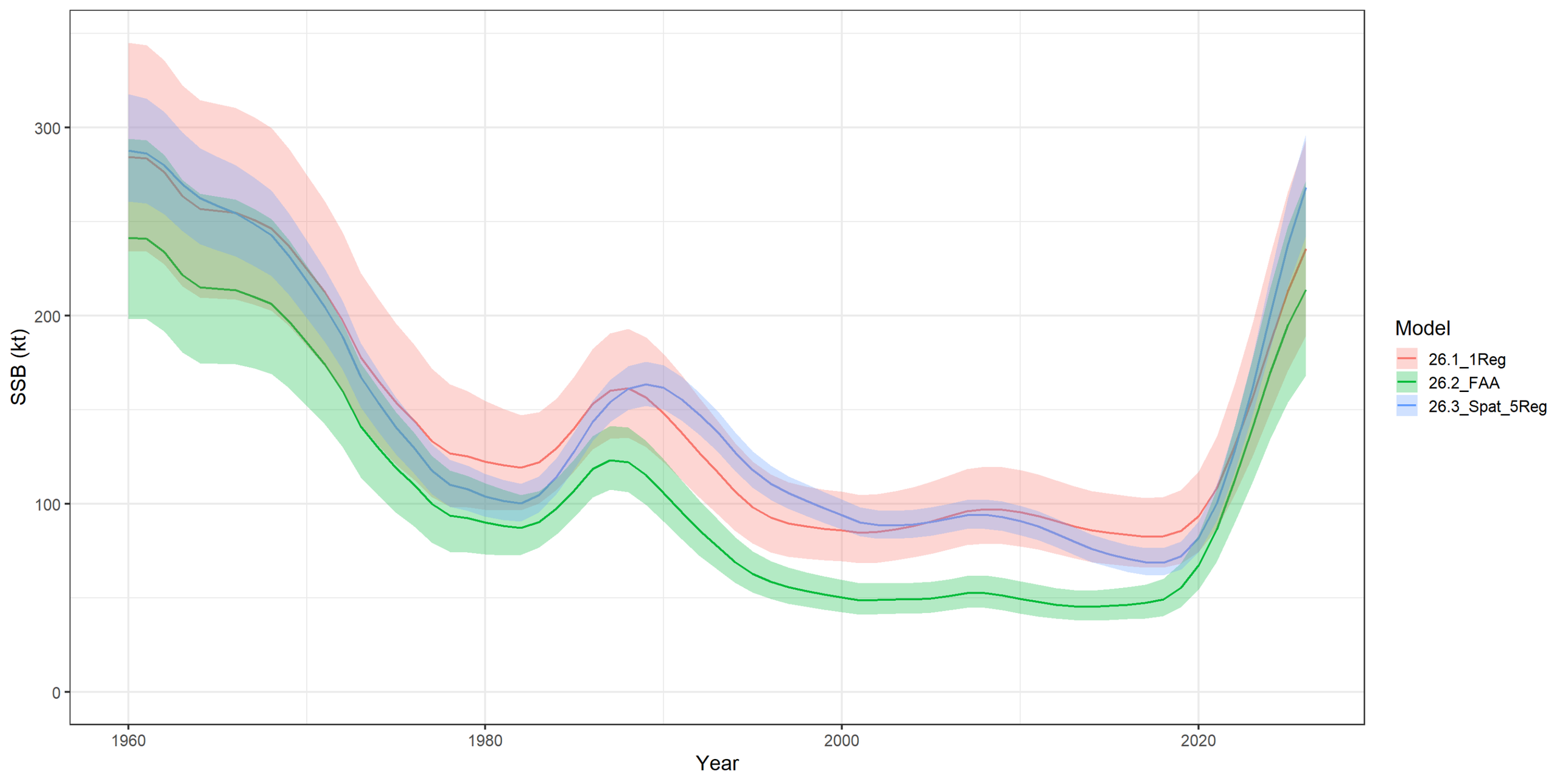

GOA LL Survey Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020


JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year


JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

Phase Plane


1995
1990
2000
2005
2010
1975
2015
1970
2020
1980
1985
2025
2026
1965
1960
1.5
1.0
0.5
0.0
F / F35%
0.0
0.5
1.0
1.5
2.0
SSB / B35%
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)


SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year
Model
26.1_1Reg
26.2_FAA
26.3_Spat_5Reg

Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.62	34.07	20.96
AI LL Survey Ages	NA	2.01	35.57	71.37
GOA LL Survey Ages	NA	0.36	67.17	24.50
JPN LL Survey Ages	NA	0.20	83.33	16.30
JPN LL Survey Lengths	NA	0.32	54.62	17.35
Fixed Gear Fishery Ages	NA	0.04	142.85	5.86
Trawl Fishery Lengths	NA	0.09	47.62	4.16

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2_FAA	1	213.866	390.7794	0.05279969	37.569	133.9653	0.07785	1.59643	0.54728	0.67818

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.088	0.088
M Male (estimated)	0.114	0.114	0.088	0.088
Average Recruitment (millions)	25.9	25.9	19.2	19.2
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	588629	586855
Projected Female Spawning Biomass (t)	219,714	241,217	226595	237749
B100% (t)	302,672	302,672	334913	334913
B40% (t)	121,069	121,069	133965	133965
B35% (t)	105,935	105,935	117220	117220
FOFL	0.102	0.102	0.093	0.093
max FABC	0.087	0.087	0.078	0.078
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	44464	44696
OFLw (t)	58,532	57,797	44333	44565
max ABC (t)	50,283	49,651	37567	37767
ABC (t)	50,283	49,651	37567	37767
ABCw (t)	50,111	49,482	37436	37636
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)
SSB (kt)
0
100
200
300
400
1960
1980
2000
2020
2040
Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

250

500

750

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

30

60

90

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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Recruitment (millions)

0

30

60

90

2010

2020

2030

2040

Year

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S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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Fishing Mortality

0.15

0.10

0.05

0.00

2010

2020

2030

2040

Year

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S1a: Author's F (FABC Ratio) Year 1, FABC After

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S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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S5: No Fishing

S6: Overfished (FOFL)

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S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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S7: Approaching Overfishing (FABC -> FOFL)

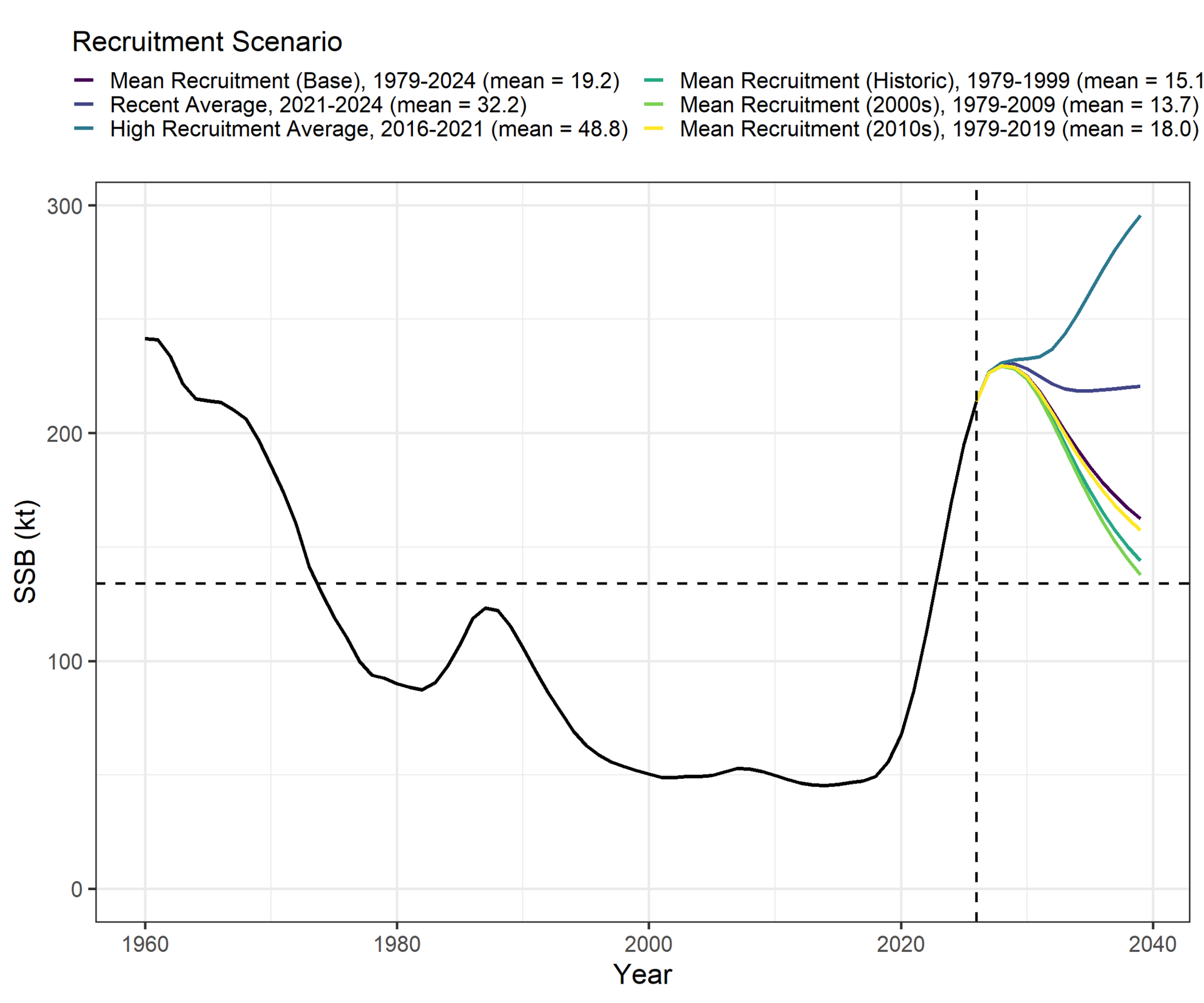
Yield (kt)

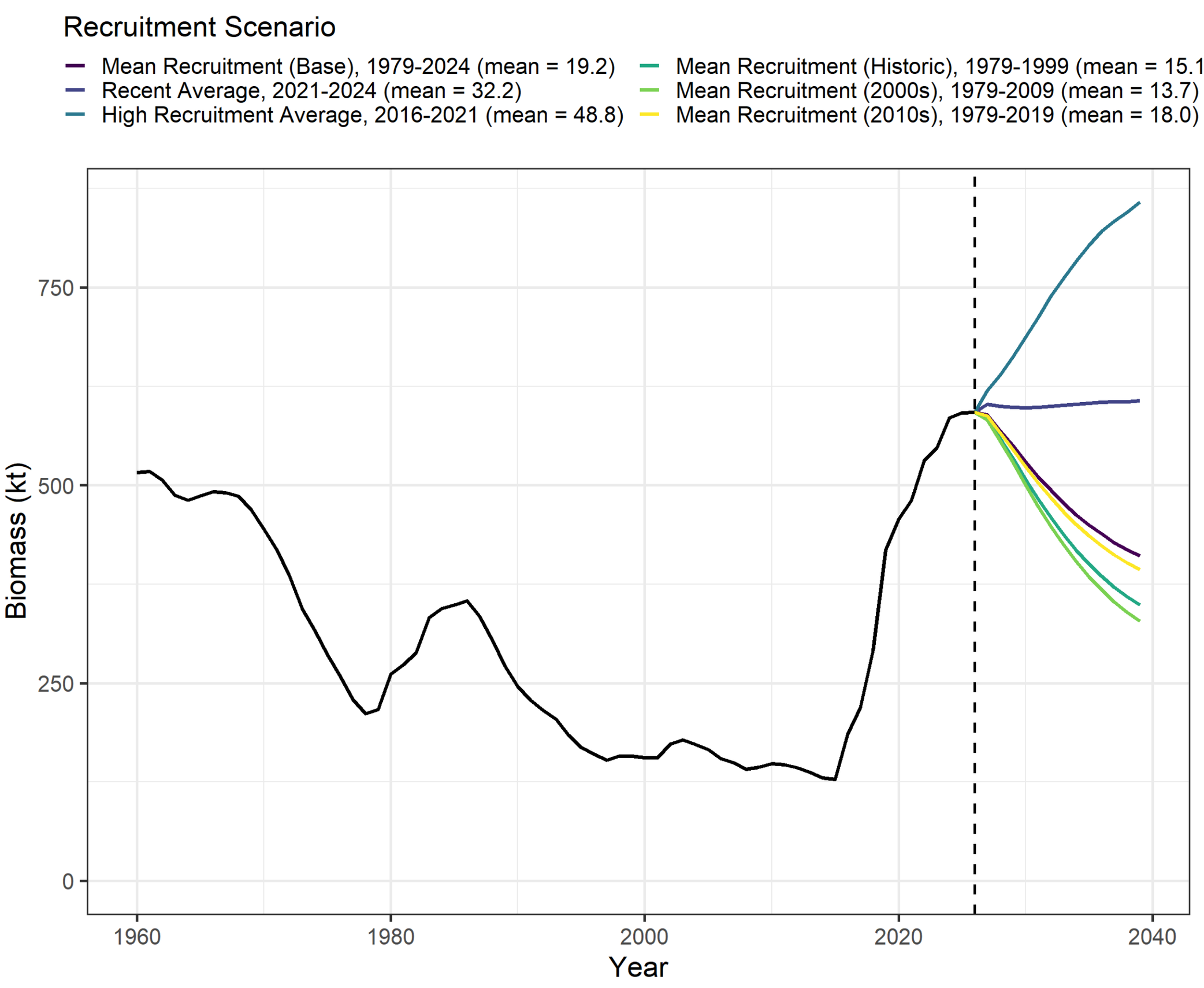
Year

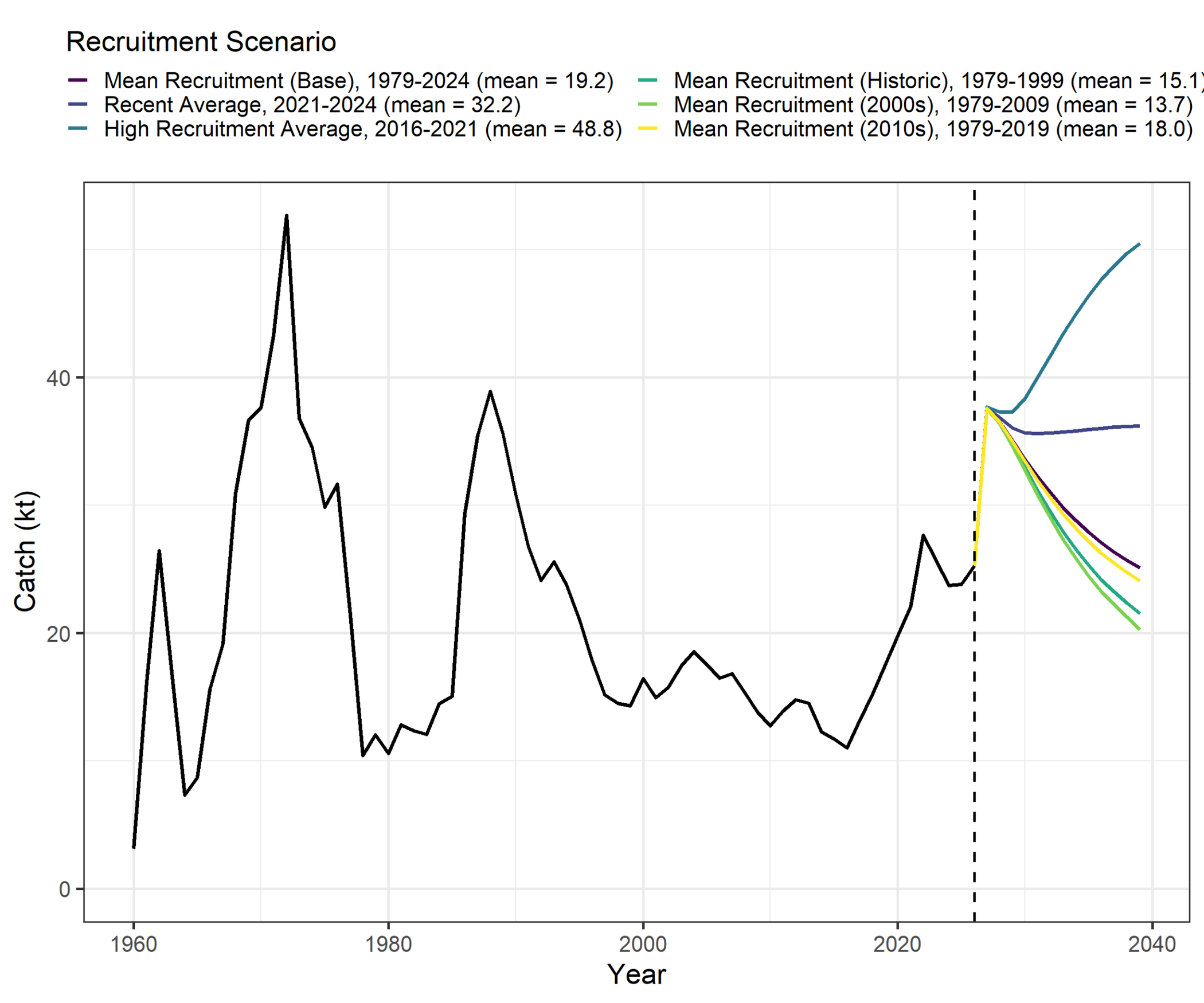
Fleet

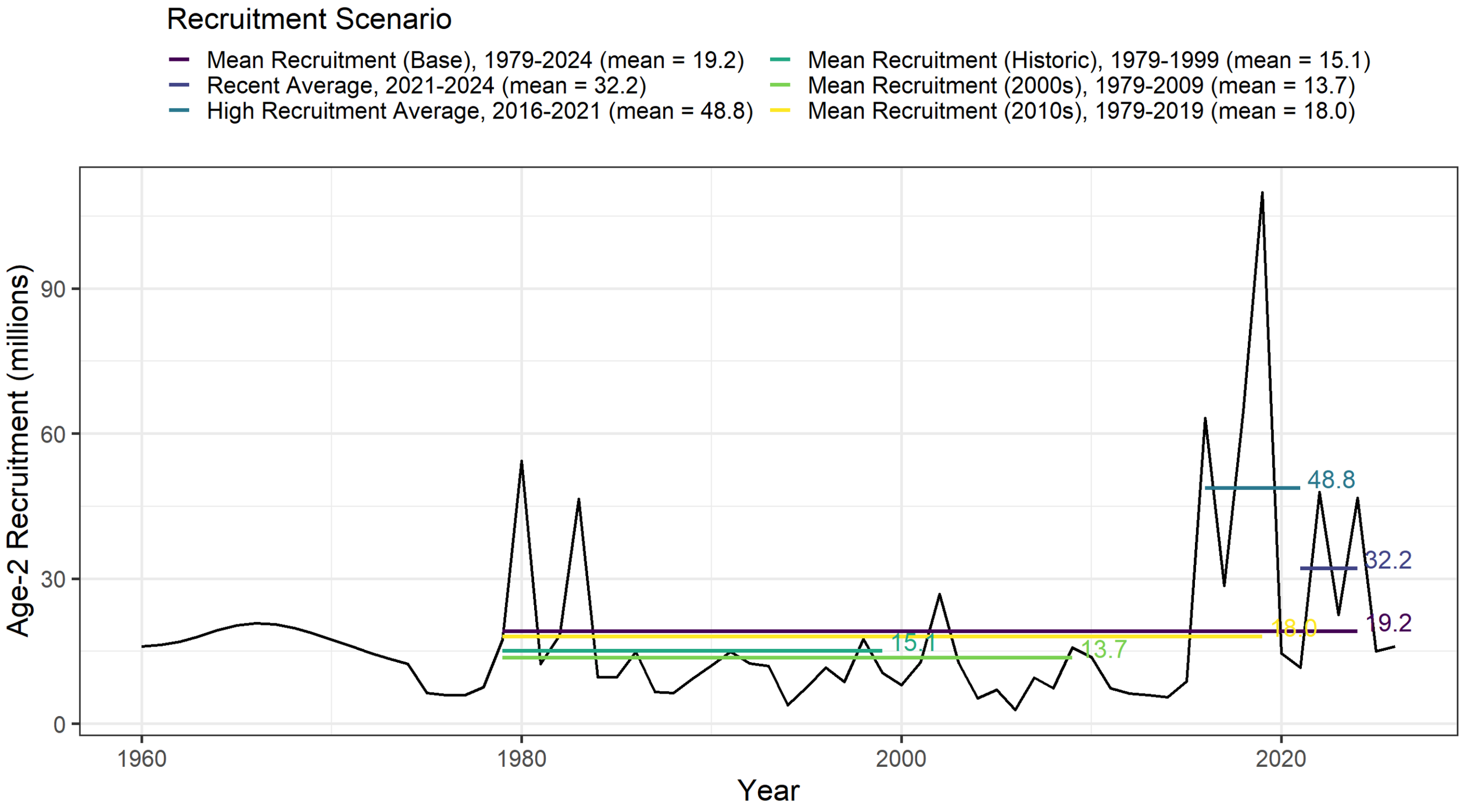
LLF

TF


Recruitment Scenario
Mean Recruitment (Base), 1979-2024 (mean = 19.2)
Recent Average, 2021-2024 (mean = 32.2)
High Recruitment Average, 2016-2021 (mean = 48.8)
Mean Recruitment (Historic), 1979-1999 (mean = 15.1
Mean Recruitment (2000s), 1979-2009 (mean = 13.7)
Mean Recruitment (2010s), 1979-2019 (mean = 18.0)
300
200
100
0
SSB (kt)
1960
1980
2000
2020
2040
Year

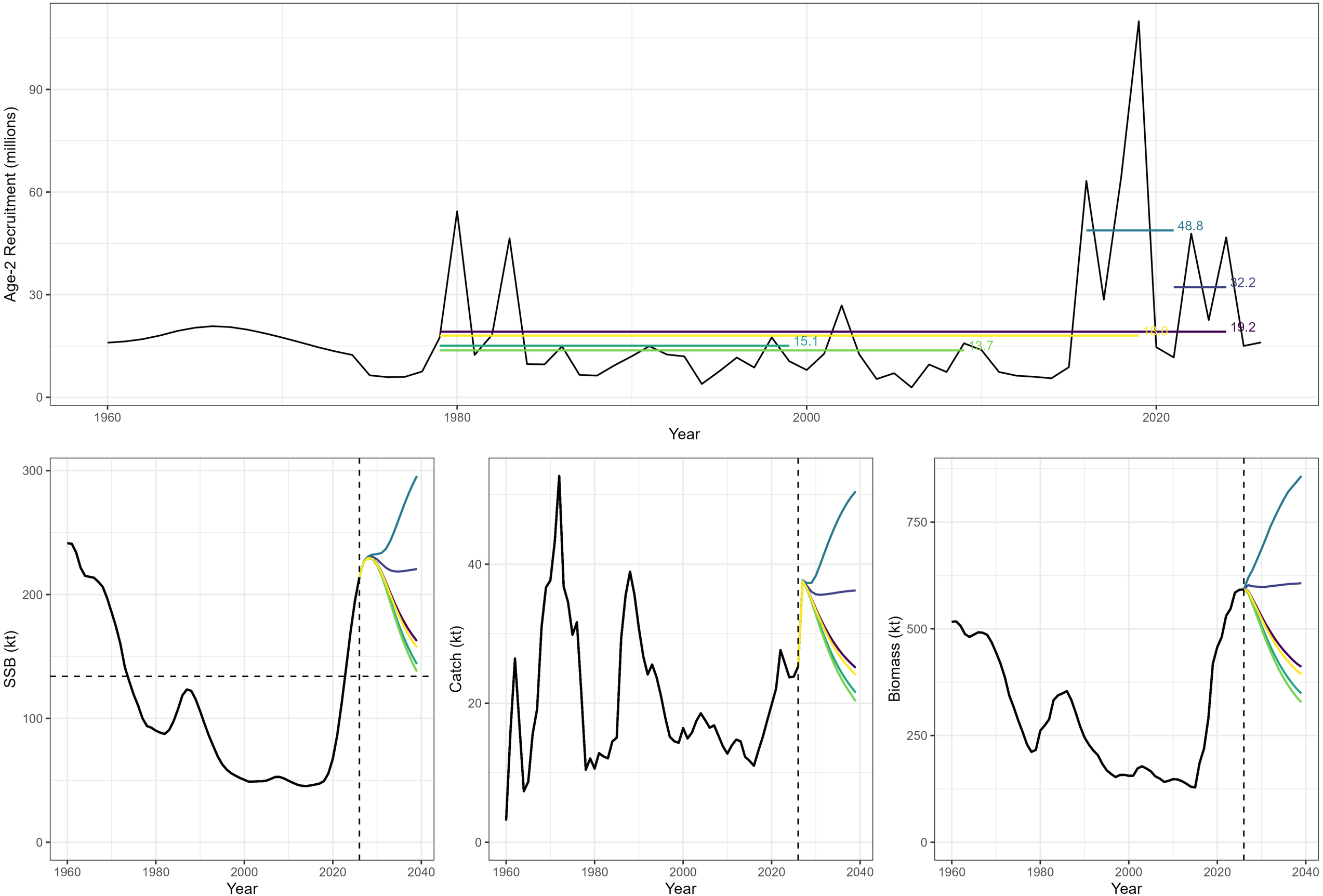

Recruitment Scenario
Mean Recruitment (Base), 1979-2024 (mean = 19.2)
Recent Average, 2021-2024 (mean = 32.2)
High Recruitment Average, 2016-2021 (mean = 48.8)
Mean Recruitment (Historic), 1979-1999 (mean = 15.1
Mean Recruitment (2000s), 1979-2009 (mean = 13.7)
Mean Recruitment (2010s), 1979-2019 (mean = 18.0)
Biomass (kt)
750
500
250
0
1960
1980
2000
2020
2040
Year

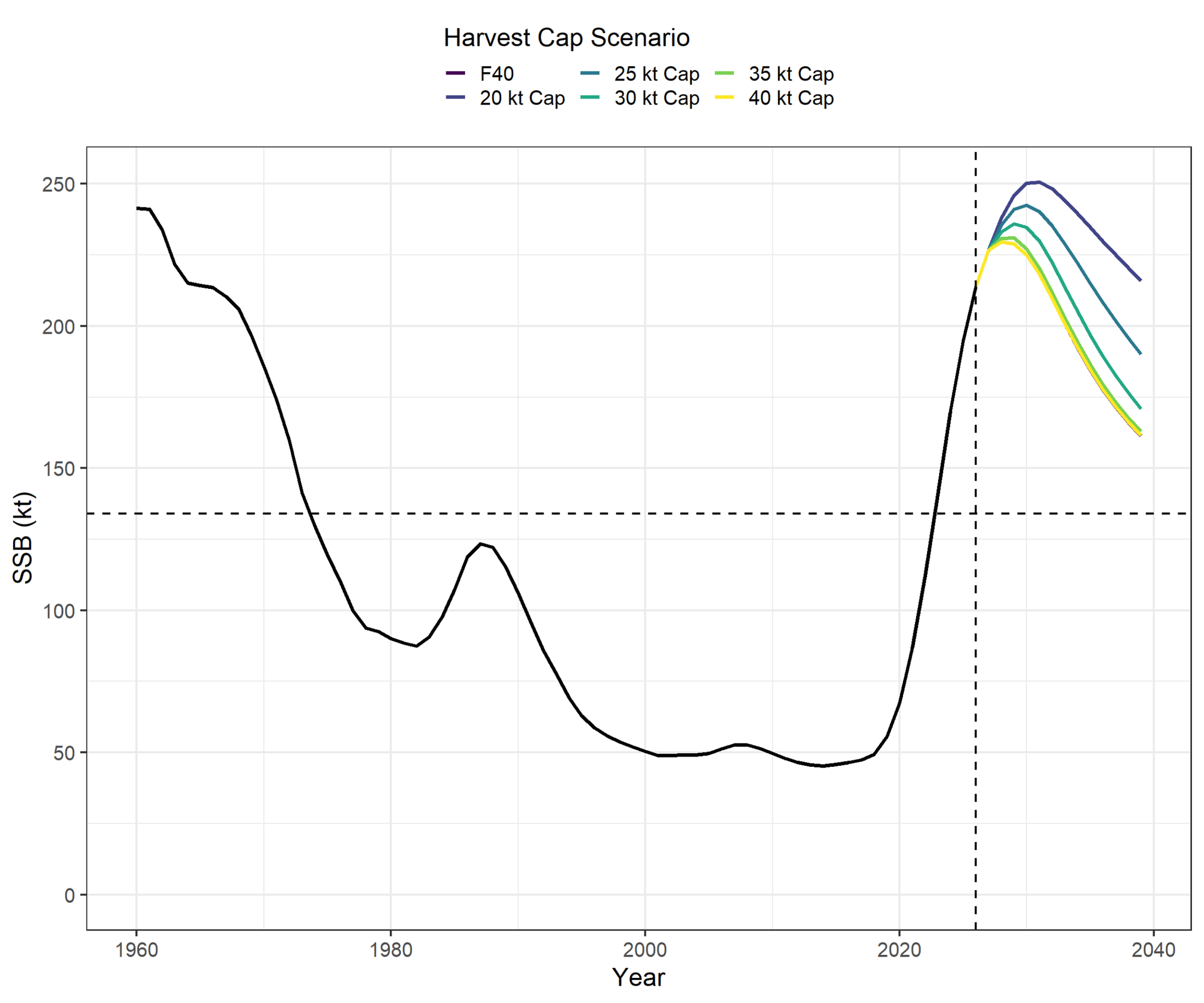

Recruitment Scenario
Mean Recruitment (Base), 1979-2024 (mean = 19.2)
Recent Average, 2021-2024 (mean = 32.2)
High Recruitment Average, 2016-2021 (mean = 48.8)
Mean Recruitment (Historic), 1979-1999 (mean = 15.1)
Mean Recruitment (2000s), 1979-2009 (mean = 13.7)
Mean Recruitment (2010s), 1979-2019 (mean = 18.0)
Catch (kt)
0
20
40
1960
1980
2000
2020
2040
Year

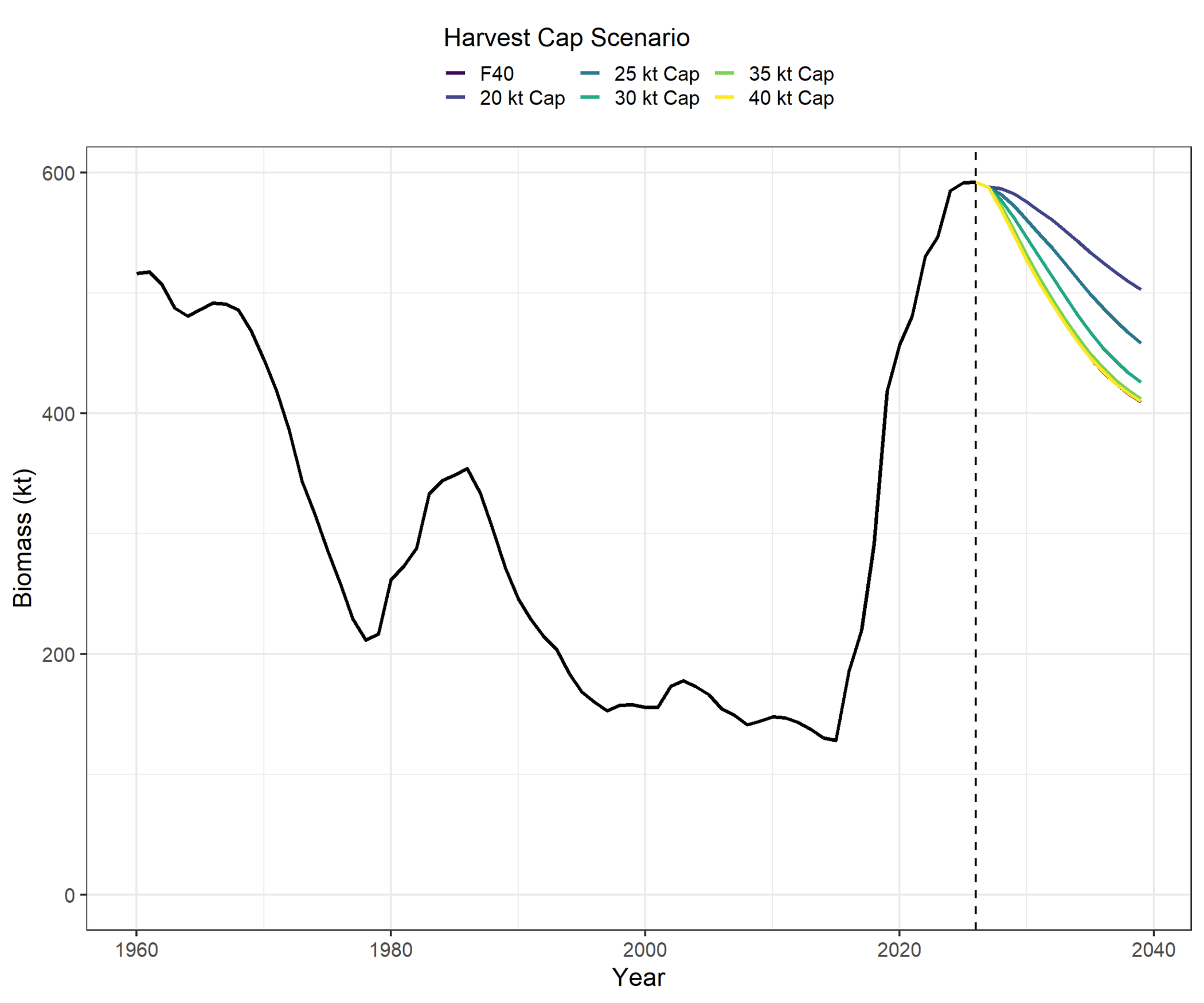

Recruitment Scenario
Mean Recruitment (Base), 1979-2024 (mean = 19.2)
Recent Average, 2021-2024 (mean = 32.2)
High Recruitment Average, 2016-2021 (mean = 48.8)
Mean Recruitment (Historic), 1979-1999 (mean = 15.1)
Mean Recruitment (2000s), 1979-2009 (mean = 13.7)
Mean Recruitment (2010s), 1979-2019 (mean = 18.0)
Age-2 Recruitment (millions)
0
30
60
90
1960
1980
2000
2020
Year
48.8
32.2
19.2
18.0
15.1
13.7

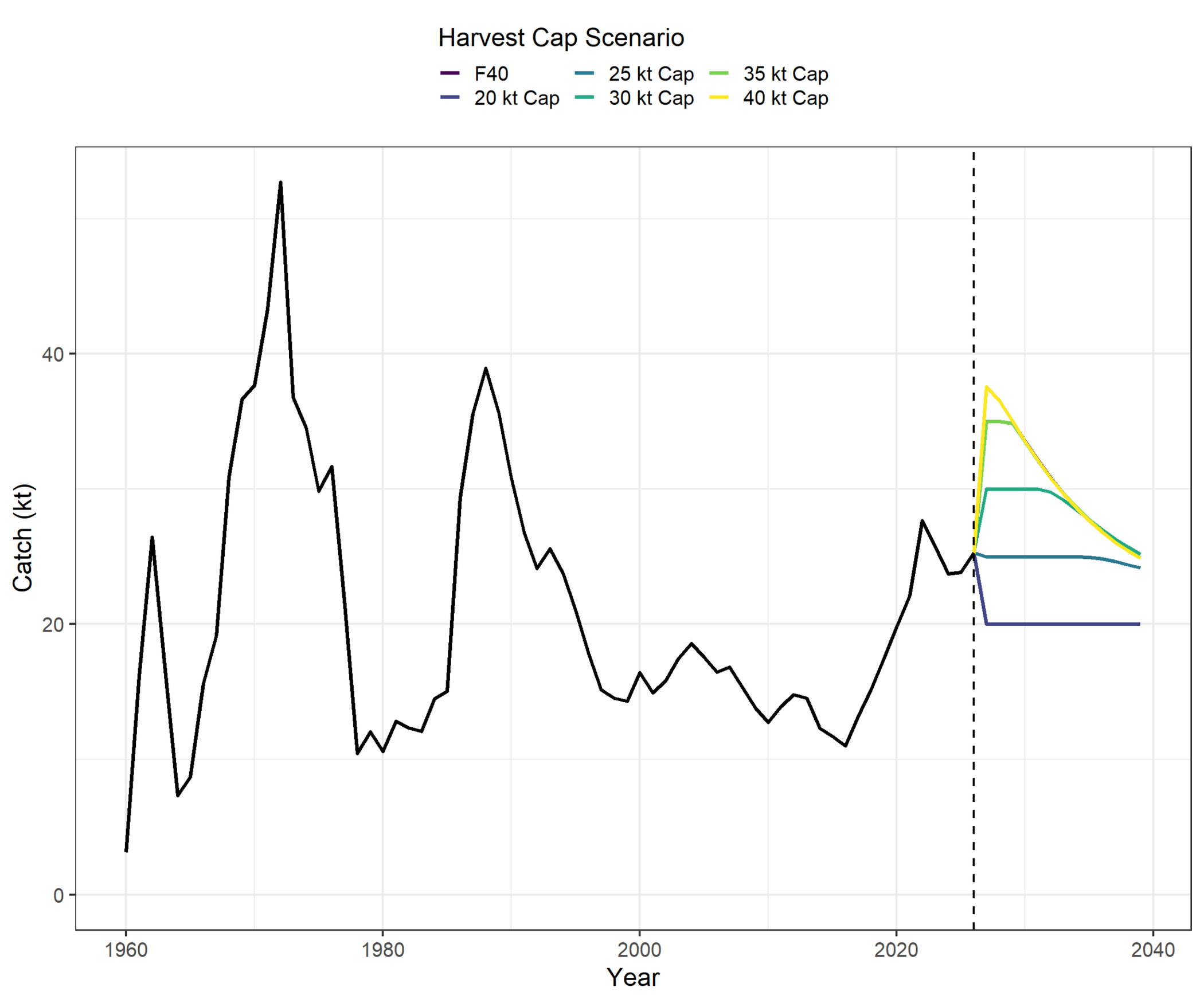
Recruitment Scenario

- Mean Recruitment (Base), 1979-2024 (mean = 19.2)
- Recent Average, 2021-2024 (mean = 32.2)
- High Recruitment Average, 2016-2021 (mean = 48.8)
- Mean Recruitment (Historic), 1979-1999 (mean = 15.1)
- Mean Recruitment (2000s), 1979-2009 (mean = 13.7)
- Mean Recruitment (2010s), 1979-2019 (mean = 18.0)




Harvest Cap Scenario
F40
20 kt Cap
25 kt Cap
30 kt Cap
35 kt Cap
40 kt Cap
SSB (kt)
250
200
150
100
50
0
1960
1980
2000
2020
2040
Year


Harvest Cap Scenario
F40
20 kt Cap
25 kt Cap
30 kt Cap
35 kt Cap
40 kt Cap
600
400
200
0
Biomass (kt)
1960
1980
2000
2020
2040
Year


Harvest Cap Scenario
F40
20 kt Cap
25 kt Cap
30 kt Cap
35 kt Cap
40 kt Cap
Catch (kt)
0
20
40
1960
1980
2000
2020
2040
Year

Harvest Cap Scenario

F40
20 kt Cap
25 kt Cap
30 kt Cap
35 kt Cap
40 kt Cap

SSB (kt)

250
200
150
100
50
0

Catch (kt)

40
20
0

Biomass (kt)

600
400
200
0

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

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S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

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S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

100

200

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

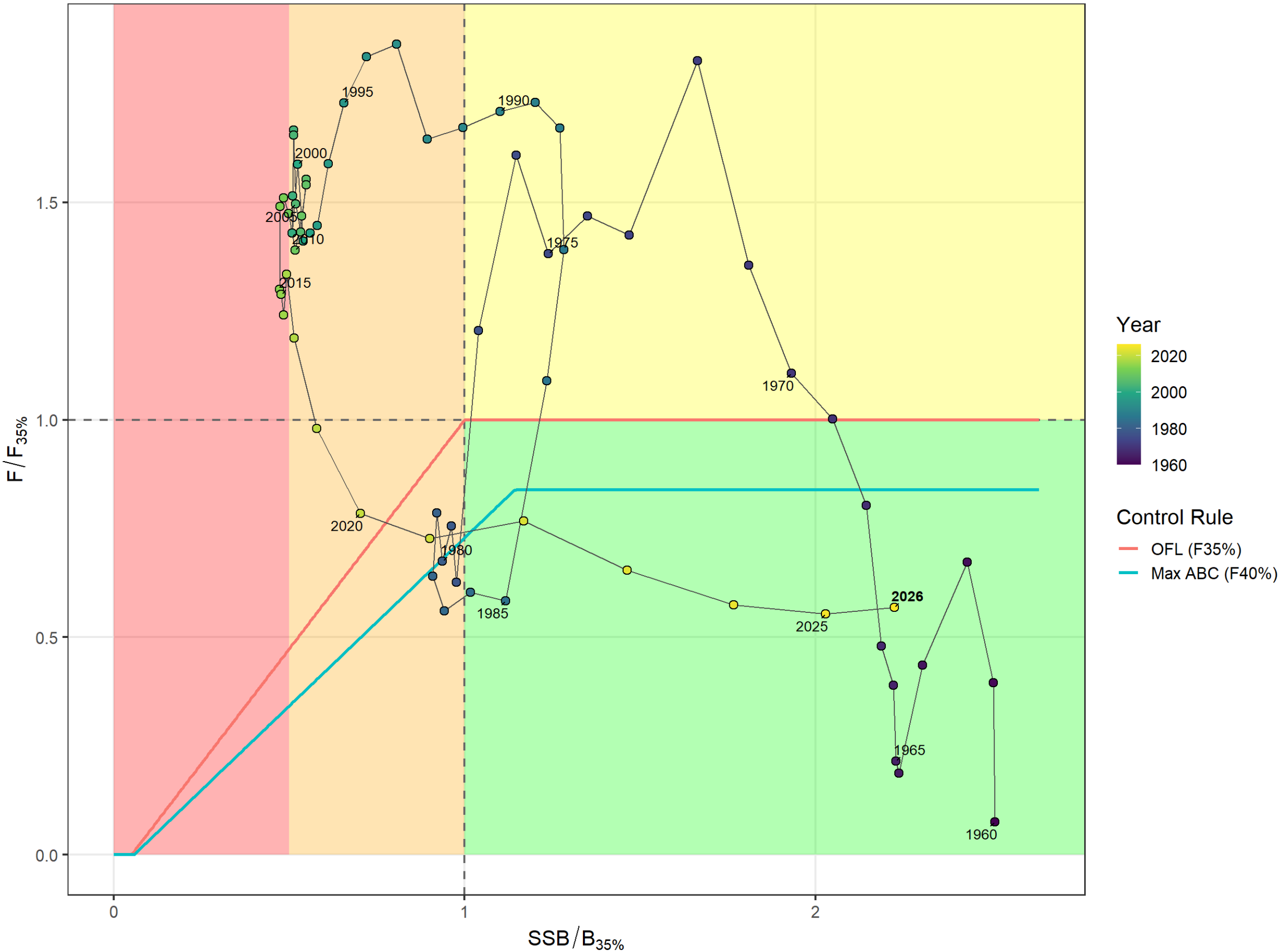
Fishing Mortality

Year

0.00
0.05
0.10
0.15

1960
1980
2000
2020
2040

Phase Plane Assuming B40 from ~2018



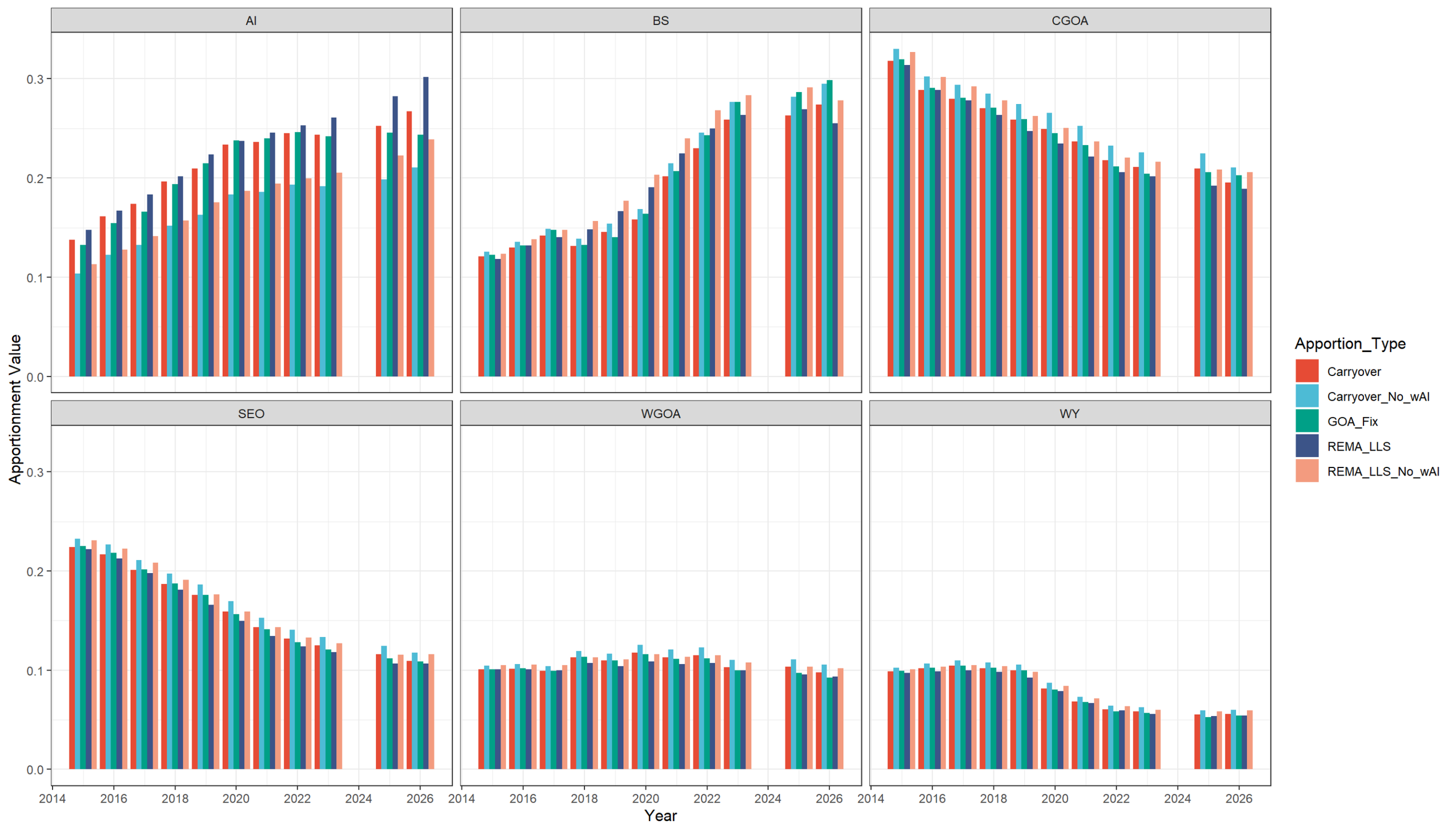
Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

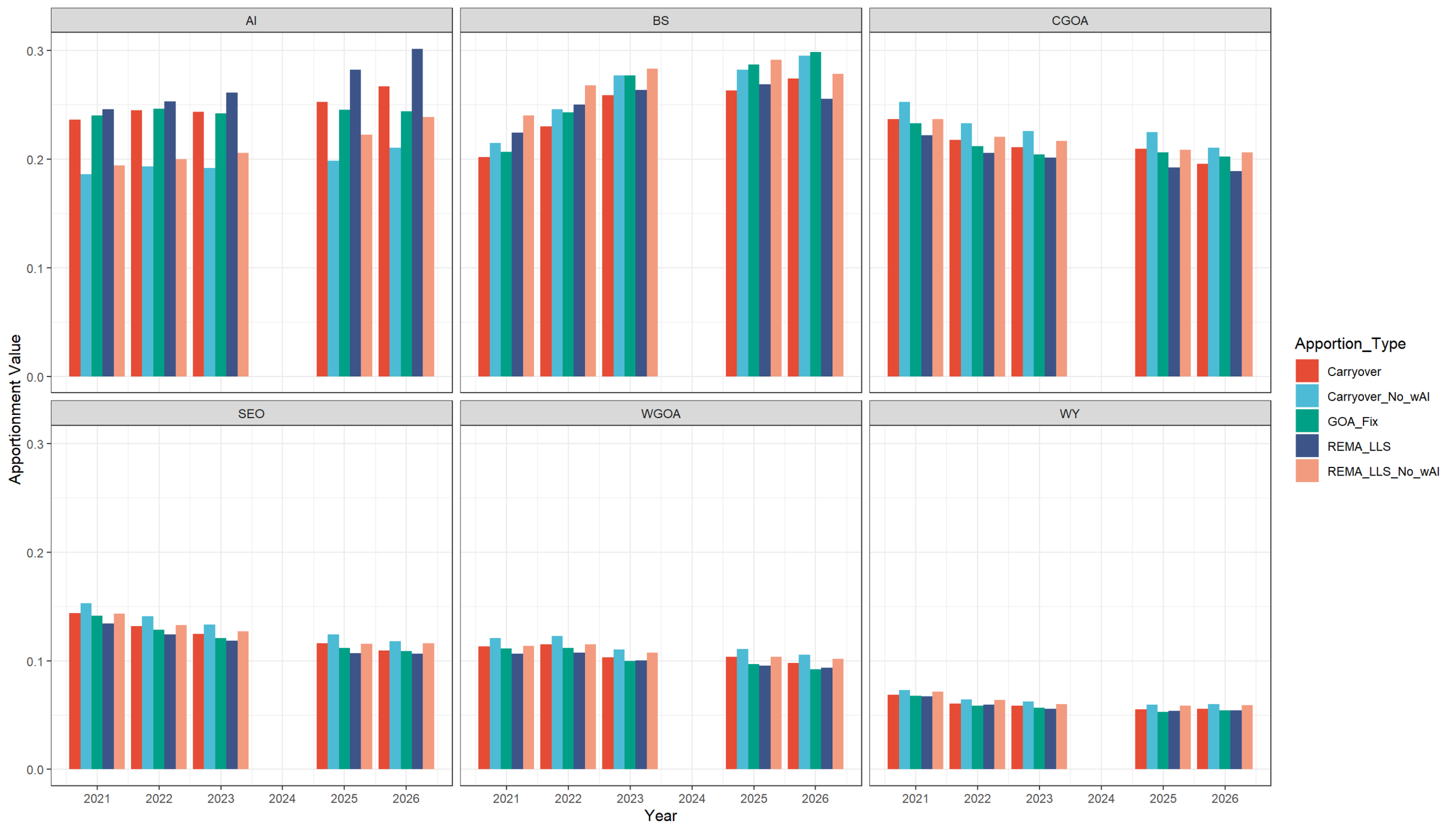
Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	11231	7487	18718	4118	7862	2246	4492	37436
2027	REMA_LLS_No_wAI	10482	8610	18344	3744	7862	2246	4492	37436
2027	Carryover	10108	9733	17595	3744	7487	2246	4118	37436
2027	REMA_LLS	9733	11231	16472	3369	7113	1872	4118	37436
2027	GOA_Fix	11231	8985	17221	3369	7487	1872	4492	37436
2028	Carryover_No_wAI	11291	7527	18818	4140	7904	2258	4516	37636
2028	REMA_LLS_No_wAI	10538	8656	18442	3764	7904	2258	4516	37636
2028	Carryover	10162	9785	17689	3764	7527	2258	4140	37636
2028	REMA_LLS	9785	11291	16560	3387	7151	1882	4140	37636
2028	GOA_Fix	11291	9033	17313	3387	7527	1882	4516	37636

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-14	-34	-17	-12	-18	-15	-20	-20
2027	REMA_LLS_No_wAI	-20	-25	-19	-20	-18	-15	-20	-20
2027	Carryover	-22	-15	-22	-20	-22	-15	-26	-20
2027	REMA_LLS	-25	-2	-27	-28	-26	-29	-26	-20
2027	GOA_Fix	-14	-21	-24	-28	-22	-29	-20	-20
2028	Carryover_No_wAI	-13	-34	-17	-12	-18	-15	-19	-20
2028	REMA_LLS_No_wAI	-19	-24	-18	-20	-18	-15	-19	-20
2028	Carryover	-22	-14	-22	-20	-22	-15	-26	-20
2028	REMA_LLS	-25	-1	-27	-28	-26	-29	-26	-20
2028	GOA_Fix	-13	-21	-23	-28	-22	-29	-19	-20

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	11231	2028	BS	NA	11291
2026	AI	NA	11421	7440	632	2027	AI	NA	7487	2028	AI	NA	7527
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	18718	2028	GOA	NA	18818
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	4118	2028	WGOA	NA	4140
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	7862	2028	CGOA	NA	7904
2026	WY	NA	2652	2652	2271	2027	WY	NA	2246	2028	WY	NA	2258
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4492	2028	SEO	NA	4516
2026	Total	57797	47008	38986	22161	2027	Total	44333	37436	2028	Total	44565	37636

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	254727	NA	18718	NA	NA
GOA	2028	252024	NA	18818	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	139483	NA	11231	NA	NA
BS	2028	138002	NA	11291	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	153559	NA	7487	NA	NA
AI	2028	151929	NA	7527	NA	NA


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI